GE Healthcare

# Discovery CT750 HD
# LightSpeed VCT
## Installation Manual

*(Book 1 of 2)*

MECHANICAL INSTALLATION

5308551-1EN
Rev. 5.0

# Book 1 of 2: Mechanical Installation

The information in this manual applies to the following GE Healthcare LightSpeed CT Scanners:

- LightSpeed VCT (with GOC6 console)
- Discovery CT750 HD also known as Lightspeed CT750 HD, High Definition, CTHD, HD, VCT HD, VCT High Definition.

The information in this manual does NOT apply to non-fixed (mobile) installations.

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# Revision History

| Rev | Date | Reason for change |
|---|---|---|
| 1 | 8/22/08 | Initial Release |
| 2 | 9/19/08 | FCTge40304, FCTge40305, FCTge40398, and FCTge40899. Updates to the System Configuration per program updates. |
| 3 | 10/08/08 | FCTge41659 |
| 4 | 10/28/08 | FCTge42654 |
| 5 | 11/21/08 | FCTge43611- *Language / Manager Sign off*<br>FCTge43087 - *Chapter 8 Load From Cold* |

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# IMPORTANT PRECAUTIONS

## LANGUAGE

| | |
|---|---|
| **WARNING**<br>(EN) | • This Service Manual is available in English only.<br>• If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.<br>• Do not attempt to service the equipment unless this service manual has been consulted and is understood.<br>• Failure to heed this warning may result in injury to the service provider, operator, or patient, from electric shock or from mechanical or other hazards. |
| **Предупреждение**<br>(BG) | • ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК.<br>• АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД.<br>• НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА.<br>• НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ. |
| **警告**<br>(ZH-CN) | • 本维修手册仅提供英文版本。<br>• 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。<br>• 未详细阅读和完全理解本维修手册之前，不得进行维修。<br>• 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。 |
| **VÝSTRAHA**<br>(CS) | • Tento provozní návod existuje pouze v anglickém jazyce.<br>• V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.<br>• Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.<br>• V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik. |
| **ADVARSEL**<br>(DA) | • Denne servicemanual findes kun på engelsk.<br>• Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.<br>• Forsøg ikke at servicere udstyret medmindre denne servicemanual har været konsulteret og er forstået.<br>• Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk, mekanisk eller anden fare for teknikeren, operatøren eller patienten. |

**WAARSCHUWING**
(NL)

- Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.
- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden voordat deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

**HOIATUS**
(ET)

- Käesolev teenindusjuhend on saadaval ainult inglise keeles.
- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

**VAROITUS**
(FI)

- Tämä huolto-ohje on saatavilla vain englanniksi.
- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

**ATTENTION**
(FR)

- Ce manuel de service n'est disponible qu'en anglais.
- Si le technicien du client a besoin de ce manuel dans une autre langue que l'anglais, c'est au client qu'il incombe de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que le manuel service n'a pas été consulté et compris
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

**WARNUNG**
(DE)

- Diese Serviceanleitung existiert nur in Englischer Sprache.
- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es aufgabe des Kunden für eine Entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch stromschläge, Mechanische oder Sonstige gefahren kommen.

**ΠΡΟΕΙΔΟΠΟΙΗΣΗ**
(EL)

- Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο.
- Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εαν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Εαν δε λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

**FIGYELMEZTETÉS**
(HU)

- Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.
- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése.
- Ne próbálja elkezdeni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

**AÐVÖRUN**
(IS)

- Þessi þjónustuhandbók er eingöngu fáanleg á ensku.
- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

**AVVERTENZA**
(IT)

- Il presente manuale di manutenzione è disponibile soltanto in inglese.
- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto
- Il non rispetto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto, alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.

**警告**
(JA)

- このサービスマニュアルには英語版しかありません。
- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

**경고**
(KO)

- 본 서비스 지침서는 영어로만 이용하실 수 있습니다 .
- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우 , 번역 서비스를 제공하는 것은 고객의 책임입니다 .
- 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오 .
- 이 경고에 유의하지 않으면 전기 쇼크 , 기계상의 혹은 다른 위험으로부터 서비스 제공자 , 운영자 혹은 환자에게 위해를 가할 수 있습니다 .

| | |
|---|---|
| **BRDINJUMS** (LV) | • Šī apkalpes rokasgrāmata ir pieejama tikai angļu valodā.<br>• Ja klienta apkalpes sniedzējam nepieciešama informācija citā valodā, nevis angļu, klienta pienākums ir nodrošināt tulkošanu.<br>• Neveiciet aprīkojuma apkalpi bez apkalpes rokasgrāmatas izlasīšanas un saprašanas.<br>• Šī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumu apkalpes sniedzējam, operatoram vai pacientam. |
| **ĮSPĖJIMAS** (LT) | • Šis eksploatavimo vadovas yra prieinamas tik anglų kalba.<br>• Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, numatyti vertimo paslaugas yra kliento atsakomybė.<br>• Nemėginkite atlikti įrangos techninės priežiūros, nebent atsižvelgėte į šį eksploatavimo vadovą ir jį supratote.<br>• Jei neatkreipsite dėmesio į šį perspėjimą, galimi sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų paslaugų tiekėjui, operatoriui ar pacientui. |
| **ADVARSEL** (NO) | • Denne servicehåndboken finnes bare på engelsk.<br>• Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse.<br>• Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.<br>• Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer. |
| **OSTRZEŻENIE** (PL) | • Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.<br>• Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.<br>• Nie próbować serwisować wyposażenia bez zapoznania się i zrozumienia niniejszego podręcznika serwisowego.<br>• Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia elektrycznego, zagrożenia mechanicznego bądź innego. |
| **ATENÇÃO** (PT) | • Este manual de assistência técnica só se encontra disponível em inglês.<br>• Se qualquer outro serviço de assistência técnica solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.<br>• Não tente consertar o equipamento sem ter consultado e compreendido este manual de assistência técnica.<br>• O não cumprimento deste aviso pode pôr em perigo a segurança do técnico, do operador ou do paciente devido a choques elétricos, mecânicos ou outros. |

**ATENȚIE**
(RO)

- Acest manual de service este disponibil numai în limba engleză.
- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării şi înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

**ОСТОРОЖНО!**
(RU)

- Данное руководство по обслуживанию предлагается только на английском языке.
- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по обслуживанию, оператор или пациент получат удар электрическим током, механическую травму или другое повреждение.

**UPOZORNENIE**
(SK)

- Tento návod na obsluhu je k dispozícii len v angličtine.
- Ak zákazníkov poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednos″ou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia skôr, ako si neprečítate návod na obluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže vyústi″ do zranenia poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, do mechanického alebo iného nebezpečenstva.

**ATENCION**
(ES)

- Este manual de servicio sólo existe en inglés.
- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

**VARNING**
(SV)

- Den här servicehandboken finns bara tillgänglig på engelska.
- Om en kunds servicetekniker har behov av ett annat språk än engelska ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

**DIKKAT**
(TR)

- Bu servis kilavuzunun sadece ingilizcesi mevcuttur.
- Eğer müşteri teknisyeni bu kilavuzu ingilizce dişinda bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kilavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyariya uyulmamasi, elektrik, mekanik veya diğer tehlikelerden dolayi teknisyen, operatör veya hastanin yaralanmasina yol açabilir.

## DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “Damage in Shipment” written on all copies of the freight or express bill before delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on http://egems.med.ge.com/edq/home.jsp
- Contact your local service coordinator for more information on this process.

Rev. June 13, 2006

## CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE’s electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE’s field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

## IMPORTANT...X-RAY PROTECTION

X-ray equipment, if not properly used, may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, GE Healthcare Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, GE Healthcare Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

## IMPORTANT...RADIOACTIVE MATERIAL HANDLING

Only employees formally trained in radioactive materials handling and this equipment are authorized by the GE Healthcare Radiation Safety Officer to use radioactive materials to service this equipment.

GE Healthcare is required to notify the applicable U.S. state agency PRIOR to any source service event involving pin source handling. See NUC/PET radioactive material guides for specific instruction or contact your EHS Specialist.

A radiation survey must be performed when a pin source has been removed and replaced. See Radiation Survey Form Instructions or contact your EHS Specialist.

Rev 2 (July 21, 2005)

## LITHIUM BATTERY CAUTIONARY STATEMENTS

**CAUTION Risk of Explosion**

**Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer’s instructions.**

**ATTENTION Danger d’Explosion**

**Il y a danger d’explosion s’il y a replacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d’un type recommandé par le constructeur. Mettre au rébut les batteries usagées conformément aux instructions du fabricant.**

## OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives. GE personnel, please use the GE Healthcare PQR Process to report all omissions, errors, and defects in this publication.

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# Table of Contents - Book 1

# *Preface*
# *Publication Conventions*

Please become familiar with the conventions used within this publication before proceeding.

# Section 1.0 Safety & Hazard Information

## 1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicated potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that is applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

### EXAMPLES OF HAZARD STATEMENTS USED

A few examples are provided below. They include paragraph prefixes and modified text styles.

**CAUTION**
**Pinch Points**
**Loss of Data**
**Sharp Objects**

**Caution is used when a hazard exists that can or could cause minor injury to self or others if instructions are ignored. They include for example:**

- **Loss of critical patient data**
- **Crush or pinch points**
- **Sharp objects**

**DANGER**
**EXCESSIVE**
**VOLTAGE**
**CRUSH**
**POINT**

**DANGER IS USED WHEN A HAZARD EXISTS THAT WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:**

- **ELECTROCUTION**
- **CRUSHING**
- **RADIATION**

**WARNING**
**ROTATING**
**EQUIPMENT**
**BARE WIRES**

**WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:**

- Potential for shock
- Exposed wires
- Failure to Tag and lockout system power could allow for un-command motion.

**NOTICE Equipment Damage Possible**

**Notice is used when a hazard is present that can cause property damage but has absolutely <u>no personal injury risk</u>.They can include:**

- Disk drive will crash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

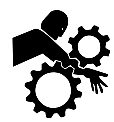
It's important that the reader not ignore hazard statements in this document.

## 1.2 Graphical Representation

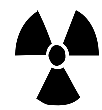
Important information will always be preceded by the exclamation point ⚠ contained within a triangle, as seen throughout this chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.

**ELECTRICAL**

**MECHANICAL**

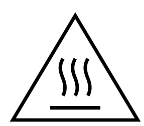


**RADIATION**

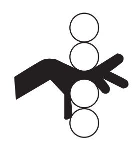


**LASER**


**HEAT**

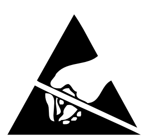


**PINCH**



Some others make you aware of specific procedures that should be followed.

**AVOID STATIC ELECTRICITY**



**TAG AND LOCK OUT**


**WEAR EYE PROTECTION**

# Section 2.0 Publication Conventions

## 2.1 General Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as Purpose, Example, Comment and Note) are used to identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

### EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

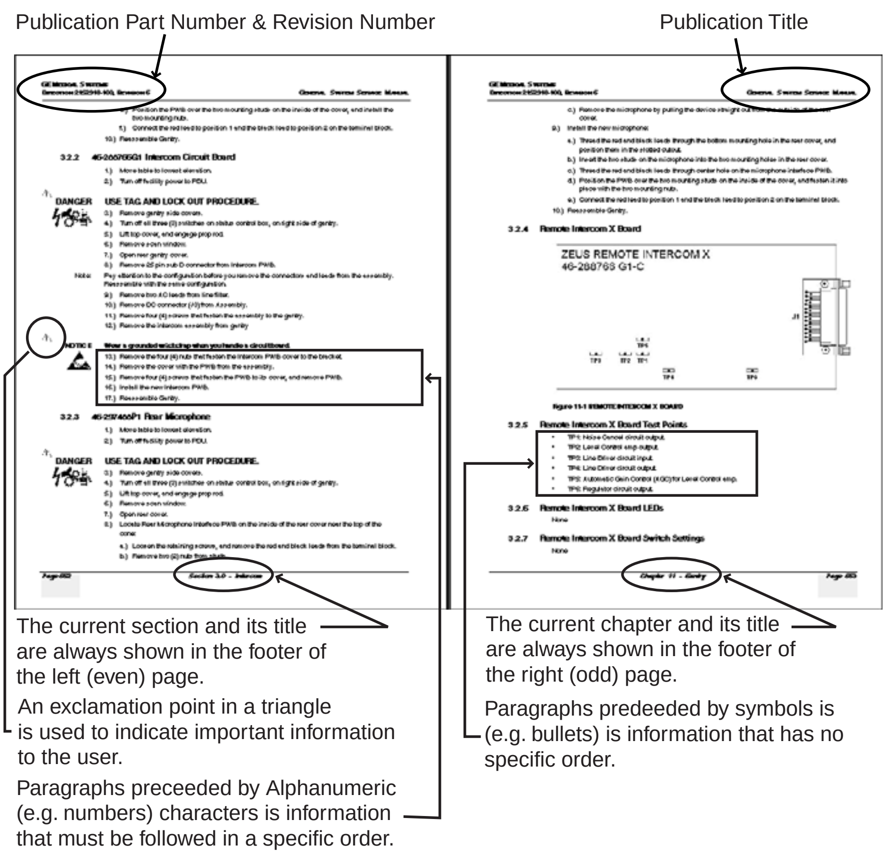
**Purpose: Introduces and provides meaning as to the information contained within the chapter, section or subsection, such as used at the beginning this chapter for example.**

Note: **Conveys information that should be considered important to the reader.**

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

*Comment: Represents "additional" information that may or may not be relevant.*

## 2.2 Page Layout



Headers and footers in this publication are designed to allow you to quickly identify your location. The document's part number and revision number appears in every header on every page. Odd numbered page footers indicate the current chapter, its title, and current page number. Even page footers show the current section and its title, as well current page number.

## 2.3 Computer Screen Output/Input Character Styles

Within this publication different character styles are used to indicate computer input and output text. Character (input, output, and variable) styles are used and applied to the text within a paragraph so as to indicate directions. Computer screen output and input is also formatted using mono (fixed width) spaced fonts.

Example: Fixed Output

This paragraph denotes computer screen fixed output. It's output is fixed from the sense that it does not vary from application to application.It's the most commonly used style used to indicate filenames, paths, and text.

Example: Variable Output

*This paragraph denotes computer screen output that is variable. Its output varies from application to application. Variable output is sometimes found placed between greater than and lesser than operators. For example: <variable_ouput>*

Example: Fixed Input

**This paragraph denotes fixed input. It's typed input that will not vary from application to application. Fixed text the user is required to supply as input.**

Example: Variable Input

***This paragraph denotes computer input that can vary from application to application. Variable text the user is required to supply as input. Variable input sometimes is placed between greater than and lesser than operators. For example: <variable_input>. In these cases, the (<>) operators are dropped prior to input. Exceptions are noted in the text.***

## 2.4 Buttons, Switches and Keyboard Inputs (Hard & Soft Keys)

Different character styles are used to indicate actions requiring the reader to press either a hard or soft button, switch, or key. Physical hardware, such as buttons and switches, are called hard keys because they are hard wired or mechanical in nature. A keyboard or on/off switch would be a hard key. Software or computer generated buttons are called soft keys because they are software generated. Software driven menu buttons are an example of such keys. Soft and hard keys are represented differently in this publication.

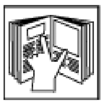
Example: Hard Keys

A power switch **ON/OFF** or a keyboard key like **ENTER** is indicated by applying a character style that uses both over and under-lined bold text that is bold. This is a hard key.

Example: Soft Keys

Whereas the computer MENU button that you would click with your mouse or touch with your hand uses over and under-lined regular text. This is a soft key.

# Chapter 1
# Position Subsystems



**NOTICE**

- **Record data collected from procedures in this chapter into Form 4879 when directed.**
- **Only use the installation manual that arrives with your system. Any other revisions of this manual may not exactly match your system.**

## Section 1.0 Installer/FE Notices

### 1.1 Shipping, Warehouse and Transportation Warning

This gantry should be moved using the shipping dollies only. Do not lift or move it using a lift truck under the gantry frame.

### 1.2 International Shipments

- Use dollies to remove the gantry from the shipping skid and to transport the gantry to the customer’s site.
- If lifting is required, instructions are in the *Pre-Installation Manual* for this system.

### 1.3 On Site Warning

This system requires a gantry bearing gap inspection *before* electrical calibration is started. See Gantry Bearing Gap Inspection, on page 51.

### 1.4 Service Actions

If the bearing inspection fails the FE opens a dispatch and does not continue with the electrical calibration procedures.

# Section 2.0 Introduction

This chapter describes how to mount, position, and level the CT scanner subsystems.

Note: Before you start the installation, make sure the site preparation complies with conditions and instructions found in the *Pre-installation Manual* for this system. Failure to comply results in excessive installation delay and potential increased, unrecoverable installation costs. This product is designed to meet specific mechanical installation standards that should be reviewed prior to installing this system.

## 2.1 Skill Set Required

**NOTICE** **All mechanical installation leaders and helpers must meet these minimum requirements.**

| **Mechanical** | | |
|---|---|---|
| **Installation Leader** | **Safety CBTs (6)** | **Helper OJT with Trainer** |
| Must be Healthcare Institute (HCI) trained as FE or mechanical contractor.<br>Must complete all training CBT courses.<br>My Learning verification. | Complete all training CBT courses.<br>My Learning verification. | Complete one training installation with trainer.<br>Complete Train-the-Trainer training<br>Pass all assessments |
| **Tools** | **Access** | **Communication** |
| Must have all required tools as listed in this manual. | Must have access to some GE Healthcare web sites and tools. | Must be able to complete all required paperwork and complete iTraks as required. |
| **Electrical** | | |
| **In-Residence HCI Training** | **Safety CBT** | |
| Must be trained to complete electrical tasks<br>Must complete HD training course in Milwaukee.<br>My Learning verification. | Complete all training CBT courses.<br>My Learning verification. | |

## 2.2 All Installers

Only people trained and experienced in the installation process of a GE CT scanner should perform the work described in this document.

A person doing the installation should have:

- Installed three (3) VCT products as a helper prior to leading an installation.
- Completed all of the above training courses and successfully passed all related tests.
- A working understanding of the mechanical installation process.
- Completed Environmental Health and Safety (EHS) training, completed Hazcom training, completed Eye & Personal Protection Safety, completed Blood Borne Pathogens training, and must understand and apply lockout/ tagout processes
- All training and records must be verifiable and available in My Learning.
- Professional communication skills needed to communicate with customer any changes or additional requirements needed to complete the installation project.
- Must be trained to complete site-ready inspections.
- Installation leader must have completed project management, leadership training, and all required training prior to becoming a project leader.
- Mechanical installation teams must have basic measurement skills, system layout skills, and the ability to read and understand blueprints.

## 2.3 Floor and Room Preparation

### 2.3.1 Preparation

The PMI notifies the installation team if all requirements are not met. It is the purchaser's (buyer) responsibility to provide an approved support structure and an approved method of mounting. General Electric is not responsible for any failure of the support structure or method of anchoring.

The system has a total floor load of approximately 3021 kg (6660 lbs). About 2291 kg (5050 lbs), including patient (227 kg (500 lbs)) is concentrated in the table-gantry assembly. For more information, refer to the *Pre-installation Manual* for this system.

### 2.3.2 Flooring

Do not place the scanner on any resilient flooring. Resilient tile or carpeting may slowly yield over a period of time and disturb the alignment of the table to the gantry. Refer to the floor template to determine locations where resilient flooring material should be removed.

Limitations include:

- No part of the floor surface within the table, gantry, or the two interface areas between table and gantry should be higher than the support areas for the table and gantry.
- The floor structure must withstand the occupied weight of table and gantry, as well as the individual contact area loading of these components.
- The method and placement of anchors or through bolts must not reduce the structural strength of the floor.
- In some circumstances, the final floor may not be installed. Refer to Chapter 8.0 in the *Pre-Installation Manual* for this system.

If you have to remove the gantry covers in order to move the gantry into the room, refer to Appendix A for the cover removal procedure. Please read the notice statement on page 169 before removing the gantry covers.

## 2.4 Overview

Note: Installation paperwork is required for all installations.

Procedures in this chapter provide detailed instructions to position, level, and anchor the gantry and table securely for operation. The system uses adjustable leveling pads to support the gantry and table. The gantry has four (4) primary leveling pads located on the gantry base. The table has four (4) pads used for leveling it.

The process you follow is:

1.) Use the room-layout template to determine the general position of the gantry and table.
2.) Move the gantry into position.
3.) Level gantry.
4.) Use the laser tool to position the table relative to the gantry.
5.) Level the table to the gantry, and anchor the system.
6.) Complete the mechanical installation section of GE Form e4879.

Note: Use the template to position the system. Use the gantry and table to locate and drill the anchor holes. Drill the anchor holes with the system in place. Refer to Section Section 4.0 on page 41 for an example of this procedure. This CT system installation procedure requires the items listed in Section Section 2.6 on page 32.

## 2.5 Pre-Installation Template

**Always** use the room layout template (two pieces), during installation (see Table 1-2, for part number). The gantry and table are not properly aligned if existing holes are used. The template shows the location of the gantry and table anchor holes.

The applicable template is shipped with the system. It is located on the middle shelf on the Lean Installation Cart. You can also order it via the web GEMS BUY, from Coakley-Tech.

| Table | Part Number |
|---|---|
| GT1700 | 5111499 |
| GT2000 | 5111499 |

**Table 1-2 Room Layout Templates**

## 2.6 Required Common Tools and Supplies

The following tools and supplies should be included in the standard CT installation tool kit. The tools listed represent the minimum tools required for installing this CT scanner.

### Wrenches

- Standard and metric combination wrench sets
- Standard and metric hex key (Allen wrench) sets
- ½ in. and 3/8 in. drive torque wrench: 0-100 N-m (0-100 ft.-lb.) Must be calibrated yearly.

## Sockets and Extensions

- 3/8 in. drive metric and standard socket set
- 1 in, 1-1/8 in, 1-1/4 in, 1-1/2 in, 1-5/8 in sockets
- 1/2 in. drive ratchet wrenches
- 3/4 in. deep well socket
- Metric hex bit set 1/4 in. or 3/8 in. drive, including 10 mm and 14 mm hex bit
- 3/8 in. drive universal joint
- 21 mm socket (optional)

## Screwdrivers

- Phillips screwdriver set (small, medium, and large)
- Straight blade screwdriver set (small, medium, and large)
- Pozi-drive 0 (S)
- Pozi-drive #1

## Drill Bits

- Complete set of standard (U.S.) drill bits (optional)
- Metric tap set (Optional)
- ½ in. masonry bit, min. 10 in. long USA; 12 in. optional (Bit must not be metric.)
- 4 in. (100 mm) hole saw with 1/4 in. (6 mm) masonry bit (to remove flooring)

## Power Tools

- 3/8 in. or ½ in. drill, cordless or electric
- Reciprocating saw (Sawzall or equivalent) and assorted blades
- Hammer drill & bit (8 in. min, 12 in. max)
- Sears shop vacuum or equivalent, with HEPA or drywall dust filter
- 25 ft. extension power cords

## Electrical Tools

- DVM capable of reading 0.5 ohms or less
- Continuity Tester
- FE Electronic Monitor Static Kit or equivalent static mat kit with ground wrist strap (P/N 2220482)
- Dale 600 or 601 Leakage meter or equivalent
- Temperature/humidity tool: Oregon Scientific Wireless Weather Station Model BAR608HGA or equivalent

## Hand Tools

- Ball-peen hammer (1 lb. or 2 lb.)
- Tongue & groove pliers (large)
- Diagonal cutting pliers, large (to cut 1/0 ground)
- Diagonal cutting pliers, small and large
- Large pry bar 460 mm – 600 mm (18 in.–24 in.)

### Recommended Levels

- Johnson Magnetic Level, model 7500M – 225 mm (9 in.)
- Johnson Professional Box Beam Level, model 9624 – 600 mm (24 in.)
- Digital level with accuracy of ± 0.1° – 225 mm (9 in.)
- Johnson Professional Box Beam Level – 1225 mm (48 in.) (Optional)

### Personal Safety Equipment

- Safety shoes*
- Safety glasses*
- Leather gloves
- Knee pads

* Required items

- LOTO kit* -- MUST have tags and appropriate lock(s)
- Hearing protection*
- 2 m (6 ft.) or 4 m (8 ft.) step ladders or equivalent

### System Cleaners

Purchase Locally:

- Windex glass cleaner or equivalent
- Scrubbing Bubbles bathroom cleaner or equivalent
- Formula 409 grease cutting cleaner or equivalent
- WD-40

### Other

Purchase locally (available at office supply stores): a China marker or wax marking pencil or equivalent, any color that is visible on the floor where the system is being installed. Permanent markers are often used if the lines will be covered by the product.

### GE Tools

- System Alignment Kit (p/n 5148193 or p/n 5272090)
  (This tool may not be available via the tool pool in some areas.)
- Tilt Alignment Tool (p/n 5309601)
- Wrench (p/n 5313542)

# Section 3.0 Delivery Procedure

## 3.1 System Transportation - Temperature Extremes

When transporting the CT system, ensure that the system is not exposed to temperatures or humidity outside the following specifications:

Temperature: 0º to +120º F (-18º to +49º C)

Humidity: 20% to 80%

NOTICE **Component freezing occurs if the CT system is exposed to temperatures below 0º F (-18º C) for a period longer than two days.**

**Allow a minimum of 12 hours for the CT system to adjust to ambient room temperature prior to installation.**

## 3.2 Stored Systems

If your system was stored for more than three months:

- Complete a visual inspection, looking for damage due to improper storage.
- Check for the latest software revisions, options, and component changes.
- Contact the OLC for support.
- Movers are required to move the equipment to the scan room.

## 3.3 Construction Site Storage

When storing the CT system at a construction site, be sure to adhere to the following storage requirements:

- Construction site packaging must be ordered and the system shipped packaged for storage.
- Do not damage or puncture the shipping crate.
- Do not remove packaging until the completion of all construction at the site and the removal of all dust created by the construction.
- Maintain a storage temperature within the range of 10º to 32º C (50º to 90º F).
- Maintain a relative humidity (non-condensing) between 20% and 70%.

## 3.4 Construction Site Installations

A construction installation describes installations at sites without an occupancy permit, or ongoing construction. In general, construction sites fail to meet the required specifications for system delivery, and GE Healthcare does not recommend such installations, as they can result in delays, increased costs, and possible damage to the system. When construction-site delivery proves unavoidable, the installation falls into one of two categories:

- Full construction site with completed radiology area.
- Full construction site with limited delivery access.

Review these categories to determine which most closely matches the condition of the planned installation site.

### 3.4.1 Construction Site with Completed Radiology Area

This type of site consists of a finished, dust-free, occupancy-ready radiology suite at a site with ongoing construction in other areas, but with no remaining construction in or around the scan suite area. At the time of delivery such sites feature:

- Dust control measures deployed in the radiology suite area.
- Scan suite access limited to a single entrance.
- Radiology suite sealed off from the remaining construction area.
- Operational HVAC, with a positive air pressure within the radiology suite.

In addition, the radiology suite at such a site **REMAIN**S in a dust-free, occupancy-ready state after delivery and throughout the remaining construction phase.

For more details, refer to the *Discovery CT750 HD Pre-Installation Manual* (Chapter 2, Section 2-1.3.1, “Full Construction Site With Completed Radiology Area”).

### 3.4.2 Full Construction Site with Limited Delivery Access

This type of site allows delivery during ongoing construction of the radiology suite area. Construction site packaging must be ordered and the system is delivered packed for construction site storage. Packaging cannot be added during the delivery.

At Full Construction sites, delivery occurs prior to site completion, but the product remains stored until the completion of a finished, dust-free, occupancy-ready radiology suite area. This system is delivered in sealed packages with dollies. Delivery to the storage area requires a lift truck or riggers. Installation work can begin **only** when the site reaches the completed, dust-free, occupancy-ready radiology suite requirement.

### 3.4.3 Construction Site Unpacking

A typical construction site package consists of 8-12 packages. Each package is plastic-wrapped in dust-free packaging. Each package must be vacuumed to remove construction dust prior to moving components into the CT scan room. This process can add approximately two hours to your installation time.

Typical components are:

- Gantry
- Table
- Console
- PDU
- UPS
- Lean Cover Cart
- Lean Install Cart
- Chair
- Service Cabinet

## 3.5 Working with the Mover

- System is shipped lean-packed in North America.
- If construction site-packed, vacuum all pieces before moving into finished room.
- If room is not completed, follow escalation process. Pre-installation escalation is the process used to consult CT Engineering, the Design Center, or EHS to resolve pre-installation issues related to siting concerns and requirements.
- Ensure that the installation lean cart is the first item moved into the room.

### 3.5.1 Delivery Dolly Options:

- Tilt Table Dolly - ordered from UMI at http://www.umi-dollyshop.com
- Gantry Mini Dolly - ordered from UMI at http://www.umi-dollyshop.com

Follow the instructions provided by your Project Manager of Installation regarding working with equipment movers. Help direct movers as to where to place equipment and which items are needed first.

Movers should move all equipment into the customer's room. Door removal and other site changes to move equipment should be done only as directed by the Project Manager of Installation.

For component sizes and weights, refer to the *Pre-installation Manual* for this system.

Note: Do not place equipment in its final location at this time. Templates must be laid first.

Note: If you have to remove the gantry covers in order to move the gantry into the room, please read the notice statement on page 169 before removing the gantry covers.

### 3.5.2 Lean Packaging

- Do not unpack this cart on the truck.
- Do not recycle or dispose of cardboard at customer site. Return all discarded packaging and unused items back onto the cart for return to GEHC for recycling.
- Packing list is included in each cart and can be used to inventory system.
- USA Installations: Return the carts including unused materials with the dollies.
- Mechanical Installers:
  - Reduced packaging / box count
  - Reduced waste at site
  - Reduced installation cycle time
  - Better organization of material at site

### 3.5.3 Inventory Process

The system is shipped Lean packed on pallets and dollies with approximately 10 units. Use the packing list to verify the number of units received.
Inside the door of each lean cart should be a detailed inventory list. Visually confirm that the items, as labeled, are present.

Note: The packing list does not detail unit contents.

### 3.5.4 Equipment Delivery Route

Prior to equipment delivery, review the delivery route with the movers. Refer to the Project Manager of Installation for any additional delivery instructions.

### 3.5.5 Floor Protection

Movers should use floor protection. Most equipment movers can provide floor protection during the equipment delivery. Installers should provide floor protection for the room.

### 3.5.6 Removing Gantry Dollies and Covers

- Gantry components cannot be removed to reduce the dimensions.
- Tilt table dollies are available from UMI at http://www.umi-dollyshop.com
- Zero clearance dollies are available from UMI at http://www.umi-dollyshop.com
- Please read the notice statement on page 169 before removing the gantry covers.

## 3.6 Damage In Transportation

Check for damage to property that may have occurred at the site during delivery, such as damage to floors, door frames or walls. If damage is found, notify the Project Manager of Installation.

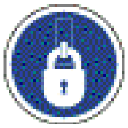
All packages should be closely examined at time of delivery. If damage is apparent, have notation “Damage in Shipment” written on all copies of the freight or express bill before delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage shall be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on http://egems.med.ge.com/edq/home.jsp
- Contact the local service coordinator for more information on this process.

Rev. June 13, 2006

## 3.7 A1 Breaker



**NOTICE**

- **All sites must have a main disconnect with Lockout/Tagout capability.**
- Non GE-supplied breakers must have/provide under-voltage protection. For more information, refer to the *Pre-Installation Manual* for this system.

Lock-out and tag-out the A1 breaker now.

| A1 Breaker | UPS |
|---|---|
| *USA 480VAC:* **E4502AE** | *HD 3-Phase Model:* **B7864PP** |
| *Europe and Asia 400VAC:* **E4502AF** | *HD 3-Phase Model* **B7864PP** |

**Table 1-3**


**Figure 1-1 Sample A1 Breaker**

## 3.8 Installation Support Kits

### HD-5133492 and VCT

A new LightSpeed VCT/Discovery CT750 HD Installation Support Kit is shipped with every system. Locate this box now and open it. All included materials are used during the installation process. Leave these items ON site.

Locate the install packet from the PMI Site-Ready Visit. Among other items, it contains:

- Room Print
- Floor Template
- IP Addresses
- Contact names
- Copy of the Site Ready Document

## 3.9 Installation Conditions

1.) A Final Site Print is required. Contact your PMI for a final site print.

2.) The room size must match the print.

Measure the room size. If it does not match the stated size, and is smaller, then check all regulatory clearances. If any regulatory clearance is less than the minimum, **DO NOT** continue. Notify the PMI to set up a site escalation.

Note: Regulatory and service clearances MUST be met to continue.

3.) A customer Anchoring Plan is required if there is anything other than a 101.6 mm (4 in.) (minimum) concrete floor. GE employees shall only install the anchors supplied with this system.

4.) Complete this section on the GE Form e4879.

5.) Do not start the installation process if the site is under construction:

- In the room
- In the scan area
- In addition, the radiology suite at such a site will REMAIN in a dust-free, occupancy-ready state after delivery and throughout the remaining construction phase.

# Section 4.0 Layout the Floor Template

## 4.1 Time & Personnel

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 2 (FE or mechanical supplier) | | | |

## 4.2 Tools and Test Equipment

- Standard Install Tool Kit
- Install Support Kit
- GE Site Print
- VCT/CT750 HD Installation Manual
- Floor Template for your system
- Chalk line
- China marker or wax marking pencil, or equivalent
- Masking Tape, or equivalent
- PPE
- Alignment Kit Laser

## 4.3 Safety

**CAUTION**

**Potential for Injury.**
**The gantry presents a variety of mechanical and electrical hazards.**
**Use appropriate safety procedures when working on the VCT/CT750 HD system.**

## 4.4 Preparation

- Use the GE print developed for your site to establish the room layout. Make sure all the operating and service clearances shown on the print are observed. Record this information on the GE Form e4879.
- Clean the area. The mounting surface must be free of any material that may interfere with the positioning and leveling of the system.
- Measure and determine ISO using the GE Site print. Using a marker, mark ISO on the floor. Use a chalk line to connect the table center line marks on the floor. This is the line on the print that runs down the center of the table through the gantry. Use this as a reference when positioning the table.

## 4.5 Procedure

1.) Lay out the two (2) pieces of the floor template (GT table: 5111499). Start with the table template, then place the gantry template over the top of the table template. Align them per the GE print.

2.) Tape the templates together, making sure that the table and gantry center lines are matched. Then tape the template to the floor.

3.) Recheck the position of the gantry in the room per the GE print. If everything matches the GE print, continue. If not, realign the templates to match the print.

4.) Make sure there are no potential clearance issues. If there are floor obstructions, such as conduits or old anchors, be sure to cut them flush to the floor to prevent the gantry from resting on them. Also, be sure there is at least 102 mm (4 in.) of clearance between any existing floor penetration and the new gantry position.

Note: There must be clear space without obstructions in order to:

- Change major components, with access to the gantry tube-change (RH) side (See Figure 1-2).

- Allow space for front and rear cover removal.

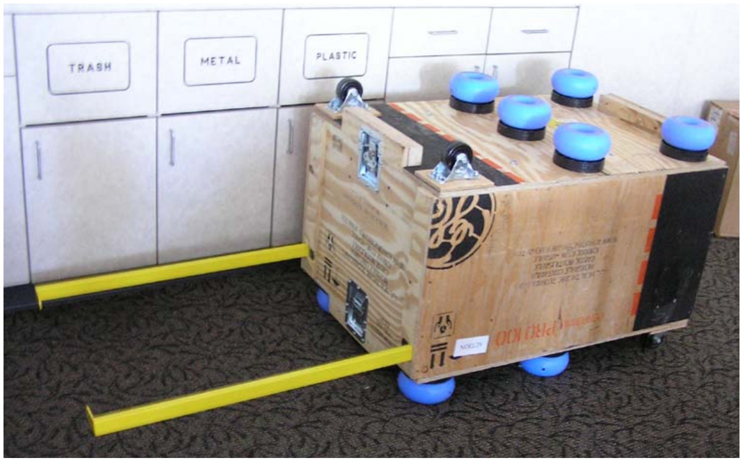



**Figure 1-2 Gantry Tube Change Cart**

5.) Prior to removing this template, check floor levelness, as shown in Figure 1-3.

6.) Position the laser from the install kit on the template behind the table base and turn it on to project a horizontal beam across the floor template area.

7.) Measure the distance from the floor to the laser line at each bolt hole location on the template and record the measurements. Use the measurements to verify the floor is within specification. The floor must meet the minimum levelness specification: 6 mm (1/4 in.) over 3.5 m (10 ft.) between the table and gantry.

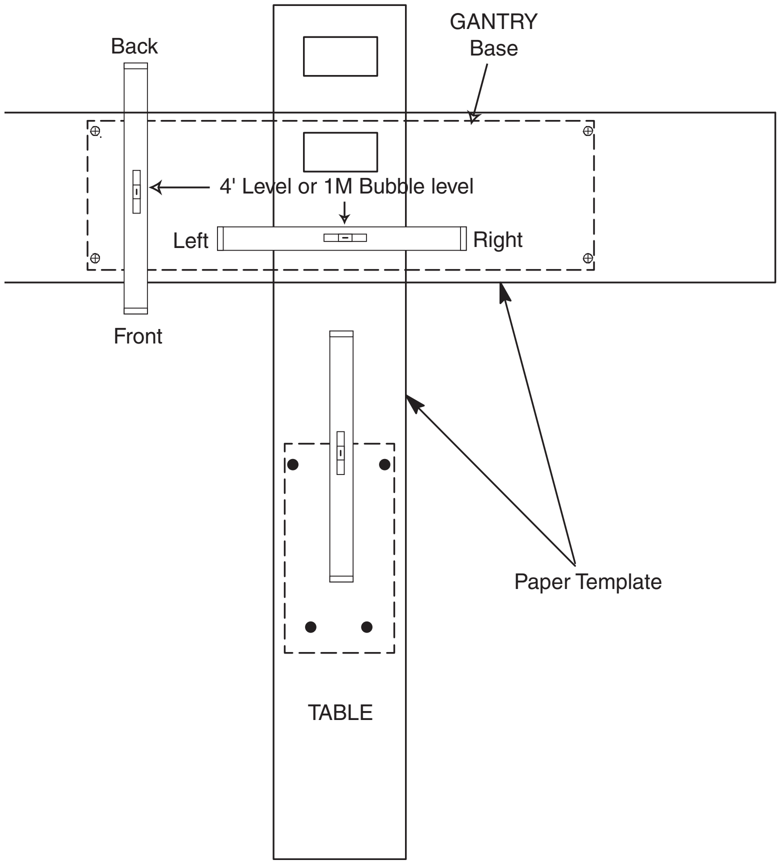
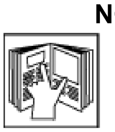



**Figure 1-3 Check Floor Level**



**NOTICE** **Positioning requires cutting eight (8) holes in the floor covering.**
**Before you drill or cut any flooring, make sure that you have discussed this issue with the customer, and that the appropriate hospital personnel have approved the location of the table/gantry.**

**Any repositioning must meet all regulatory requirements to be completed.**

- Check that the floor meets the levelness specification. Follow the escalation procedure if the floor does not meet the floor specification.
- If the floor is not level, the system does not meet the table ISO specification. The distance from the table cradle to the floor cannot be greater than 1005 mm (40 in.).

8.) Check with the customer for approval of the gantry/table placement.

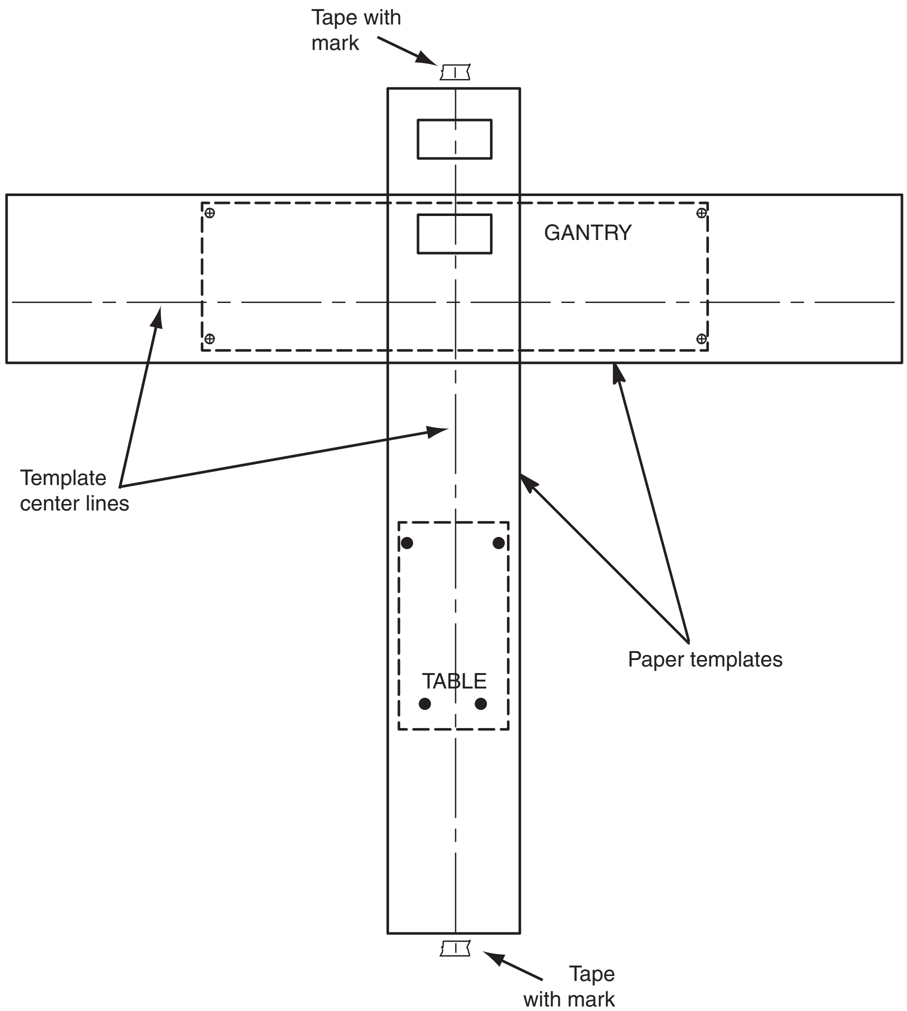



**Figure 1-4  Center Line with Marks**

9.) Use a center punch to mark hole centers for each of the eight (8) leveling pad/anchor locations per Figure 1-4. Before moving on to the next step, see Step 11 and its note for an alternative method.

**CAUTION**

**Potential for personal injury.**

**Use appropriate safety procedures when drilling the floor holes, especially if there is lead under the floor.**

**Appropriate PPE is required when working with hazardous materials.**

10.) Remove the floor template.

11.) Cut tiles (or other resilient flooring) around all holes punched in the template for the gantry and table.

Note: A fast way to remove flooring is to use a 4 in. hole saw with a 1/4 in. masonry bit to cut through the flooring at each leveler pad location.

12.) Some sites require sealing of the floor penetrations after the flooring is removed. If this site does, use RTV or other sealant to seal the floor covering as necessary.

**NOTICE** **All documentation in this manual is based on mounting the table/gantry on a 102 mm (4 in.) concrete floor.**

13.) Snap a chalk line using the marks that were made on the tape at the ends of the table template.

# Section 5.0 Install the Gantry

## 5.1 Time and Personnel

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 2 (FE or mechanical supplier) | | | |

## 5.2 Tools and Test Equipment

- Standard Install Tool Kit
- Install Support Kit
- GE Site Print
- VCT/CT750 HD Installation Manual
- Gantry Adjuster Tool, P/N 2107863
- Spanner Wrench, P/N 2110003
- PPE

## 5.3 Gantry Preparation

Note: Locate and install any required floor protection now.

### 5.3.1 Access Greater than 28 in.

Remove all the transportation packaging from the gantry, except for the dollies.

### 5.3.2 Access Less than 28 in.

Measure from the wall or object protruding from the wall to the gantry side cover. The gantry left side cover must be installed for this measurement. When finished, the gantry cannot be closer than 14 in. to the wall or object protruding from the wall.

1.) Remove all the transportation packaging from the gantry, except for the dollies.

2.) Remove the blue dolly from the left side of the gantry and install the limited access dolly so that the gantry can be positioned closer to the left side wall.

   a.) Remove the three (3) M14 hex bolts that secure the gantry to the dolly.

   b.) Replace the removed dolly with the shipped black gantry-positioning dolly, and reinstall the three (3) M14 hex bolts.

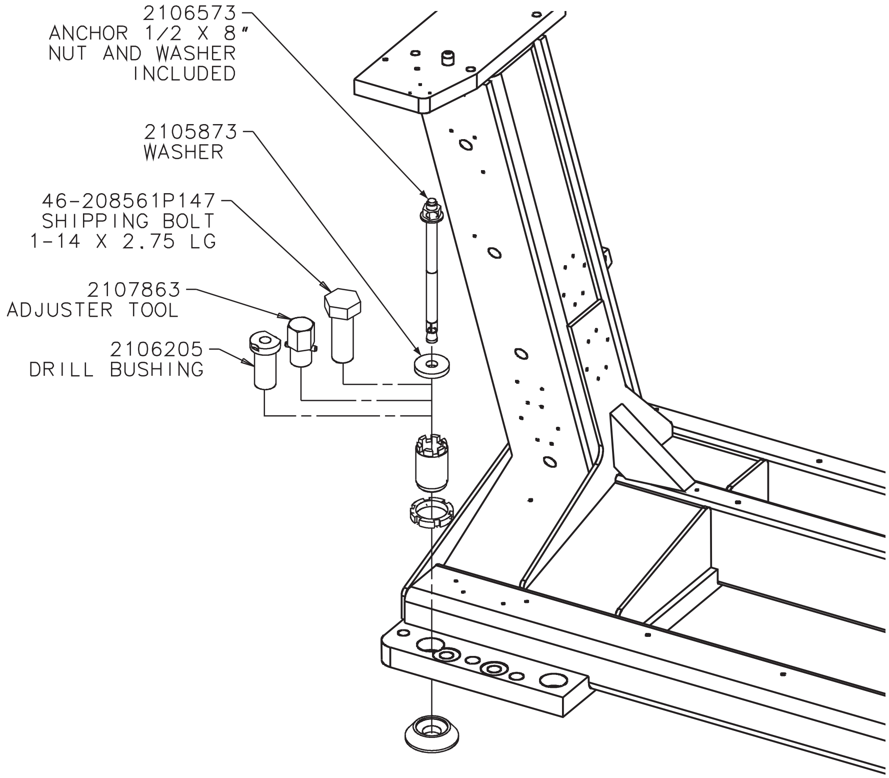
   c.) Raise the gantry so that it is once again off of the floor.

   The gantry can now be moved up to 14 in. from the wall, measured from the wall or object protruding from the wall to the gantry side cover. Only use the supplied, limited-access dolly for this procedure.

Note: If this procedure cannot be completed, follow the site escalation procedure established for your area.

## 5.4 Procedure

1.) Position the gantry over the floor cutouts appropriately.
   a.) Locate the four (4) leveling pads, and position each beneath its associated adjuster. A ½ in., ratchet and 1-½ in. socket are required to loosen the gantry shipping bolts.
   b.) Use the dollies to evenly lower the gantry, until it is just off of the floor (approximately ¾ in. or 20 mm). Use a ½ in. ratchet to raise and lower the dollies.
   c.) Carefully rotate the gantry into the correct position over cutouts in the floor.
   d.) Use floor protection as appropriate.




**Note:** Adjusters are used at each anchor location. Anchor hole ID is 1" (2.5 cm). Void between adjuster and anchor must be filled according to local building codes for seismic application.

**Figure 1-5 Gantry Base Installation Hardware**

2.) Loosen the locking rings with the spanner tool. Turn gantry levelers by hand until each leveler is down ¾ in.or 20 mm) from the bottom of the gantry frame.

3.) Leave the locking rings loose so you can fine-tune the leveling pads to compensate for slight variations in the floor surface.

4.) Position the gantry so that the adjusters are centered over their respective holes marked earlier into the floor cutouts.

Note: Due to the large plug head size, it may be easier to route the gantry 3-phase power cord under the rear gantry rails before lowering the gantry. If not, feed it from the PDU side of the cable. (This cable can be found on the Lean Cart with the power cables.)

5.) Using a ½ in. ratchet, gently lower the gantry until it rests on the floor over the marked areas. When finished, the gantry should now be resting on all four leveling pads.

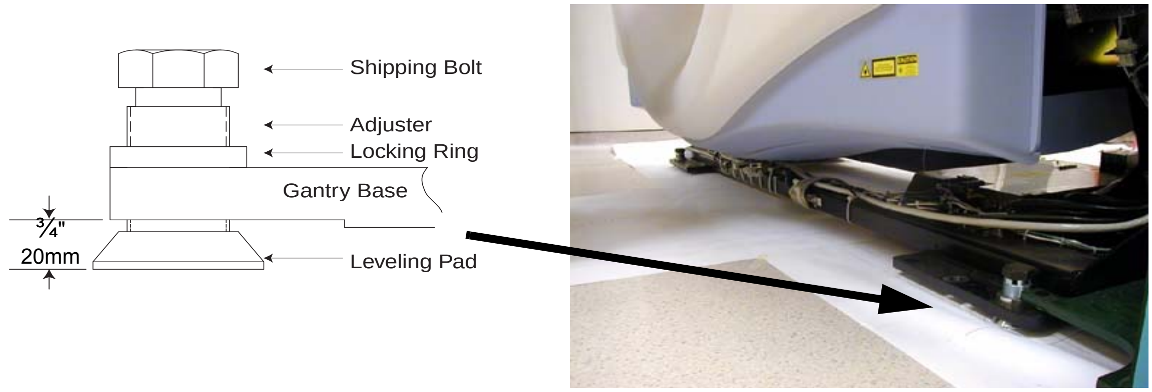



**Figure 1-6 Gantry and Table Base Leveling Pads (Starting Positions)**

**NOTICE** **Gantry dollies weigh approximately 113.5 kg (250 lbs.) each. Use caution when removing the dollies to avoid damaging the floor covering.**

6.) Using a 14 mm hex socket, remove the dollies from the gantry by removing the three dolly bolts found at both ends of the gantry (Figure 1-7).

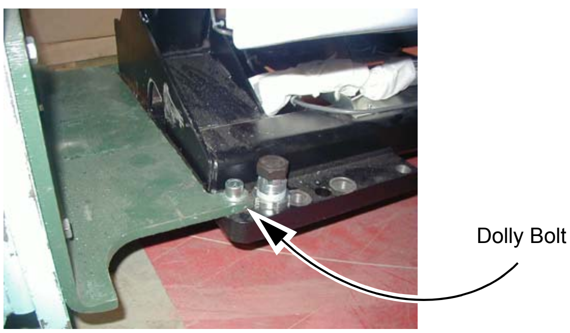



**Figure 1-7 Gantry Dolly Bolts**

7.) If needed, remove the four (4) gantry shipping bolts, using a 1½ in. socket.

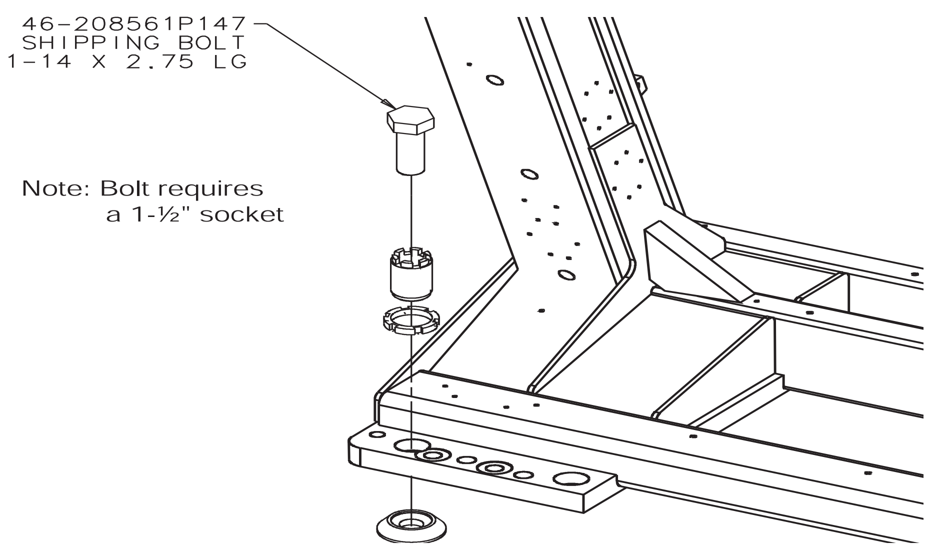



**Figure 1-8 Gantry Shipping Bolts**

# Section 6.0 Level the Gantry

This procedure is designed for gantries that are mounted on 102 mm (4 in.) concrete floors that meet all of the pre-installation flooring requirements.

The gantry uses two (2) bubble levels that are permanently mounted to machined surfaces on the stationary base to determine when it is level.

The bubble levels are located on both ends of the gantry stationary base near a point where the rotating structure pivot-mounts to the base structure. (See Figure 1-9.)

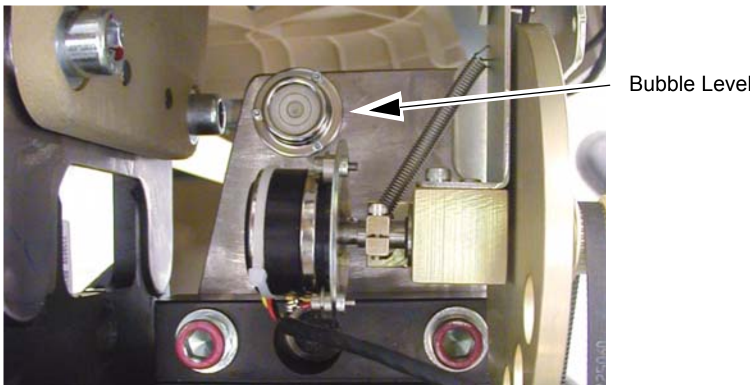



**Figure 1-9 Gantry Bubble Level**

## 6.1 Time and Personnel

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 2 (FE or mechanical supplier) | | | |

## 6.2 Tools and Test Equipment

- Standard Install Tool Kit
- Install Support Kit
- GE Site Print
- VCT/CT750 HD Installation Manual
- Gantry Adjuster Tool, P/N 2107863
- Spanner Wrench, P/N 2110003
- PPE

## 6.3 Procedure

1.) Loosen all adjuster lock rings (use a spanner wrench or large tongue and groove pliers).

2.) Systematically turn each of the gantry's adjusters (locations 1, 2, 3 and 4 in Figure 1-11) until both bubble levels are centered left-to-right and front-to-back (see Figure 1-10).

   a.) Begin by turning each adjuster no more than 1 turn at a time.

   b.) Use the adjuster tool, 1-1/8 in. socket, and the ½ in. drive ratchet to turn each adjuster. (Refer to Figure 1-5, on page 46.)

   c.) Level the left side from front-to-back by turning adjusters #1 and #2.

   d.) Level the right side from front-to-back by turning adjusters #3 and #4.

   e.) Level the side (right or left) that is higher with respect to the other side. Turn both adjusters on a side equally until that side is level. The side should now also be level.

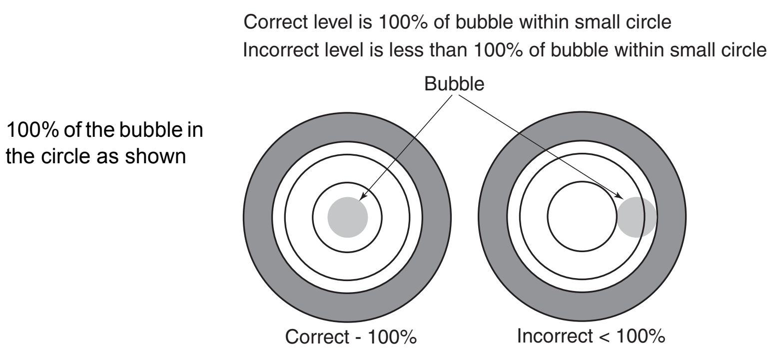



**Figure 1-10 Bubble Level Centering**

Note: Gantry level is critical to system operation!

3.) When the bubble levels are centered (Figure 1-10), each of the four (4) leveling pads should be carrying a portion of the gantry weight. Distribution of the gantry weight prevents the base frame from rocking during normal operation.

**CAUTION DO NOT leave any adjuster un-loaded or floating.**

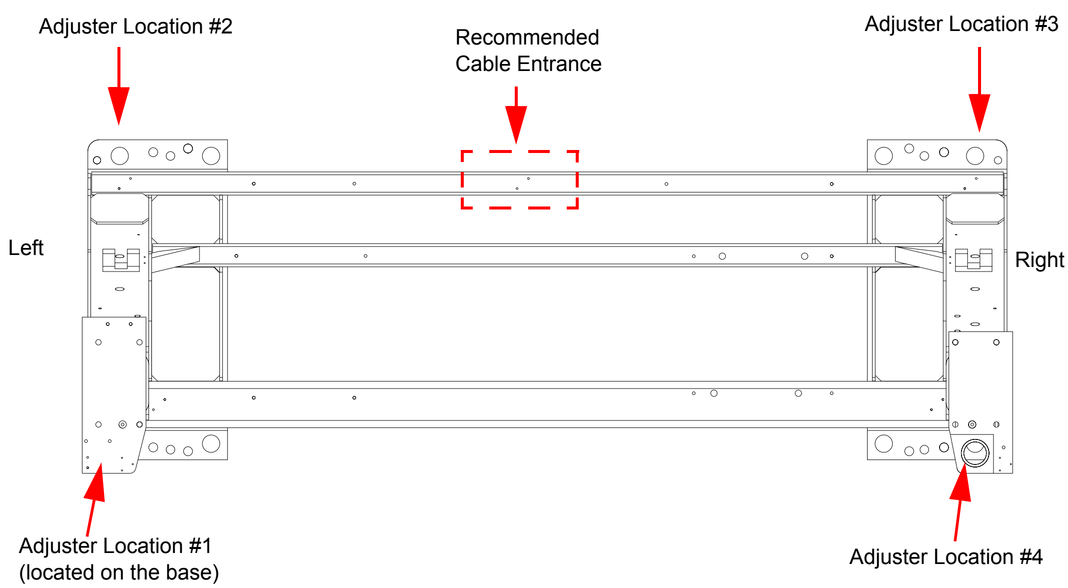



**Figure 1-11 Gantry Base Adjuster Locations — Top View**

4.) If there are floor obstructions, such as conduits or old anchors, be sure to cut them flush to the floor to prevent the gantry from resting on them. Also, be sure there is at least 102 mm (4 in.) of distance from any existing floor penetration to the new gantry anchor positions.

5.) Using a measuring tape, measure and mark two spots 673 mm (26.5 in.) from the gantry base frame (under the front cable tray) out toward the table floor cutouts, as illustrated by X marks at the bottom of the dotted lines in Figure 1-12.

6.) Snap a chalk line using the two 673 mm (26.5 in.) marks. This line should be parallel to the gantry and 673 mm (26.5 in.) from the front of the gantry base. See Figure 1-12. In a later section, you will move the table against the 673 mm (26.5 in.) mark and center it over a table center line (to be drawn at that time.)

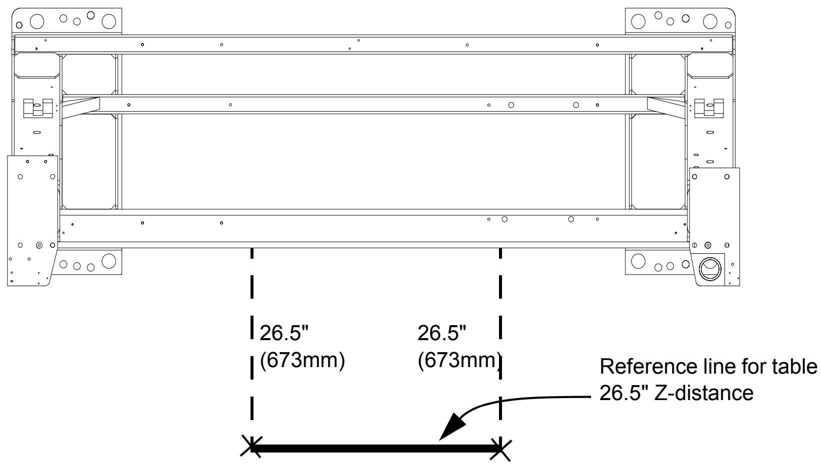



**Figure 1-12 Table Base Reference Line and Marks**

# Section 7.0 Gantry Bearing Gap Inspection

All CT systems require a gantry bearing gap inspection before starting electrical calibration.

All international gantries are shipped in a wooden shipping crate that should not be removed until it arrives at the installation site. This shipping container is designed to reduce the risk of shipping damage.

The back cover needs to be removed to gain access to the gantry bearing.

## 7.1 Time and Personnel

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 2 (FE or mechanical supplier) | 15 min | 15 min | 5 min |

## 7.2 Tools and Test Equipment

- Standard tool kit
- Inspection document
- 2.5 mm Allen wrench
- Rear cover dollies (Qty = 2)
- Flashlight

## 7.3 Preparation: Damage Indicators

If this is a Rotek bearing, a mark similar to that shown in Figure 1-13 is visible on the inside edge of the black-colored bearing assembly.

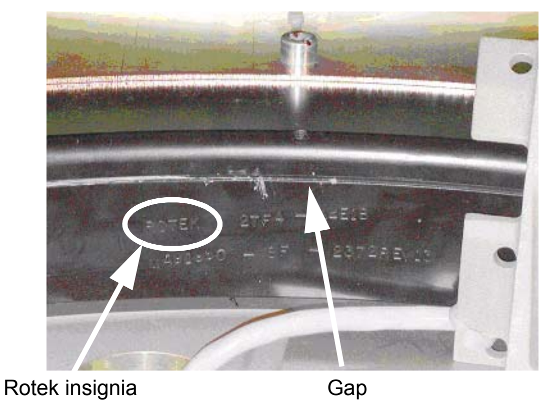



**Figure 1-13 Gantry Bearing - Rotek Label**

The mark has a serial number in the same format as:
ROTEK 2TF4-44E1B-MA91960-8F-2372-REV13.

The gap to inspect is shown in Figure 1-14 next to the serial number.

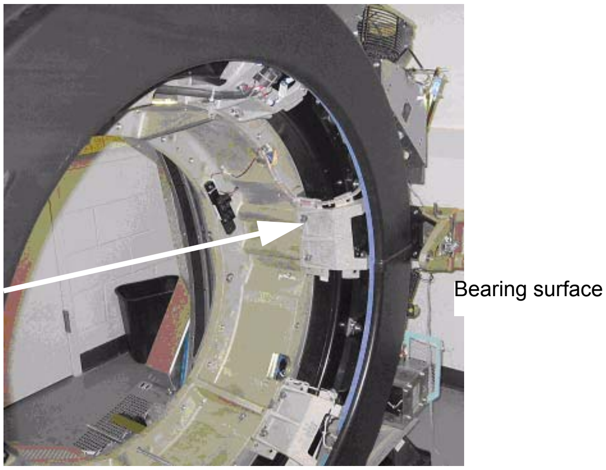



**Figure 1-14 Gantry Bearing**

On most systems, a change in the bearing gap does not cause the gantry to make unusual sounds, unless the gap is severe. If the gantry is badly damaged and the gap is severe, it can cause operation issues. Some systems are shipped with shock indicators that must be returned to Milwaukee.

A severe failure may be seen during installation as a problem rotating the gantry.

## 7.4 Procedure

1.) Remove the scan window following the procedure in Appendix A, Section 1.3 on page 167.
2.) Remove the top and rear gantry covers, following the procedures in Appendix A, Section on page 170 and Appendix A, Section 1.4 on page 174.
3.) Use a 2.5 mm hex wrench as a tool to measure the gap at the positions shown in Figure 1-15. The location of gantry components does not matter. Measure four (4) locations 90 degrees apart from each other.

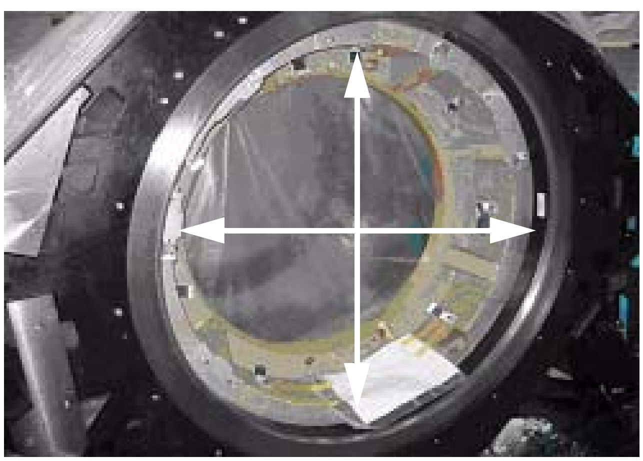


**Figure 1-15 Inspection Locations**

4.) If the 2.5 mm easily fits without effort in the gap, the gap is out of spec. Figure 1-16 shows a gap that is too large in the left picture. Figure 1-16 shows a gap that is good in the right picture. Notice that the hex wrench does not fit in the gap in Figure 1-16 (left picture), but does in Figure 1-16 (right picture).

Note: Do not use force when putting the wrench in the gap. Either it slips in or it doesn't.

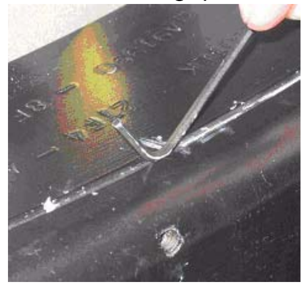


**Figure 1-16 Gap too large (left) Gap is good (right)**

## 7.5 Finalization

### 7.5.1 Mechanical Installers

If the Bearing Gap Inspection passes, complete the sign-off on the GE Form e4879, Installation Data verification form, that this inspection was completed.

If the Bearing Gap Inspection fails, contact your site FE.

### 7.5.2 FE Service Action, if Required

If the Bearing Gap Inspection fails, the mechanical installer notifies the site FE that the inspection failed.

The site FE should:

1.) Open a bearing inspection dispatch.

2.) Follow the inspection procedure described in this section.

3.) Record the bearing inspection results.

If no damage is found, close this dispatch and continue with the electrical calibration procedures.

If the system is damaged, go to the Equipment Delivery Quality web site and follow their instructions.

To enter a damaged in shipping claim, go to this web site:
http://egems.med.ge.com/edq/home.jsp

### 7.5.3 FE Inspection Completion

1.) After the Gantry Bearing Inspection passes, complete the opened service dispatch with the following information:

- Gantry Serial Number
- Gantry Type
- System ID
- Site Name
- Installation date
- Was the gantry transported to the site in the shipping crate? (Yes/No)
- Was the gantry lifted or hoisted, were riggers used, or was the gantry delivered via flatbed wrecker? (Yes/No)
- Number of locations that fail the gap inspection if any: _____

2.) Close the service dispatch.

Should any follow-up be required after this inspection, the site engineer will be contacted directly by CT Engineering.

# Section 8.0 Install Gantry Alignment Laser and Bracket

## 8.1 Time and Personnel

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 2 (FE or mechanical supplier) | 15 min | 15 min | 5 min |

## 8.2 Tools and Test Equipment

- Standard tool kit (P/N 5148193)
- Laser Alignment Kit
- 9" level
- Tape measure
- Masking tape

## 8.3 Procedure

**NOTICE** **Use caution while removing the gantry scan window.**

1.) Rotate the gantry by hand until the collimator face plate is at the 5 o’clock position.

Note: With power OFF, the gantry movement is tight.

DO NOT pin the gantry during this alignment process.

2.) With the top and back gantry covers removed, locate the two M10 bolt holes as shown in Figure 1-17. These bolt holes are used to attach the laser tool to the gantry.

- The bolts can be installed using an 8 mm Allen wrench. Be careful not to bump the alignment light; the mounting space is tight near the alignment light. Tighten bolts until both are snug.
- Do not drop bolts or the bar on the collimator faceplate. Attach the bar as shown in Figure 1-17.
- Using a minimum 223 mm (9 in.) level placed on the attached bar, level the bar by rotating the gantry.

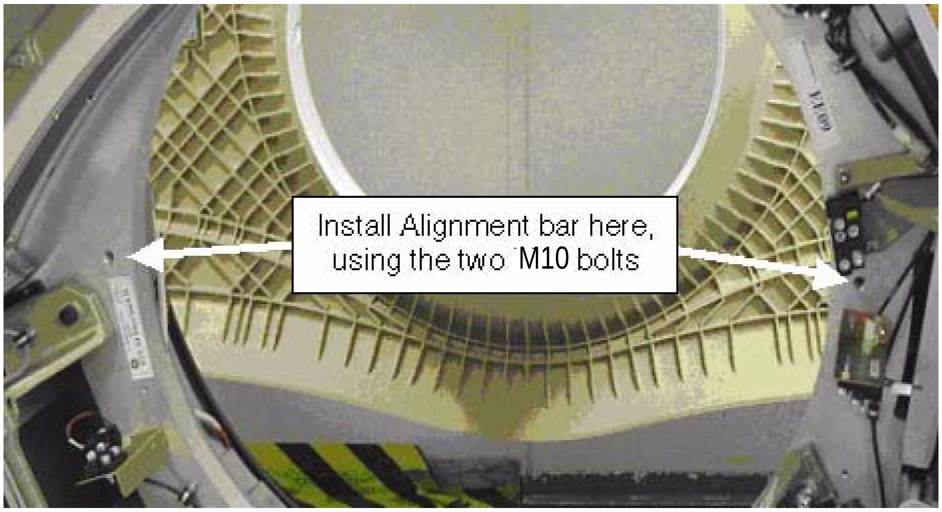



**Figure 1-17 Alignment Bar Installation Location**

**CAUTION**

**Potential for injury.**

**DO NOT look into the laser.**

**Use appropriate safety procedures when working with lasers.**

3.) Attach the laser centering plate onto the laser mounting bar as shown in Figure 1-18. The plate is attached from under the alignment bar using two fixed locators and two thumb screws.

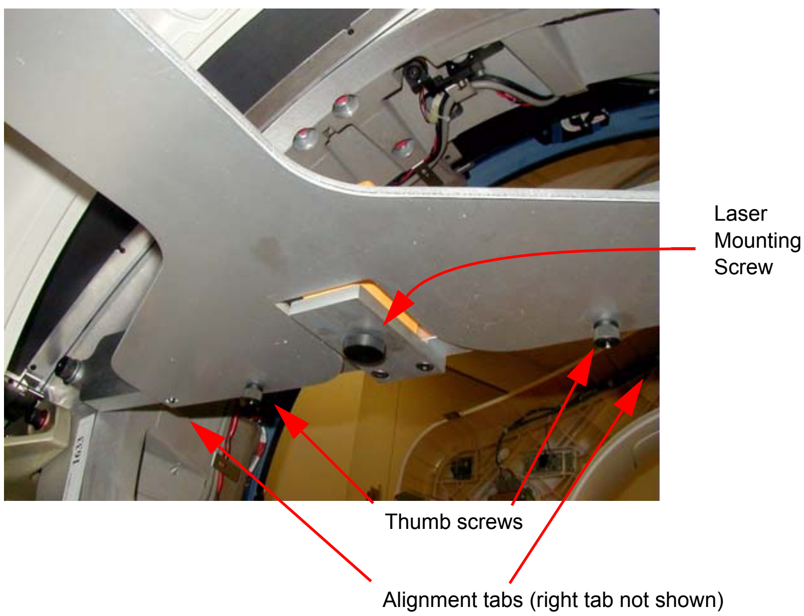



**Figure 1-18 Attach Laser Center Plate**

4.) When done, insert the laser and turn on the laser using the controls on the back.
If the laser is loose when mounted, use a 2 in. piece of Velcro loop (fuzzy) section and attach it to the alignment plate over the attachment screw. Remount the laser and it should fit snugly without moving.

5.) When pressed, the **ON** button steps through four different beam profiles and "Self-Leveling Off". Press the **ON** button until the "|" beam shows. It is used for this operation.

| Times pressed | Function | Notes |
|---|---|---|
| 1 | — | Self-leveling on |
| 2 | \| | |
| 3 | + | |
| 4 | Self-leveling off | Do not use |

**Table 1-4 Laser On/Off Button**

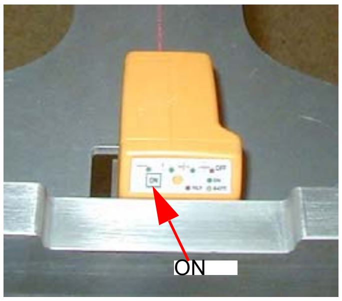



**Figure 1-19 Laser ON button**

6.) Align the laser by carefully rotating the laser base assembly so that the "I" beam shines through the center of the alignment sight mounted on the end of the alignment plate.

Note: The laser beam may be wider depending on the battery life.

7.) Use the locking screw on the bottom of the alignment bar to secure the laser to the bar, as shown in Figure 1-20. When done, the laser should fit snugly without moving on the mounting bracket.

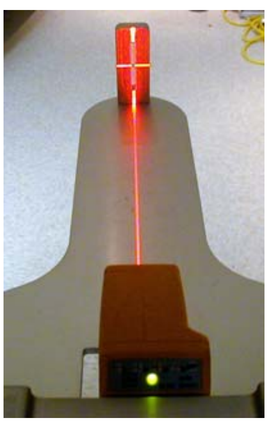


**Figure 1-20 Laser Centering**

**CAUTION** When tightening, the laser may move. Use caution to prevent any movement, as this can result in drilling the table anchor holes in the wrong location.

8.) After the laser is centered, notice that the laser beam also appears on the back wall. Place a piece of masking on the wall and carefully mark a line on the tape where the laser appears. This line is later used in the table alignment. This line is also useful in determining if the laser unit moves during the alignment process.

9.) Remove the alignment centering plate and store it in the alignment case.

10.) Using a chalkline, mark a table center line on the floor along the laser light shining on the floor.

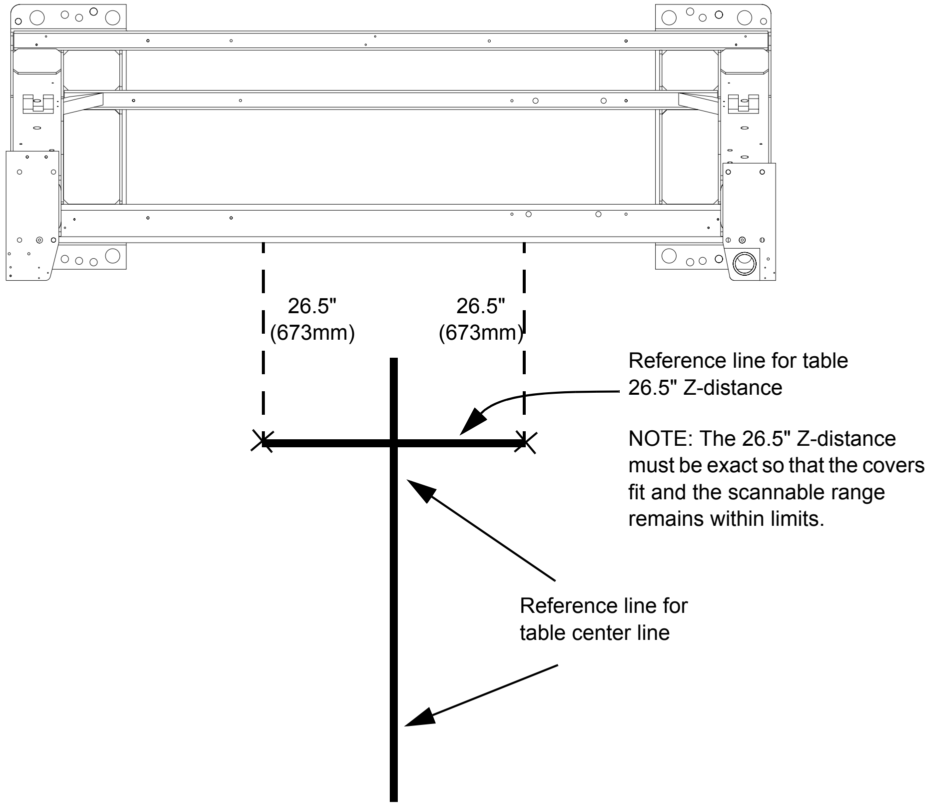



**Figure 1-21 Table Center Line**

11.) Recheck the table-to-gantry reference line for 26.5 in. (± 0.25 in.) Z-distance. Refer to Figure 1-21.

12.) Turn off the laser but do not remove.

# Section 9.0 Table Installation

Note: **Review the table installation and leveling video for this procedure.**

## 9.1 Time and Personnel

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 2 (FE or mechanical supplier) | | 1.5 hours labor on-site | |

## 9.2 Tools and Test Equipment

- Standard Install Tool Kit
- 1-½ in., 1- in., ¾ in.sockets
- 1-5/8 in. (40-41 mm) socket
- 10 mm and 14 mm hex socket bits

**CAUTION** **Potential for Injury.**
**GT 2000 Table on dolly length is 3 m (2997 mm) [118 in. (9 ft.-10 in.)].**
**Exercise caution when moving the table on the dolly.**

## 9.3 Procedures

**CAUTION** **Potential for Injury.**
- **Table will tip if not anchored or attached on the dolly.**
- **Make certain that Table is adequately secured to the dolly.**

1.) Remove all the transportation packaging and boxes from the table, except the dollies (see Figure 1-22). Leave a layer of packing material on the cradle to protect the cradle from damage. (It can be removed during laser alignment of the table.)

**Figure 1-22 Remove Table Packing**

2.) The GT table on dollies is approximately 2997 mm (118 in.) long and may require additional room to maneuver.

3.) Using the table centering and distance locator marks made earlier, wheel the table to its approximate position relative to the gantry.

4.) Locate the table leveling pads inside the table in the back and on the side in the front. Preset leveling pad heights to 20 mm (3/4 in.) (see Figure 1-23).

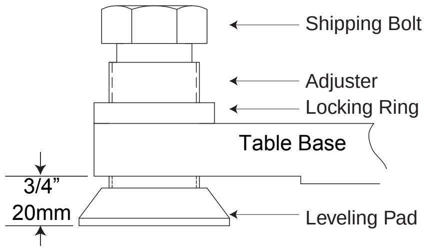



**Figure 1-23 Table Base Leveling Pads (Starting Positions)**

5.) Use a 1-5/8 in. socket and ½ in. ratchet to loosen the shipping bolt. Loosen the locking rings if present.

6.) Use a 1-1/8 in.socket with the adjuster tool, if needed to lower the adjuster.

7.) Use the dollies to evenly lower the table until it rests on the leveling pads using a ½ in. ratchet on each end.

Back Front

**Figure 1-24 Adjusters and Lock Rings**

## 9.3.1 Remove Table Covers

1.) If installed, remove the table right side cover, as shown in Figure 1-26.

a.) Remove the two screws on each end of the underside of the long side cover of the table.

b.) Slide each cover forward to unlatch, lift upward slightly to disengage the latches, and remove the side cover. Doing this procedure requires patience and practice to remove and replace this cover.

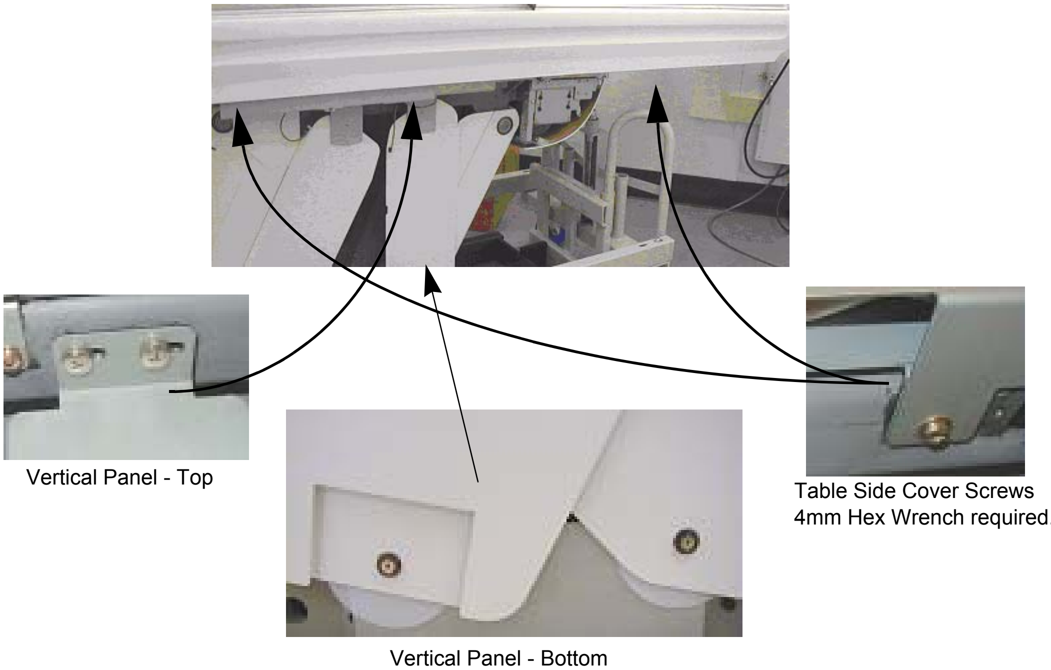



**Figure 1-25 Table Covers**

2.) Make sure that all four of the table levelers are on the floor. The table should set on the four levelers with the dollies still installed.
3.) Carefully center the four levelers over the floor cutouts.
4.) Check that the front table base center line is on the chalk table center line.

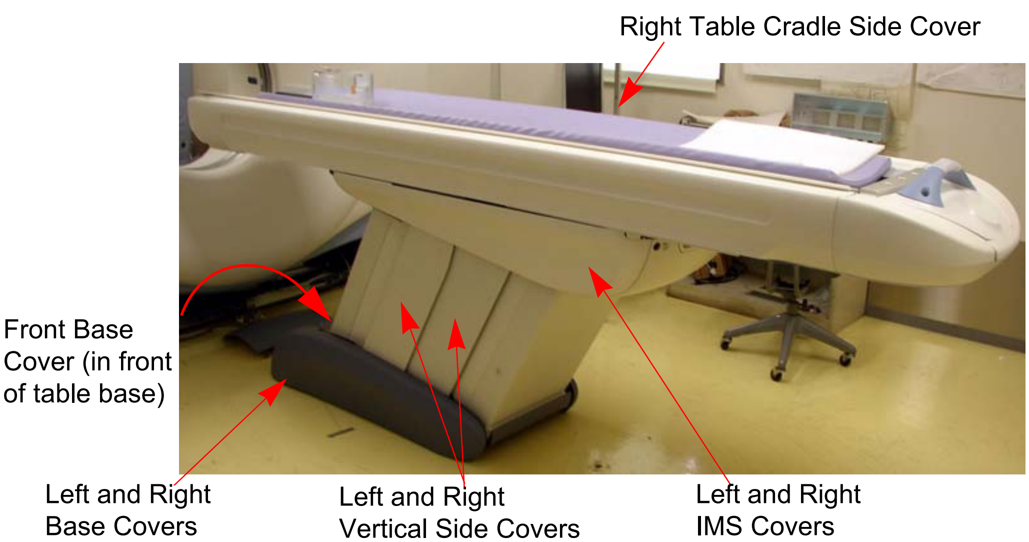



**Figure 1-26 Table Covers (table shown with all covers installed)**

### 9.3.2 Remove the Accessory Rail Strip

1.) Remove the accessory mounting strip attached on each side of the cradle using a small, flat blade screw driver. The nylon screws are inserted inside the accessory rail on the cradle.
2.) Place the accessory strips on the floor and reinstall the nylon screws into the accessory rail for safe keeping.

**Figure 1-27 Accessory Rail Screw**

### 9.3.3 Gantry Tilt Zero Alignment Check

**Manufacturing Shipping Specification:** Must be 90 degrees ± 0.14 degrees

#### 9.3.3.1 Tools and Test Equipment

- Digital Level with accuracy of .1 degrees

#### 9.3.3.2 Procedure

1.) Place the digital level on the rear corner of the gantry as shown in the photo.

**Figure 1-28 Checking Gantry Alignment**

2.) Check that the gantry is at 90 degrees +/- .1 degrees.

a.) Note: If the tilt is greater than .1 degrees recheck the alignment after power is applied and the gantry tilt is corrected.

### 9.3.4 Install the Table Cradle Laser Alignment Plates

1.) Locate the aluminum accessory tray mounting plate with the three holes on the rear of the cradle. Fit the rear alignment target into the two mounting holes as shown in Figure 1-29. Use the adjustment screw to adjust the fit as needed. See Figure 1-29. When you are finished, the fit should be snug, without play.


**Figure 1-29 Cradle Rear Laser Alignment Tool**

2.) Check that the table base is centered over the table center line, and the base is on the 26.5 in.line (± 0.25 in.) made on the floor.

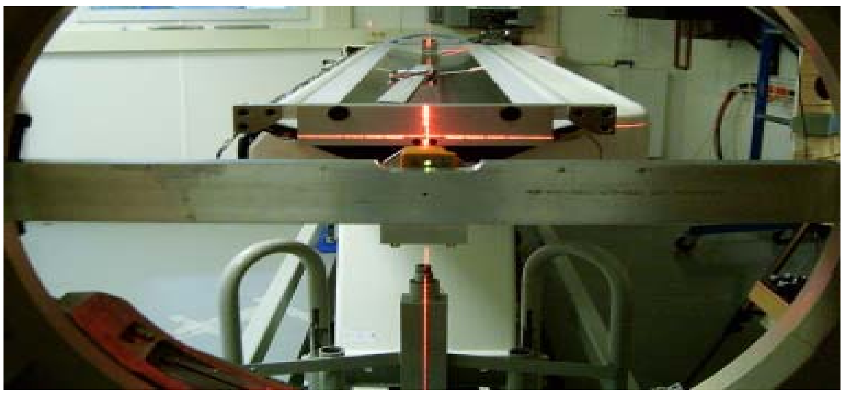


**Figure 1-30 Rear Laser Alignment Tool - Installed**

3.) Lower the table to the floor using the dollies, making sure to maintain the 26.5 in. (± 0.25 in.) distance.

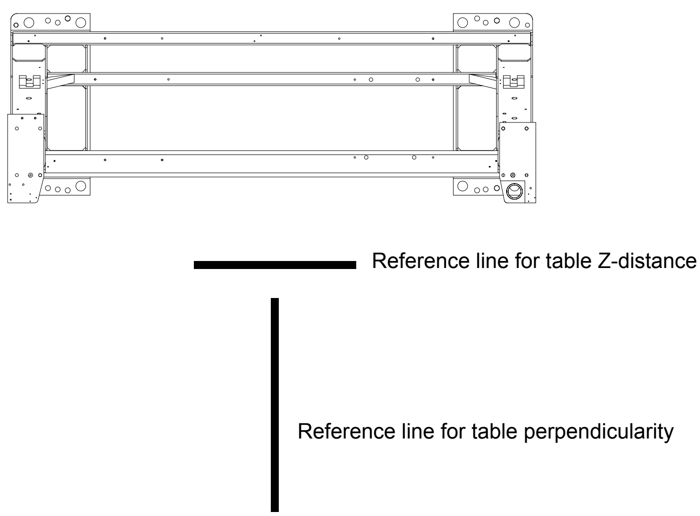



**Figure 1-31 Two Reference Lines**

# Section 10.0 Level the Table

## 10.1 Time and Personnel

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 2 (FE or mechanical supplier) | | 0.5 hours labor on-site | |

## 10.2 Tools and Test Equipment

- Standard Install Tool Kit
- ¾ in., 1-¼ in., 1-½ in., and 1-5/8 in. sockets
- 8 mm, 10 mm, and 14 mm hex socket bits
- Laser Alignment Kit
- Johnson Professional 4 ft. level

**Figure 1-32**

## 10.3 Alignment Preparations

### 10.3.1 Conditions

- Before you start, turn on the laser and check that the beam is still on the mark placed on the wall. If not, reset the laser.
- If the mark is not present, use a measuring tape and place a 102 mm (4 in.) piece of masking tape on the cradle at the 1000 mm and on the laser line.
- The gantry must be at zero degrees within 0.16 degrees from gantry zero.
- Table base to cradle alignment location is 980 mm or 1005 mm from the center of cradle to the floor.

### 10.3.2 Specifications

- Mechanical base alignment must be perpendicular within 0.14 degrees from gantry zero (± 1.5 mm), as measured in this procedure.
- Table cradle travel (X axis) must be perpendicular within 0.14 degrees from gantry (Y axis) zero (± 1.5 mm) as measured in this procedure.
- Table cradle must be level in all directions ± 0.0625 in. (centered within the lines on a Johnson Professional level).
- All table adjusters should be preset to 20 mm (3/4 in.) down from the table base to make adjustment easier. Based on floor levelness and your experience, a different preset height may work better. One thread must be showing above all locking rings when leveled.
- Table cannot be higher than 1005 mm from floor to cradle.

## 10.4 Procedures

**NOTICE** **Avoid leaning on the cradle during this procedure.**
**DO NOT pin the gantry during this alignment process.**
**This procedure is for systems mounted on 102 mm (4 in.) concrete floors only!**

Note: If the floor covering was not properly removed with the glue removed, or the levelers were not centered over the floor cutouts, the leveler may become trapped against the edge of the floor covering, causing the table to become unlevel. If this happens, move the table and enlarge the 102 mm (4 in.) floor cutout for the table. Glue removal is important and aids in moving the table to its final location in accordance with the floor levelness specification.

1.) Have the table side panels removed and a ratchet, 1-1/8 in. socket, and a 2 ft. level ready to use.

2.) Turn on the laser's "I" beam (vertical beams) by pressing the **ON** button 2 times.

Note: Step 3 through Step 7 are for perpendicular positioning of the cradle to the gantry.

3.) The table on the dollies should be resting on the floor, and the laser beam visible on the cradle. The laser light should now shine down the cradle onto the rear vertical target. Moving the table on the dollies by raising and lowering makes it easier to center the table right-to-left.

Note: When using the table dolly to move the table, be sure that the shipping bolts are still attached to the adjuster leveler feet. This prevents the adjuster levelers from gripping on the floor adhesive, making it difficult to move.

4.) Move the table so that the base is roughly centered over the scan center line, the front edge of the table base is on the 26.5 in. (± 0.25 in.) line, and the table is resting on the floor. Check that the leveling feet are centered in the cutout circles.

5.) Carefully move the table so that the cradle front center line and the back target are aligned. You may need to raise the table to move it. When aligned, lower the table to the floor.

6.) If not already done, measure 1000 mm from the front of the cradle, and place a piece of tape under the laser center line. Carefully mark a line along the laser line when the table is centered.

7.) The laser beam should now connect the cradle front centerlines, the 1000 mm cradle center line, onto the rear alignment tool vertical center, and finally onto the alignment centering mark placed on the wall, when properly centered along the center line. Use the centering alignment line on the wall to be sure the laser is still centered. If the alignment line on the wall is NOT on the original mark, readjust the laser and repeat the above steps (see Figure 1-33).

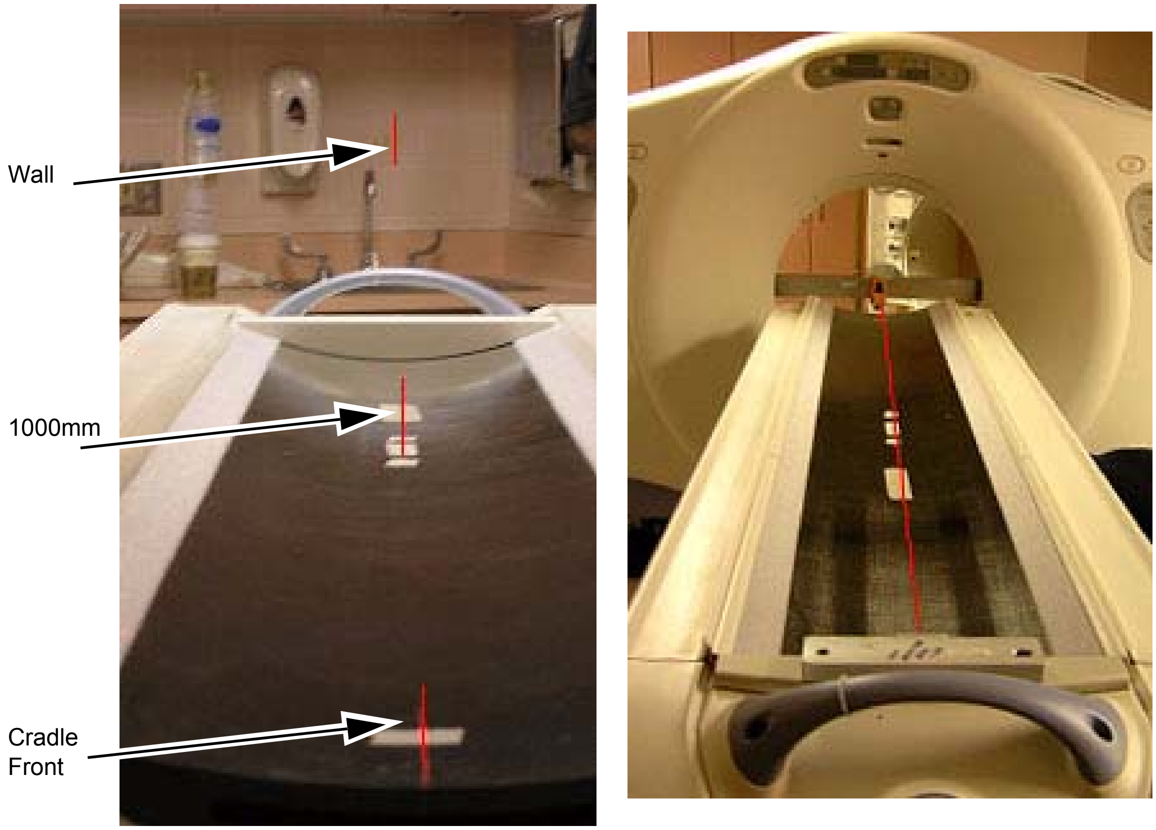



**Figure 1-33  Alignment Laser Marks - Table & Wall**

Note: Step 8 through Step 10 are for front-to-back and side-to-side leveling of the cradle.

8.) The table should be completely on the floor and resting on all 4 levelers.

**CAUTION**

**Potential for Injury.**

**In the ship position, the table tips easily!**

**DO NOT lean on the table! The shipping bracket should still be in place.**

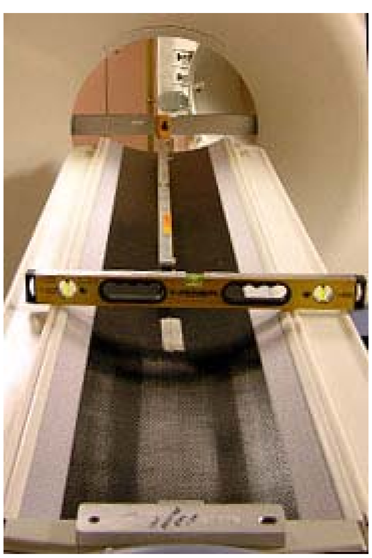


**Figure 1-34  Alignment Laser Marks - Table & Wall**

9.) Raise or lower the table as needed using the front and rear levelers and level the cradle horizontally in the front and back locations (refer to Figure 1-34):

a.) First, horizontally across the cradle at the front of the cradle.

b.) Second, horizontally across the cradle at the 1000 mm mark on the cradle.

c.) When the cradle is roughly level in both locations, leave the level on the cradle horizontally.

10.) Level the cradle front to back using a 4 ft.level placed in the center of the cradle. These two leveling actions often conflict and a few iterations may be required.

This process is complete when:

- The cradle is still centered on the front, mid, and rear marks.
- The cradle is leveled horizontally (across table) front and back (with the bubble centered between the lines.)
- The bubble is centered between the lines in the Y direction along the length of the cradle.
- The laser is still centered on the wall center line.
- The table is still on the 26.5 in. line and the levelers are not resting on the flooring.
- The laser is the same as in Step 7.

Note: The leveling process may take several iterations of Step 1 through Step 10. Patience and accuracy is required to properly complete this process.

11.) When completed, turn off the laser tool.

Note: Do not remove the table dollies.

## 10.4.1 Cradle/Table Parallel Check

### 10.4.1.1 Tools and Test Equipment

- Tape measure
- Masking tape

### 10.4.1.2 Procedure - Initial Setup

1.) Confirm that there is a centerline on the table at 1000 mm measured from the black dot on the front of the table.

2.) If the line is not present, turn on the laser and using masking tape from the installation support kit, add the centerline.

3.) This line should be centered with the line on the front of the cradle and the rear table line.

4.) This reference line will also be used to determine x-ray on table alignment. Do not remove.

5.) Remove the alignment laser and bar assembly from the gantry.

**CAUTION Do not lean on the table during this procedure.**

.

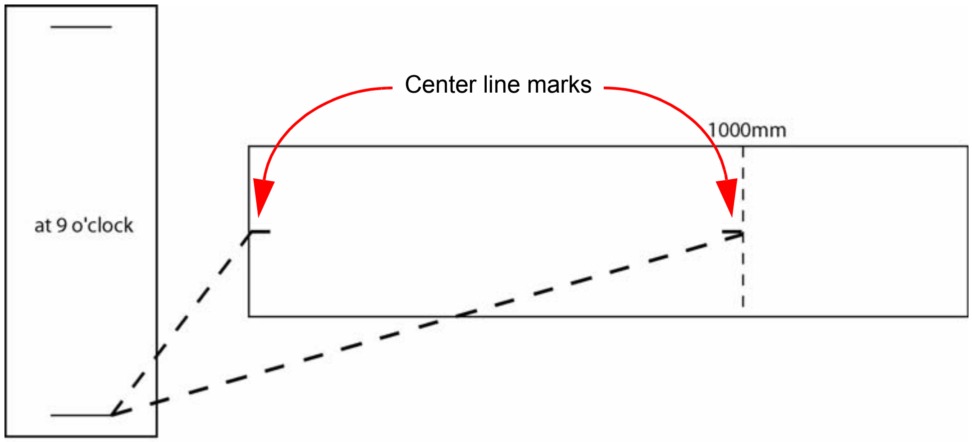



**Figure 1-35 First Cradle/Table Parallel Check**

**Figure 1-36 Collimator Face Plate to Front of Table–Overview**

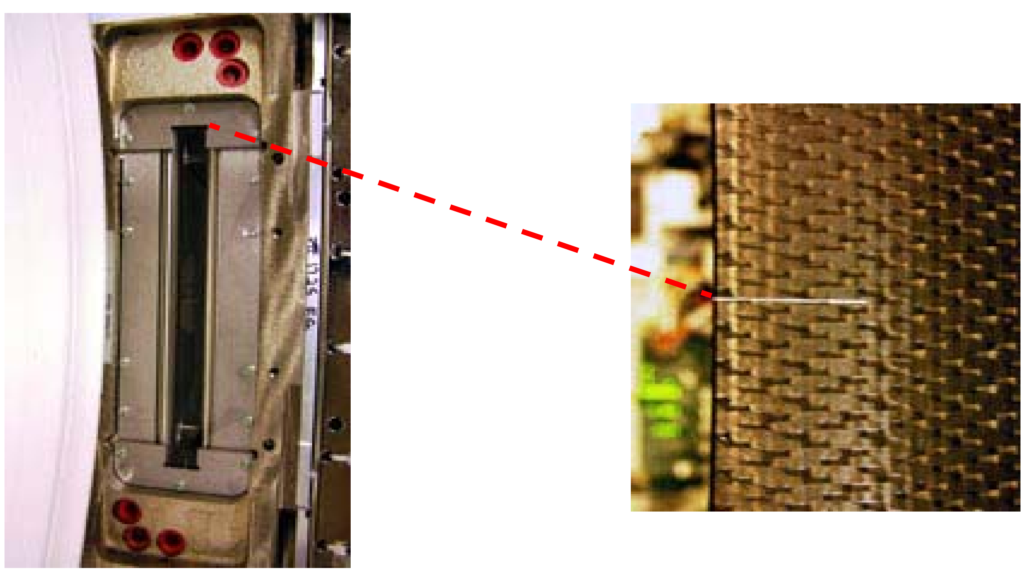


**Figure 1-37 Collimator Face Plate to Front of Table–Detail**

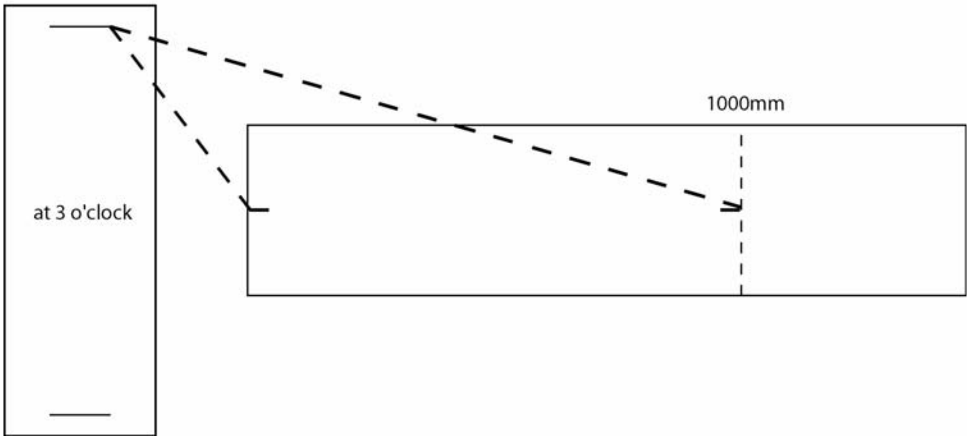



**Figure 1-38 Second Cradle/ Table Parallel Check.**

| Measurement | A: 3 o’clock | B: 9 o’clock | Difference: \| A-B \| |
|---|---|---|---|
| Test #1 | | | |
| To front of cradle | __________ | __________ | ________ ____ |
| To 1000mm mark | __________ | __________ | ________ ____ |
| Test #2 | | | |
| To front of cradle | | | |
| To 1000mm mark | | | |
| Test #3 | | | |
| To front of cradle | | | |
| To 1000mm mark | | | |

**Table 1-5 Alignment Worksheet (record your data here)**

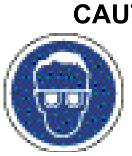
# 10.5 Finalization

## 10.5.1 Tighten the Lock Rings

1.) Re-check gantry bubble levels.

2.) Re-check that each of the eight adjuster is loaded by attempting to turn it.

**CAUTION** **Eye protection is required when using a hammer and chisel.**



3.) Tighten the lock rings at all locations with the spanner, where possible. Use a hammer and chisel to tighten the lock rings only where you cannot use the spanner.

4.) At least one full round of thread must be visible above the lock ring at all adjuster locations.

Note: The table may not have lock rings; nevertheless, one full round of thread must be showing above the base (see Figure 1-39).

**Figure 1-39 Table base with no lock ring**

# Section 11.0 Drilling the Table Anchor Holes

**WARNING** **POTENTIAL FOR PATIENT INJURY.**

**IMPROPERLY SECURED TABLE MAY TIP, DISLODGING PATIENT.**

**PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING SYSTEM OPERATION.**

## 11.1 Notes to Mechanical Installers

### Note 1: Basic Anchoring Information

GE-provided floor anchors are designed for use ONLY on concrete floors that meet the 102 mm (4 in.) concrete floor requirement. Supplied floor anchors must be installed by a trained contractor, and set to a minimum depth of 3 in. at each anchor point. ANY anchors having more than 1 in. of thread showing above the nut when torque is set to 75 ± 6 N-m (55 ± 5 ft.-lb.), shall have a second anchor installed in the closest adjacent hole. This is because the minimum anchor engagement length in the concrete was not met. The second anchor shall be installed to the standard depth and torque specification. **Do not cut anchor bolts that extend longer than the 1 in. limit.**

### Note 2: Alternate Anchoring

If at least four anchors cannot be set for the gantry, and at least four anchors for the table using the alternate anchor holes, then the installer must inform the PMI that the minimum anchoring cannot be met. Additionally, the customer’s structural engineering contractor must be engaged to determine the anchoring method, set the anchors, and certify that their anchoring meets the stated GE minimum load requirement and torque specification.

### Note 3: Non-Concrete Floors

All other anchoring methods—on floor types other than the concrete minimum—must be determined at the customer's expense by a structural engineering contractor. The anchoring and method must be certified by the customer’s contractor to meet the stated GE minimum load requirement and torque specification.

### Note 4: GE Notification

It is not the role of mechanical contractors or installers (FEs) to determine acceptable methods to install or anchor equipment on non-4 in. (102 mm) concrete floors. The PMI or appropriate GE contact person shall be notified that the facility's floor type DOES NOT MEET the installation mounting requirement for the installation procedure (described in this Installation Manual), and therefore the table-gantry mounting process CANNOT continue.

## 11.2 Time and Personnel

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 2 (FE or mechanical supplier) | | 0.5 hours labor on-site | |

## 11.3 Tools and Test Equipment

- Standard Install Tool Kit
- Hammer Drill
- ½ in. x 12 in. Drill Bit (Metric equivalent must not be used)
- ½ in. Drill Bushing (shipped in install support kit)
- Vacuum with HEPA or drywall dust filter
- Vacuum Hole Attachment – to clean debris from the holes
- PPE

## 11.4 Procedures

### 11.4.1 Driling the Anchor Holes

Note: The gantry rear cover should still be removed and the table should still be on the dolly.

1.) Make sure that all table and gantry levelers (four each) are firmly on the concrete floor.

**NOTICE Potential for Equipment Damage from Dust** **To prevent damage due to the dust created during drilling, you must cover all electronic assemblies in the table base prior to drilling.**

2.) Locate the hammer drill and ½ in. X 12 in. drill bit. The ½ in. bit is used to drill all eight (8) table and gantry anchor holes. You must use the drilling bushing to drill all table and gantry holes. All primary holes can be drilled with the gantry front covers installed.

3.) Use a piece of tape to mark the drill bit depth of 190 mm (7-1/2 in.) from the tip of the ½ in.masonry drill bit.

4.) Use the ½ in. bit to drill all eight (8) anchor holes to a depth of 190 mm (7-1/2 in.), as measured from the top of the drill bushing. Review Figure 1-40 and Figure 1-41 prior to drilling.

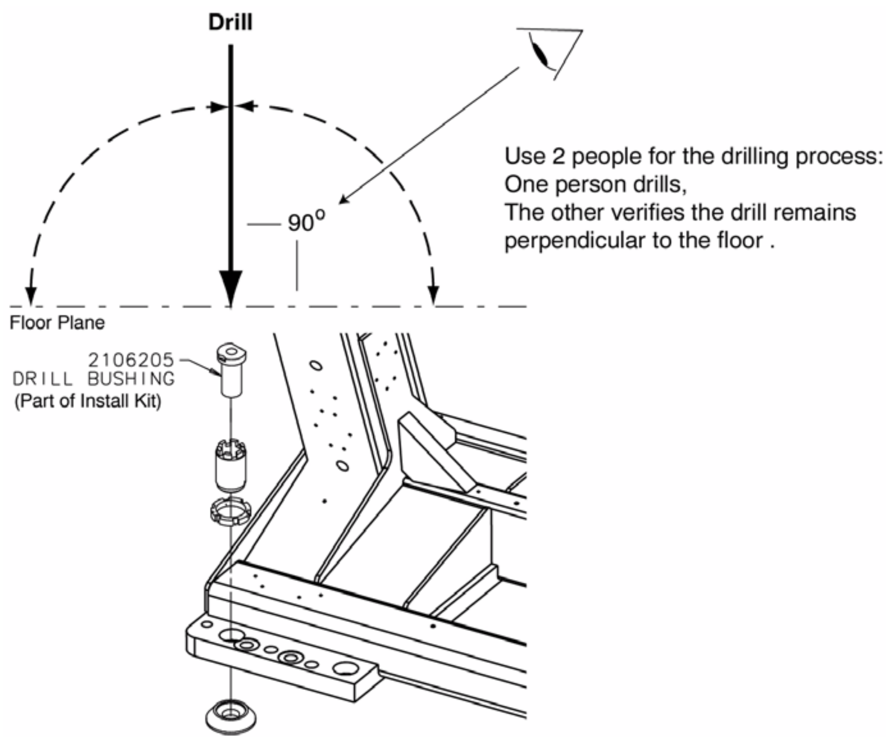



**Figure 1-40 Drilling Position**

5.) Place appropriate protection to prevent damage and dust contamination to electronic assemblies.

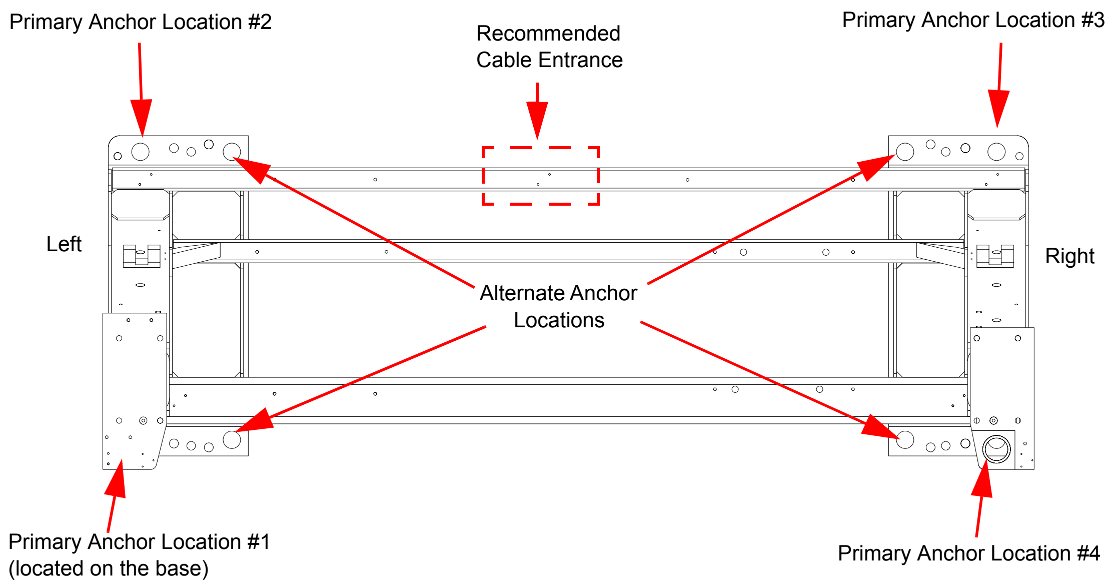
6.) Place the drill bushing inside each adjuster, to keep the hole vertical and centered within the adjuster.
- Use the drill bushing to center the anchor holes in all adjuster locations, to provide maximum lateral alignment capacity when you center the cradle on isocenter during subsequent system testing.
- Take care not to injure yourself on the gantry cover brackets.

7.) Drill the holes perpendicular to the floor.

**Important - Follow these guidelines when drilling anchor holes:**
- While one person drills the holes, position a second person to watch the relationship between the drill bit and floor. Make sure the bit remains absolutely perpendicular to the floor throughout the drilling operation.
- Always use the mechanical guide when drilling.
- Stop drilling every 15 or 20 seconds and clear the hole of debris. This lets the drill bit cool and helps to prevent binding of the drill bit.
- Vacuum while drilling to keep gantry and table as free of dust contamination as possible. Place the funnel tip or long extension tip inside the hole.
- **A drywall dust filter must be used on the vacuum.**
- Drill each hole until the mark on the drill bit is even with the top of the drill bushing. All holes must be a minimum of 190 mm (7.5 in.) deep, as measured from the top of the adjuster to the bottom of the hole (see Figure 1-42, on page 75). Use an upside-down anchor to check the hole depth.

8.) Recheck the depth of all holes by inserting an anchor backward into the hole. You should have 13 mm (½ in.) or less showing. Re-drill if needed.

9.) When finished drilling and clearing the anchor holes, vacuum the debris from the inside of each of the holes and from the surrounding (floor) area.



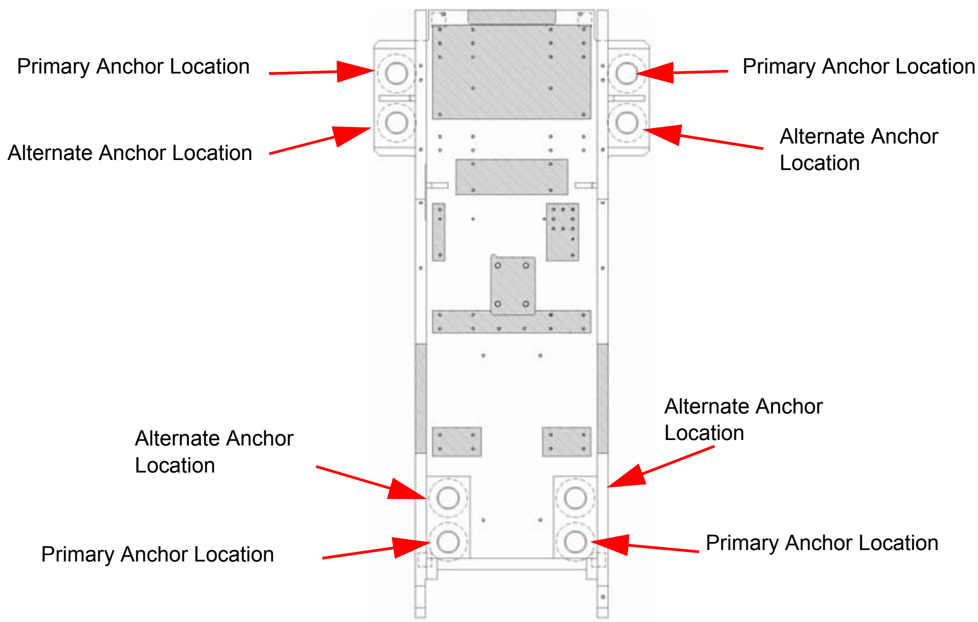




**Figure 1-41 Anchor Locations**

Note: If alternate location(s) are used to anchor the table or gantry, you must move the respective leveler(s) and pad(s) to the new alternate location(s) and re-drill.

### 11.4.2 Alternate Table and Gantry Anchor Holes

If you cannot use one of the adjuster anchor holes due to structural interference, such as reinforcement bars in the concrete, you must use one of the alternate anchor locations, as shown in Figure 1-41. You must also move the respective leveler(s) and pad(s) to the new alternate location(s) and re-drill.

Move the adjuster to the alternate anchor hole.

- The gantry requires a minimum of four (4) anchors, one (1) in each corner.
- The table requires a minimum of four (4) anchors, one (1) at location.

If you must use an alternate anchor hole in the gantry, you must remove the gantry covers to drill the holes. See Appendix A Gantry Cover Removal and Dolly Setup for gantry cover removal.

**WARNING**

**POTENTIAL FOR PATIENT INJURY.**

**IMPROPERLY-SECURED TABLE MAY TIP, DISLODGING PATIENT.**

**PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING SYSTEM OPERATION.**

It is the purchaser's responsibility to provide an approved support structure and mounting method for all floor types other than those listed. General Electric is not responsible for any failure of the support structure or method of anchoring, including seismic requirements and/or through-bolting.

Note: GE is not responsible for anchoring methods other than those listed in the *Pre-Installation Manual* for this system.

Provided floor anchors are designed for use ONLY on concrete floors that meet the 102 mm (4 in.) concrete floor requirements.

| Mounting Requirements | Gantry | Table |
|---|---|---|
| **Minimum Floor Thickness:** | 102 mm (4 in.) | 102 mm (4 in.) |
| **Recommended Drilling Depth:** | 95 mm (3-¾ in.) | 95 mm (3-¾ in.) |
| **Average Anchor Embedment:** | 89 mm (3-½ in.) | 89 mm (3-½ in.) |
| **Minimum Anchor Embedment:** | 76 mm (3 in.) | 76 mm (3 in.) |
| **Available Alternate Anchor Locations:** | Yes | Yes |
| **Shipped Anchor Size:** | 203 mm (8 in.) | 203 mm (8 in.) |
| **Alternate Anchoring Methods:** | Yes (see notes, above) | Yes (see notes, above) |
| **FLoor Levelness Requirement:** | 8 mm (0.315 in.) over 3 m (10 ft) | 8 mm (0.315 in.) over 3 m (10 ft) |

**Table 1-6 Gantry and Table Mounting Requirements**

### 11.4.3 Installing the Anchors

**Recommended**: Use “Hilti Kwik-Bolt II” anchors P/N 2106573 (½ in. dia. x 8 in. long) as shipped with the VCT/CT750 HD system for this procedure.

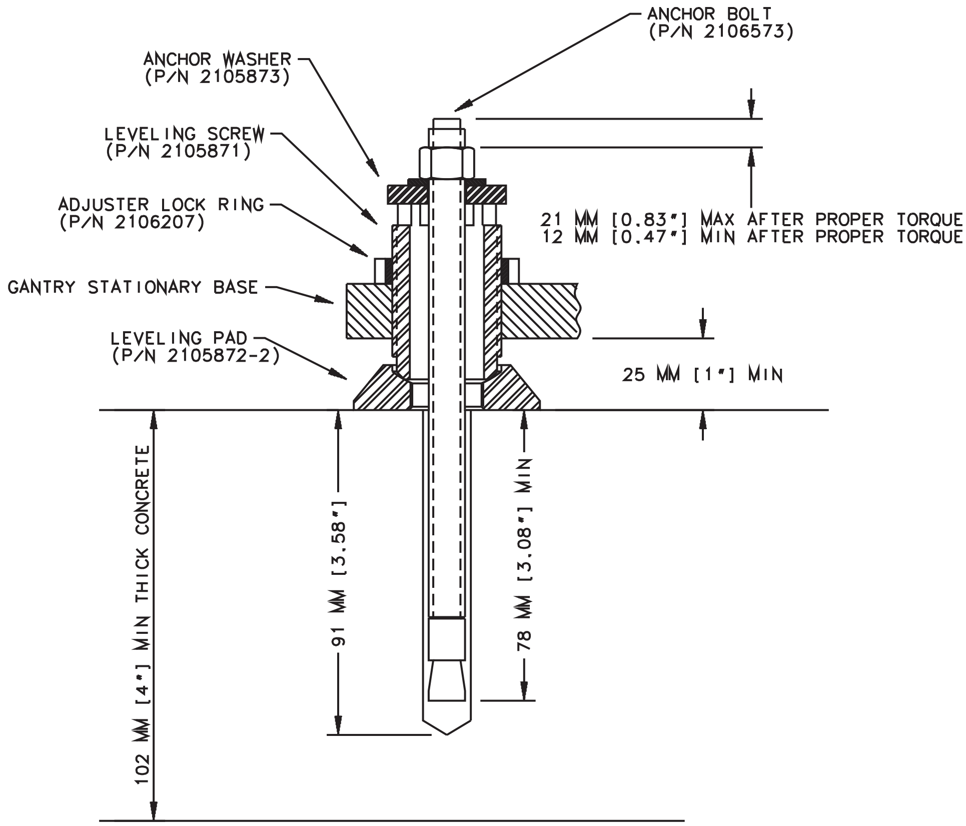


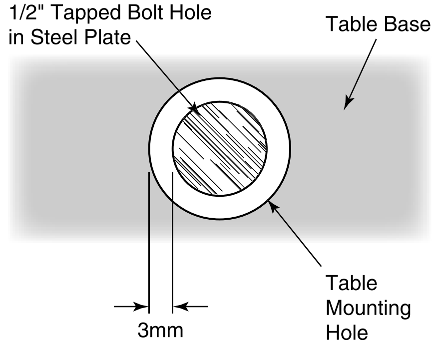

**Figure 1-42 Table Base Anchor Assembly**



**Figure 1-43 Correct Table Base Anchor Assembly (left) vs. Incorrect Anchor Assembly (right)**

1.) Assemble the anchors before you install them. (refer to Figure 1-42).
   a.) Remove the nut and washer from the anchor.
   b.) Add a ¼ in. thick washer (PN 2105873) under the regular anchor washer.
   c.) Reassemble the anchor washer and nut and position nut so the top is flush with the threads of the anchor.

2.) Use the anchor seating tool to hammer anchors into the holes.
3.) Adjust all eight (8) anchor bolts until tight.
4.) No more than 25 mm (1 in.) should show above the nut.
5.) Do not cut off the anchor.
6.) There should be at least one full round of thread showing above the lock ring.
7.) Complete this section on GE Form e4879.




Bolt centering is important to provide ± 3mm of adjustment for electrical alignment. Always use the drilling centering tool when drilling all bolt holes.

**Figure 1-44 Center tapped holes under mounting holes in table base**

# 11.5 Finalization

## 11.5.1 Alignment Recheck

Note: Alignment is critical. Recheck carefully.

1.) Turn on the alignment tool and recheck alignments. The table alignment must be the same as in the Cradle/Table Parallel Check ( Section 10.4.1 on page 66). If re-leveling is required, repeat this procedure. Using the bubble levels, make adjustments to maintain the required alignment.
2.) Once alignment is verified, torque all mounting bolts.
Tighten the location #1 through #7 anchors and torque to 75 ± 6 N-m (55 ± 5 ft.-lb.)
3.) Mark the laser locations at 1000mm and at 2000mm.
4.) Using the 1mm wire, place the wire over the mark at each location, and confirm that the laser is still on the wire.
5.) Remove the laser tools.
6.) Reinstall all the removed table panels and hardware.

Note: If you cannot replace the lower table cover because the floor interferes, adjust all of the table and gantry levelers by half-turn increments to raise the table/gantry until the lower table covers clear the floor. Then return to the alignment sections to level the gantry, level the table, and tighten the locking rings, respectively.

### 11.5.2 Remove the IMS Shipping Lock

1.) Remove the right top cover of the table.

2.) A yellow shipping lock is located on the right side of the table.

3.) Unscrew the two bolts and remove the shipping lock.

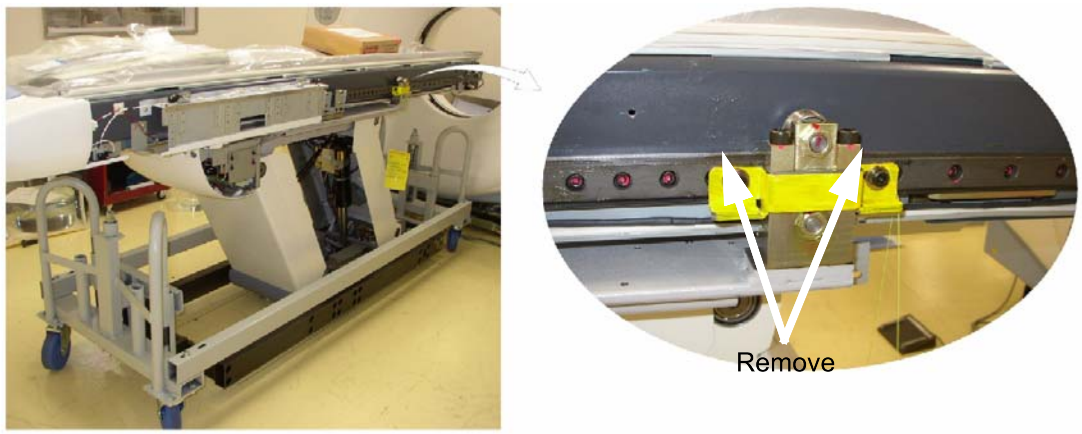



**Figure 1-45 IMS Shipping Lock**

4.) Re-install the table side cover. See section on Table Cover Installation.

# Section 12.0 Removing Table Shipping Dollies

## 12.1 Time and Personnel

(FE or mechanical supplier)

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 1 (FE or mechanical supplier) | | 20 min. labor on-site | |

**Table 1-7**

## 12.2 Tools and Test Equipment

- Standard Installation Tools Kit

## 12.3 Preparation

- All table mechanical alignment procedures are completed.
- The table is on the floor with at least one anchor in place.

## 12.4 Procedure

Refer to Figure 1-46 for the location of items in the table dolly.

1.) Remove the two, long side (stabilizer) rails using the quick disconnect pins. There is one pin on each end of the bar.

2.) Carefully slide the bar out and place the bars on the side, out of the traffic area.

Note: The table should be resting on the floor. You may need to lower or raise the dolly to remove the dolly ends.

3.) Using the quick release pins, remove each end of the dolly.

4.) Slide the dolly off of the long attachment bar on each side.

5.) Using the quick release pins, remove the two long attachment bars that are attached to the front and rear table attachment points. Place the bars on the side, out of the traffic area.

6.) Use the 19 mm wrench to remove the bolts on each side of the smaller front table attachment bracket.Remove the bracket.

7.) Use the 19 mm wrench to remove the bolts on each side of the larger front table attachment bracket and transporter base. Remove the bracket.

8.) Reassemble the dolly for transportation.

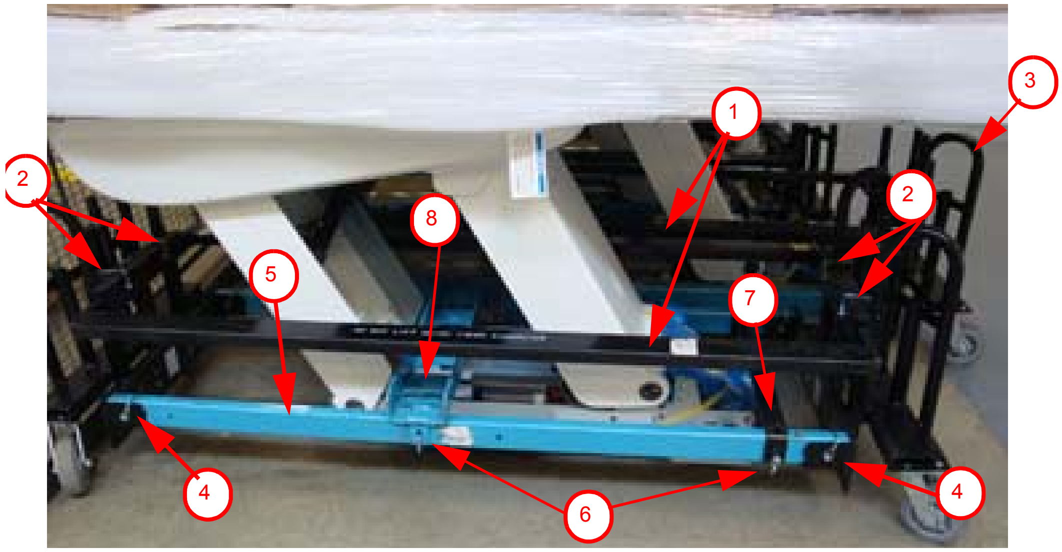



**Figure 1-46 Table Shipping Dollies**

| Item | Description |
|---|---|
| 1 | Side Rails (two rails, black, one each side) |
| 2 | Quick Release Pins (for side rails, two on each side) |
| 3 | Dolly Ends (two, one each end) |
| 4 | Quick Release Pins (for dolly ends, two each side) |
| 5 | Table Dolly Lifting Tube (two, blue, one each side) |
| 6 | Quick Release Pins (for dolly lifting tube, two each side) |
| 7 | Front Attachment Bar (black, one; 19 mm table base bolt located under the bar, not shown) |
| 8 | Back Attachment Bar (blue, one; see Figure 1-47) |
| 9 | Table Base Bolt (see Figure 1-47) |

**Table 1-8 Description of Table Shipping Dollies**

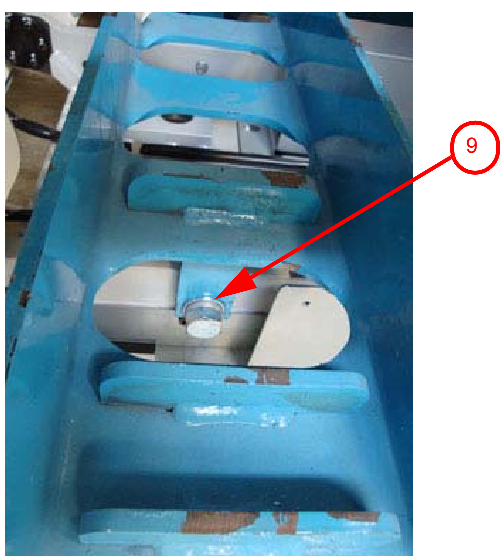



**Figure 1-47 Close-up of table base connection**

# Section 13.0 Rear Entry Cable Box

If you cannot bring up the cables to the gantry inside the gantry base, use a rear entry cable box (B7850TC).The box is supplied with the system. Gantry rear entry is preferred for cables. The base access area is limited.

1.) Attach the rear entry cable box frame to the gantry base using four (4) screws that are shipped with the kit (see Figure 1-48). The assembly can be made to fit floor entrance conduit or surface floor duct.

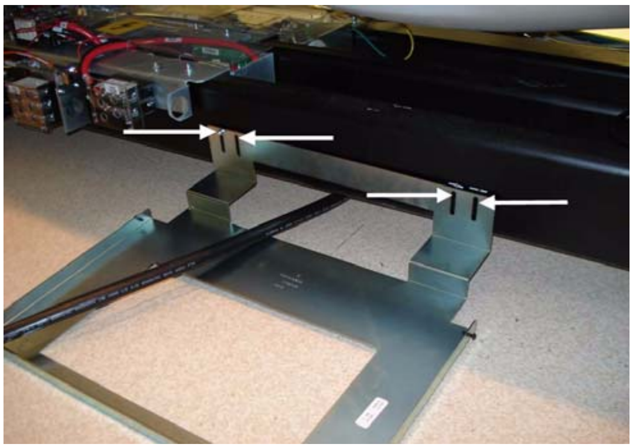


**Figure 1-48 Rear Entry Cable Box**

2.) There are three pairs of spacers shipped with this cover. Select the pair that is most appropriate for this site, based on the hardware. The three types are: Solid metal, Precut L-shaped metal, and Solid plastic - Can be cut

3.) Cut filler plates to fit the cover box into the opening, if necessary.

4.) When finished, it should look like the lower picture in Figure 1-48.

## 13.1 Rear Entry with Surface Floor Duct

An OSHA ramp is required. The ramp must have 1’ run of slope for each 1" rise in height.

# Section 14.0 Install Table Footswitch Assembly

## 14.1 Time and Personnel

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 1 (FE or mechanical supplier) | | 1.0 hour labor on-site | |

## 14.2 Tools and Test Equipment

- Standard Install Tool Kit

## 14.3 Procedure

After table positioning is completed and the anchors are installed, install the footswitch assembly as shown in Figure 1-49.

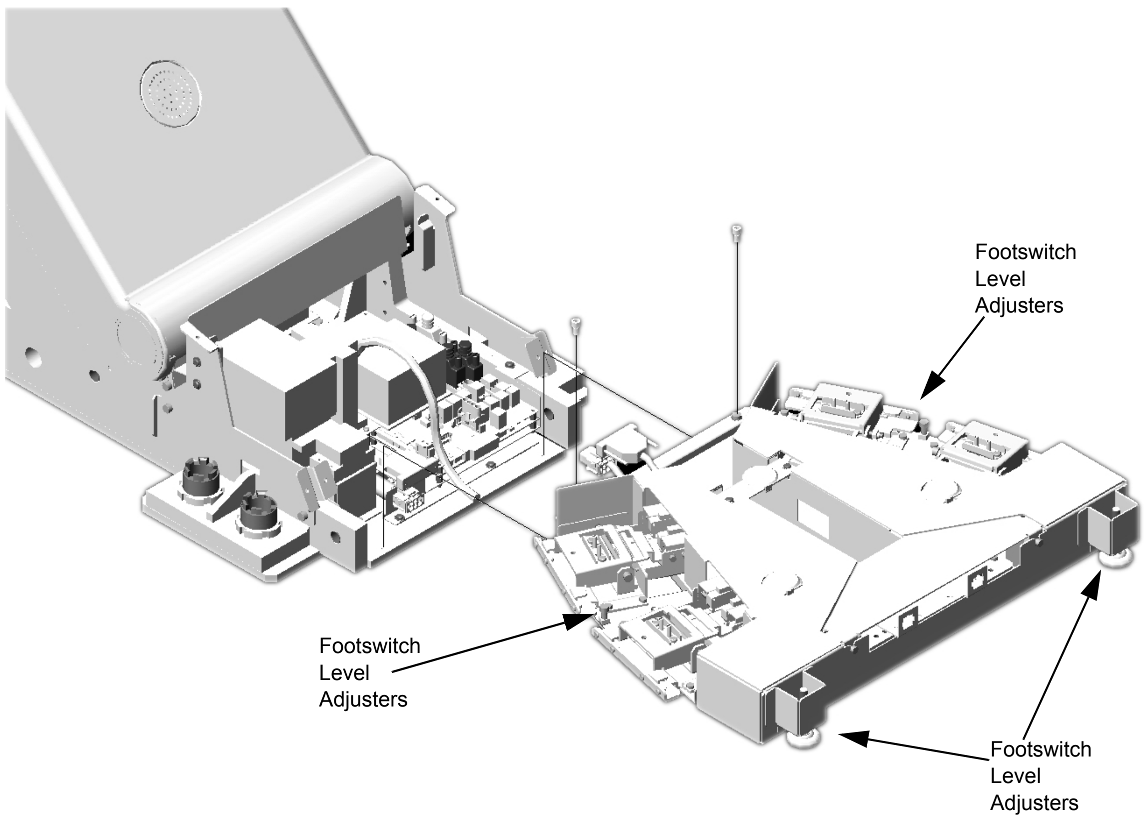



**Figure 1-49 Footswitch Assembly Installation**

1.) Pop off foot pedal screw cover tabs.
2.) Remove foot switch covers.
3.) Remove 3 Phillips screws that secure the assembly cover.

4.) Remove the footswitch assembly cover.

5.) Using two (2) M6 bolts, attach the footswitch assembly to the table base.

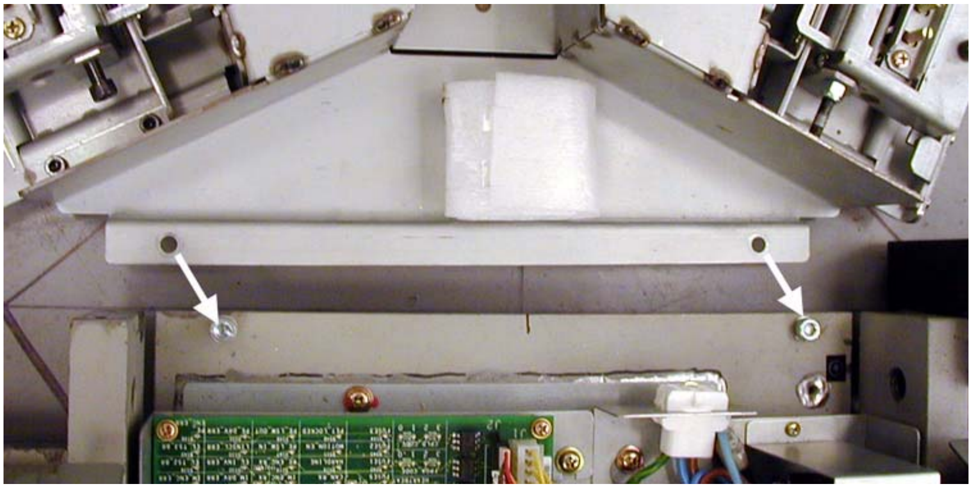
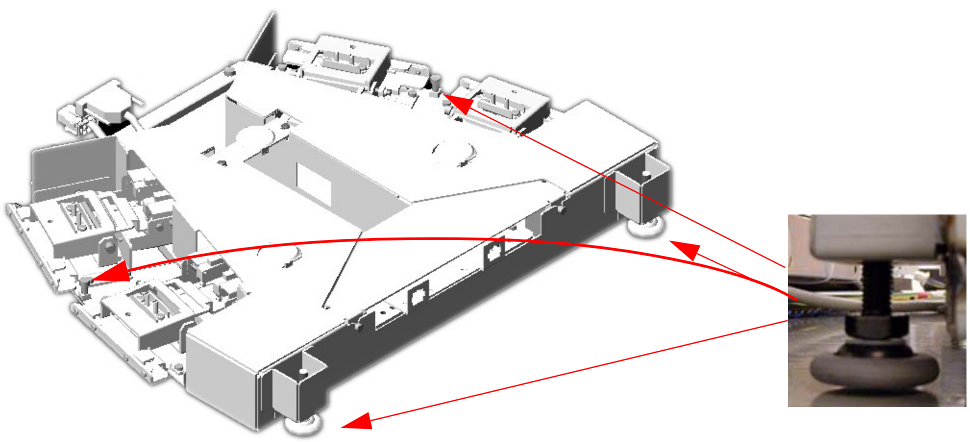


**Figure 1-50  Attach Footswitch**

6.) Level the footswitch assembly using the three (3) level adjusters. Two are on the gantry side and one is in the middle. Use a 9 in. level to check the levelness in all directions.



7.)

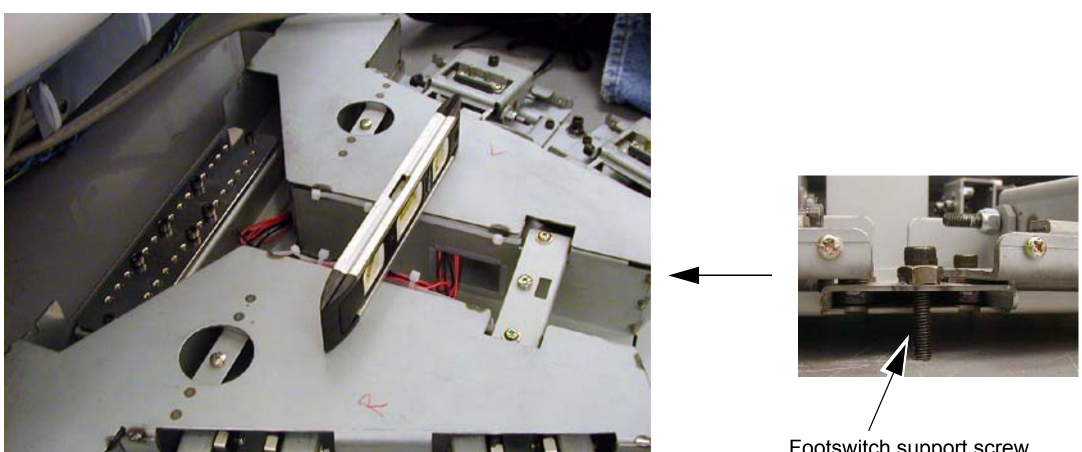



**Figure 1-51  Level Footswitch**

8.) Cut the tie-wraps from around the cables in the gantry base and route the power cables from the gantry as shown in Figure 1-52.

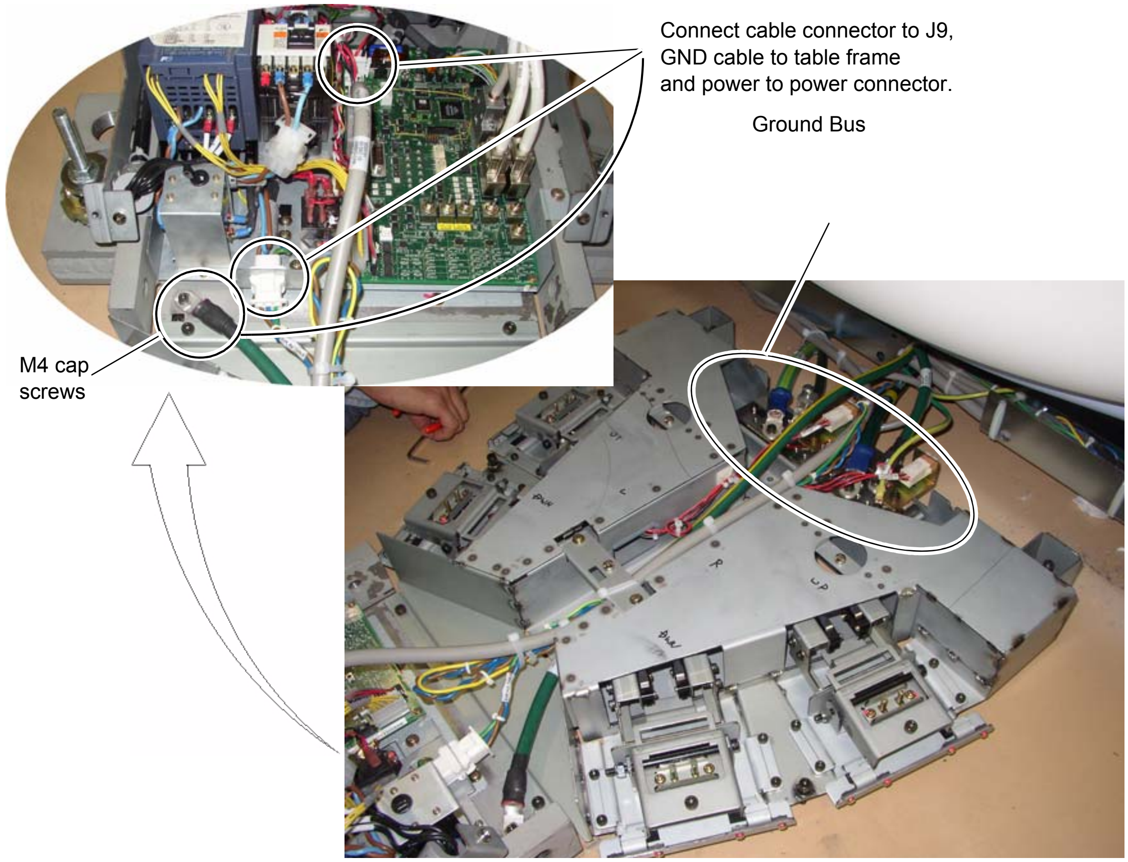


**Figure 1-52 Footswitch Assembly Cable Wiring**

9.) Connect the ground bus connector plate.

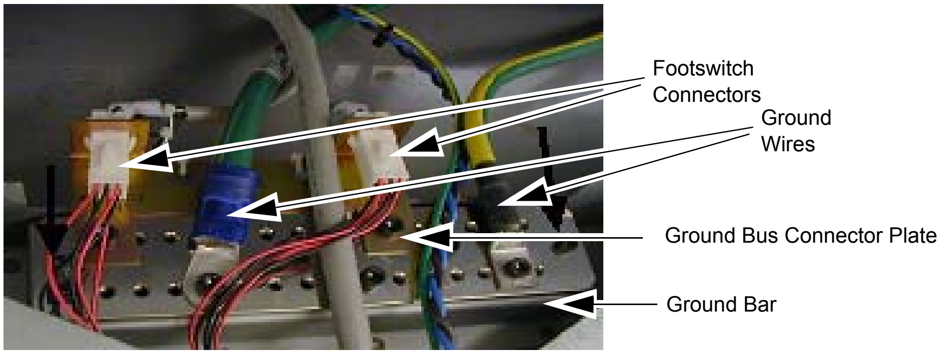



**Figure 1-53 Footswitch Ground/Bus Bar**

10.) Install the footswitch pedal bracket onto the installed ground bus bar.

11.) Connect the ground wires (not all shown in Figure 1-53) to the installed ground bus: Torque to 7.9Nm (5.8lb-ft)

Table#2

Gantry#1/0

Console#2

PDU#1/0

12.) Install all footswitch covers after work is completed. See Chapter 4 Section 1.7.4.

# Section 15.0 Remove Gantry Tilt Bracket

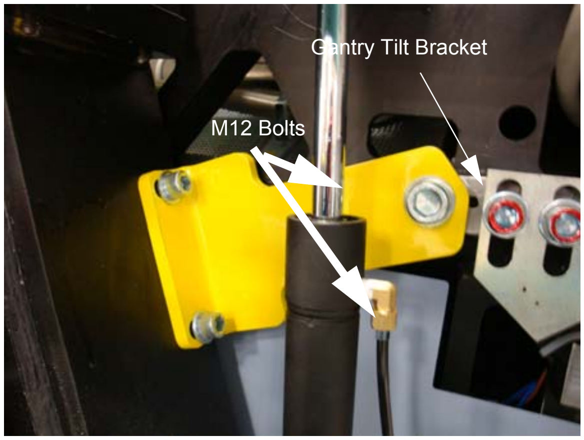



**Figure 1-54 Gantry Tilt Bracket Removal**

**CAUTION**

**Potential for personal injury.**

**If there is a load on the tilt bracket, removal may cause the gantry to suddenly tilt all the way back due to a possible lack of hydraulic pressure.**

**If tilt bracket is not loose, stop and put the M12 bolts back in and tighten the tilt bracket back in place.**

## 15.1 Time and Personnel

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 1 (FE or mechanical supplier) | | labor on-site | |

## 15.2 Tools and Test Equipment

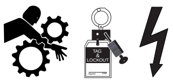
- 10 mm Hex wrench
- 14 mm Hex wrench

## 15.3 Procedure

1.) Remove the M12 bolts using a 10 mm Hex wrench (see Figure 1-54).
2.) Loosen the M16 bolt 1-2 turns and check the gantry tilt bracket; it should be loose to the touch.
   - If the tilt bracket is loose, continue with step 3.
   - If the tilt bracket is NOT loose: check the hydraulic connections for leaks or lack of fluid. You must wait until the system can be energized to use the tilt controls to relieve the load on the tilt bracket prior to removal. Do not use force to remove the bracket.
3.) If the bracket feels loose, remove the M16 bolt using a 14 mm Hex wrench.
4.) Remove the bracket.
5.) Reinstall the gantry rear covers and the scan window. (Refer to Appendix A Gantry Cover Removal and Dolly Setup, on page 167.)
6.) Store the brackets in the service cabinet.

# Section 16.0 Position the Power Distribution Unit

**WARNING**



**LOCKOUT/TAGOUT IS REQUIRED BEFORE PERFORMING THIS TASK. USE THE SUPPLIED LOTO KIT.**

**Note:** **Connecting the primary incoming power (steps 2 through 5, below) is performed by the customer's electrical contractor. The electrician needs to provide a reducing bushing to attach the flexible conduit to the PDU**

## 16.1 Time and Personnel

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 1 (FE or mechanical supplier) | | Labor on-site | |

## 16.2 Tools and Test Equipment

Hex wrench set

## 16.3 Procedure

1.) Roll the PDU into position on its permanently mounted casters. Leave at least 15.5 cm (6 in.) between the PDU and back wall to allow cooling air to circulate.

| Connection or Wall Box | AWG # | Connection From | Connection To PDU | Installed & Checked |
|---|---|---|---|---|
| TS1 | #1 | PDB-A | TS1-1 | |
| | #1 | PDB-B | TS2-2 | |
| | #1 | PDB-C | TS3-3 | |
| | #1/0 | GND | N/G<br>**(Do NOT connect anything to neutral point.)** | |

**Table 1-9 Contractor Connections**

2.) Run the main input power conductors and ground though flexible metal conduit (attached between the PDU chassis and room duct-work) so you can move the PDU away from the wall during service.

**Figure 1-55 Flexible Conduit for PDU Power**

3.) Locate the hole cover plate in Box 1 and attach the flexible metal conduit to the PDU.
4.) If present, remove the TS1 panel front cover.
5.) Strip the wires to fit securely on the power block.
6.) Observe incoming phases (L1, L2, and L3) and insert bare leads into power block.
7.) Insert vault ground into PDU vault ground lug.
8.) Tighten all fasteners securely and replace the TS3 front panel.

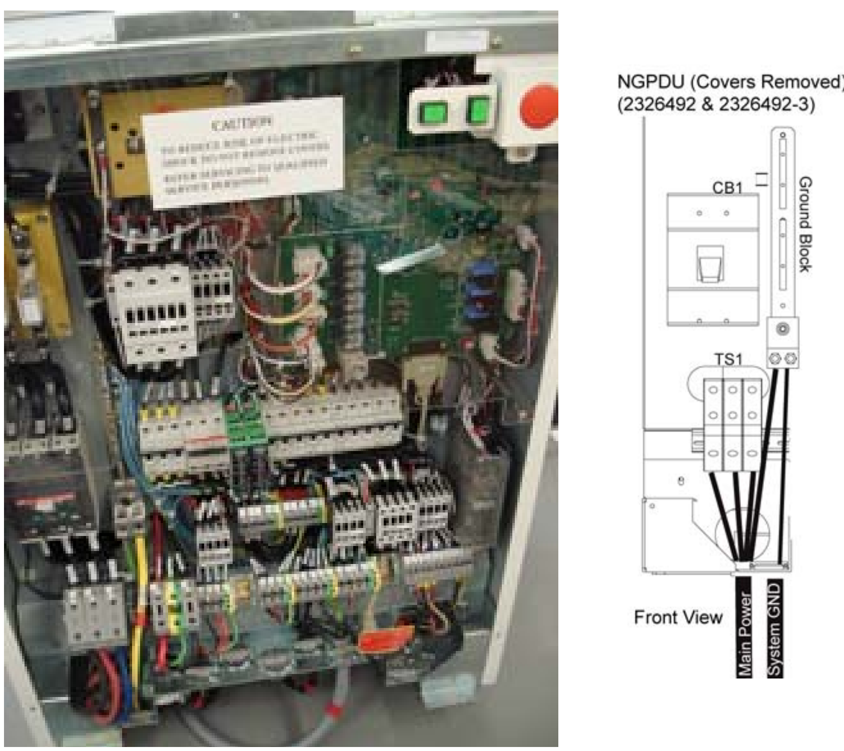



**Figure 1-56 PDU Area Locations**

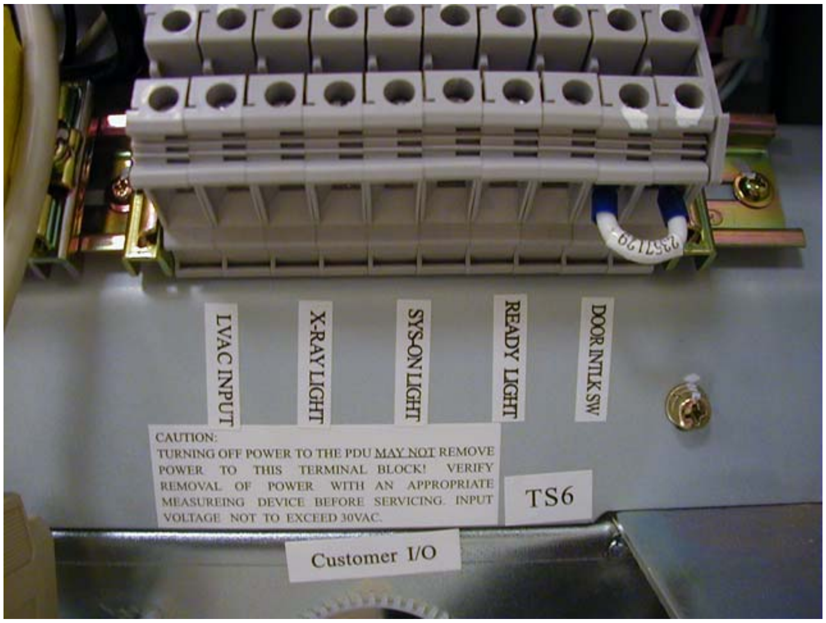



**Figure 1-57 PDU Power Connections**

**WARNING**

**WORK WITH THE ELECTRICAL CONTRACTOR TO BE SURE EXTERNAL POWER SOURCE IS TURNED OFF.**

| CONNECTION OR WALL BOX | AWG # | CONNECTION FROM | CONNECTION TO PDU | INSTALLED AND CHECKED |
|---|---|---|---|---|
| A1 | #1 | Load - T1 | TS-1 L1 | |
| | #1 | Load - T2 | TS-1 L2 | |
| | #1 | Load - T3 | TS-1 L3 | |
| | #1/0 | GND | TS-1 GND **(Do NOT connect anything to neutral point.)** | |
| WL (Warning light) *See Figure 2-39, on page 130.* | #14 | LV Source -1 | TS6 1 | |
| | #14 | LV Source -2 | TS6 2 | |
| | #14 | X-Ray ON Light -1 | TS6 3 | |
| | #14 | X-Ray ON Light -2 | TS6 4 | |
| | #14 | Sys-ON Light -1 | TS6 5 | |
| | #14 | Sys-ON Light -2 | TS6 6 | |
| | #14 | Ready Light -1 | TS6 7 | |
| | #14 | Ready Light -2 | TS6 8 | |
| DS (Scan Room Door Switch) | #14 | Door SW-1 | TS6 9 | |
| | #14 | Door SW-2 | TS6 10 | |

**Table 1-10 Contractor PDU Connections**

# Section 17.0 Install Operator Console (GOC6)

## 17.1 Time and Personnel

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 1 (FE or mechanical supplier) | | labor on-site | |

## 17.2 Tools and Test Equipment

## 17.3 Procedures

### 17.3.1 Unpack the Console

**Note: If console is construction site packed, vacuum the dust and debris before removing packaging.**

**Figure 1-58 Console boxed on skid**

1.) Remove all items from the console.
2.) Remove all packing materials and discard.

3.) Place the step-board under the front edge of the skid and step on it to raise the front edge of the skid as shown in Figure 1-59.

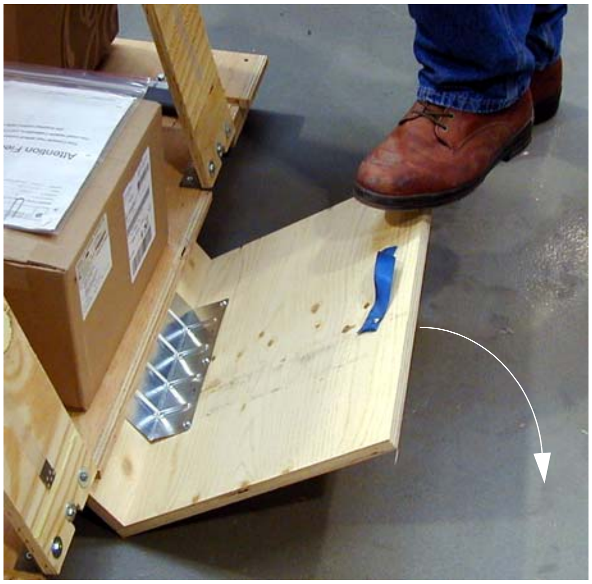


**Figure 1-59 Step-board used to raise front edge of skid**

4.) Remove the two front cushions from the bottom of the skid (see Figure 1-60).


**Figure 1-60 Cushion on bottom of skid**

5.) Remove the Seismic Brackets from each side of the console (see Figure 1-61). Save the brackets for use later if you need to mount the console to the floor.

**Figure 1-61 Removing the Shipping/Seismic Brackets**

6.) Lift up on the strap on the front of the step-board (Figure 1-59) to lower the skid. Remove the step-board.

7.) Ensure the console stabilizers are in line with the notched portion at the front of the skid. This allows enough clearance to smoothly roll the console down the ramps (see Figure 1-62).

**Figure 1-62 Console ready to be unloaded**

8.) Move the console to the installation location.

9.) Attach the two pads to the console stabilizers (see Figure 1-63).

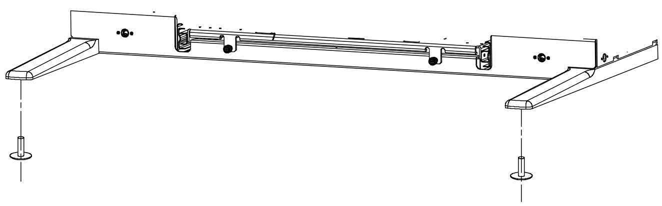


**Figure 1-63 Console Stabilizer Pads**

10.) Screw the pads all the way into the legs

### 17.3.2 Remove Console Covers

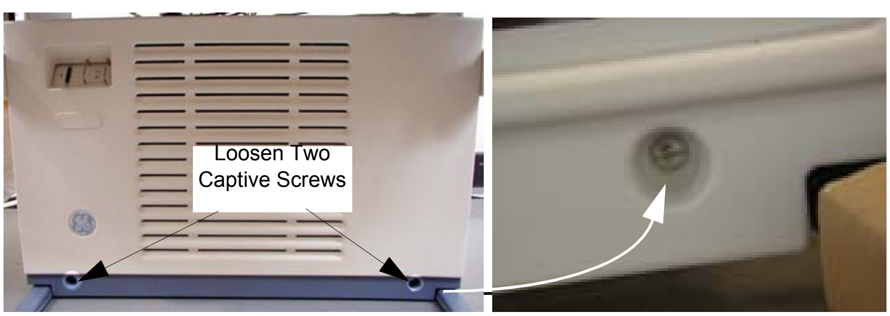



**Figure 1-64 Front Cover Removal - Global Consoles**

1.) From the service desktop, shut down the system

2.) Turn off console power.

3.) Loosen two the captive screws at the bottom of console.

4.) Rotate the bottom of the cover outward and upward until you can lift it free of the console at the top.

### 17.3.3 Adjust Table Top Height and Position Console

1.) The console normally arrives with the bottom hole on the monitor top aligned with the fourth bolt from the bottom (see Figure 1-65).


**Figure 1-65 Global Console tabletop adjustments**

2.) For optimum operator comfort, align the bottom hole of the keyboard table top with the fifth hole from the bottom on the console.

Note: Your site may have different requirements and you may have to adjust the monitor top and/or keyboard table top up or down from this position.

- Always select a table top height that permits the operator to see the patient on the table.
- Keep the one bolt-hole relationship between the monitor top and the keyboard table top.
- Fasten the keyboard table top one (1) hole lower than the console table top (Figure 1-65).

3.) Insert washers on front bolts.

4.) Install the console side covers.

Side covers are left and right specific.

5.) Move the console into its final position, maintaining a 15 cm (6 in.) minimum distance between the console and the wall.

### 17.3.4 Attach Keyboard Table Top

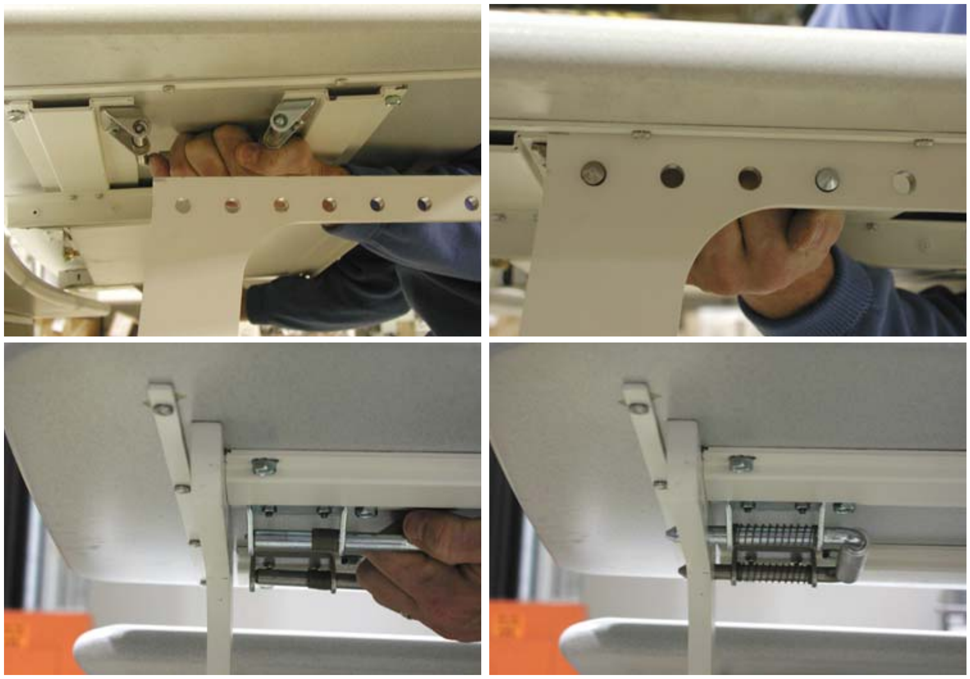


**Figure 1-66 Keyboard Table Top Placement**

1.) Locate the spring-loaded latches at either end of the underside of the keyboard table top.
2.) Pull the latches inward (toward center of table top).
3.) Position table on the brackets so that the latches align with the holes.
4.) Release the latches, making certain that the pins fully engage the bracket holes (pins should protrude beyond outside edge of bracket).

# Section 18.0 Seismic Mounting

Before proceeding with seismic mounting for any of the components in this section, be sure to allow sufficient space to unbolt and move the component from its mounted location for service.

- You may need to remove all four mounting bolts.
- If removing the component requires lifting, use an appropriate-sized pry bar to lift each corner of the component.
- Two installers may be required to safely complete this task.
- 5/8 in. anchor for IBC 2007 standards or as provided in the seismic kit.

## 18.1 Time and Personnel

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 1 (FE or mechanical supplier) | | Labor on-site | |

## 18.2 Tools and Test Equipment

- Seismic Mounting Kit (P/N R4390JC)

## 18.3 Procedures

### 18.3.1 Console

If site specifications require seismic mounting to the floor, use 5/8 in (15 mm) bolts, or those supplied in the seismic mounting kit, to mount the seismic brackets that shipped with the console. Refer to Figure 1-67 for mounting hole locations, and mount the console so it can be easily removed for service.

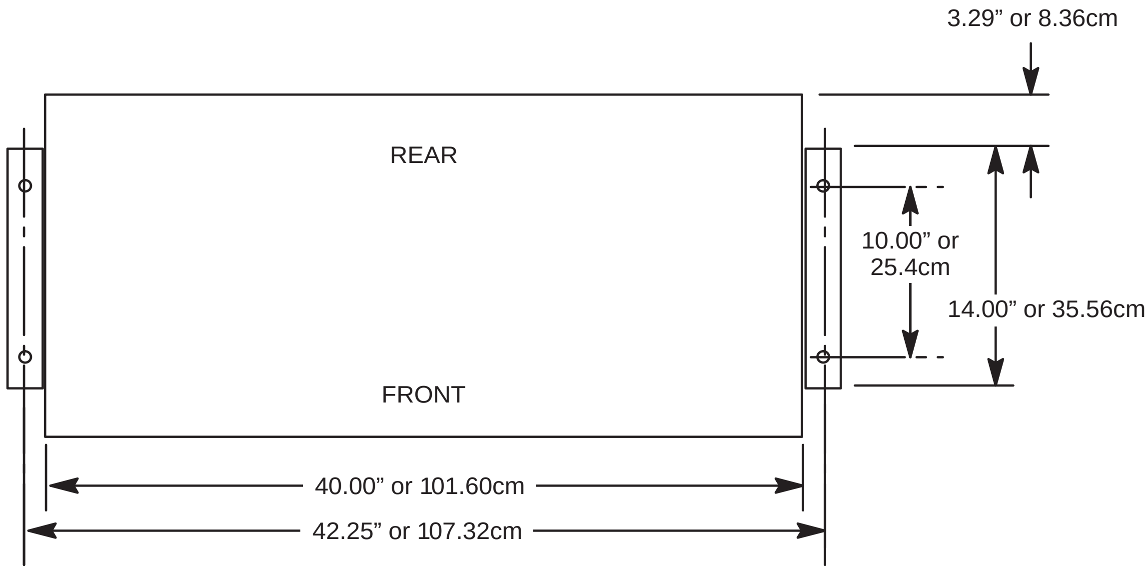



**Figure 1-67 Seismic Console Mounting Hole Locations**

### 18.3.2 NGPDU

**CAUTION** **The PDU is very heavy and may present a crush hazard if proper precaution and tools are not used.**

If site specifications require seismic mounting, use 5/8 in. (15mm) bolts, or those supplied in the seismic mounting kit, and the seismic brackets and anchors that shipped with the NGPDU. Refer to Figure 1-68 for mounting hole locations, and mount the PDU so it can be easily removed for service.

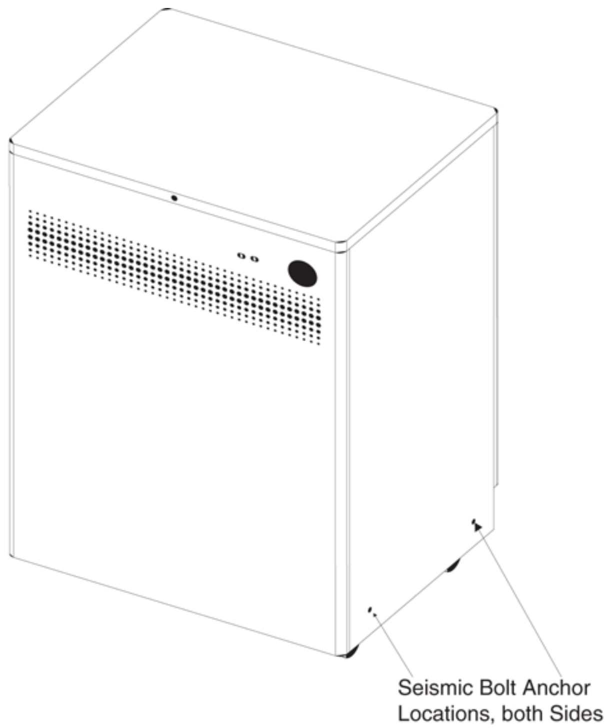



**Figure 1-68 Seismic PDU Mounting Hole Locations**

## 18.4 Uninterruptible Power Supply (UPS)

**CAUTION** The UPS is very heavy and may present a crush hazard if proper precaution and tools are not used.

If the site has a UPS (shown in Figure 1-33), refer to the 3 Phase UPS Installation manual, Direction 5116426-100. If site specifications require seismic mounting, use 5/8 in. (15mm) bolts, or those supplied in the seismic mounting kit, and the seismic brackets that shipped with the UPS. Refer to the manufacturer's installation instructions to locate the seismic mounting holes, and mount the UPS so it is easily removed for service.

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# *Chapter 2*
# *Power, Ground & Interconnect Cables*

**NOTICE** **Potential for Data Loss and/or Equipment Damage**

**To prevent potential data loss and equipment damage, please do the following:**

- **Record data collected from procedures in this chapter into Form F4879 when directed.**
- **Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.**

## Section 1.0 Introduction

Site use of conduit, floor duct, wall duct, or a raised computer floor, as well as the individual component layout determines the system cable sequence. If your site has floor or wall ducts that will interfere with placement of the table/gantry, it may be important to have the movers unload the cable boxes (8 & 9) first and run those cables while others unload the subsystems.

- **Try to run the system cables after the contractor completes the contractor supplied wiring.**
- **All ground wires and other contractor wiring should be complete to the point of equipment placement.**

**NOTICE** **Potential for Equipment Damage**

**Do not store excess cable in the bottom of the PDU or Gantry.**

Do not store excess cable behind or under the installed components (table, PDU, gantry or console). Check with the site electrical contractor, before hiding excess in conduits or cable ducts.

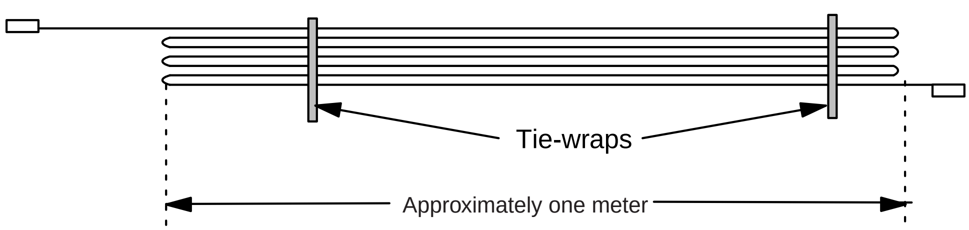



**Figure 2-1 Excess Cable Storage Configuration**

- **Keep signal and control cables away from power cables and power wiring. When you lay cables in a raceway, locate the signal cables in a separate section of the raceway, or a separate conduit.**
- **Check all connections for tightness.**
  - Use suitable tools and judgment.
  - Check all visible connections, especially ground connections.
- **Check for reasonable cable routing.**
  - Take into consideration necessary take-up distances for equipment maintenance, etc.
  - Try to complete as neat a job as possible.

## 1.1 System Component Identification

Identify all system cables by the system component designators listed in Table 2-1. Each end of a system cable has a label, and may have a color near the connector, (refer to Table 2-2) to indicate the component and the jack identifier of the component. All cables are located on the lower right shelf of the lean cart.

| Designator | System Component |
|---|---|
| CT2 | Gantry |
| CT1 | Patient Table |
| PM | Power Distribution Unit |
| OC | Operator Console<br>(Console Computer) |
| WL | X-Ray ON Warning Light |

**Table 2-1 System Component Identifiers**

## 1.2 Cable Color Identifiers

The ends of the cables may be marked with a piece of blue, yellow, red, or orange colored tape to help with the cable installation. Table 2-2 lists the subcomponent, and corresponding color.

| Subcomponent | Color |
|---|---|
| Gantry | Blue |
| Table | Yellow |
| PDU | Red |
| Console Computer | Orange |

**Table 2-2 Cable Color Identifiers**

If your system is not a lean packed system, locate the boxes with the system cables. They should match the cables in the following table.

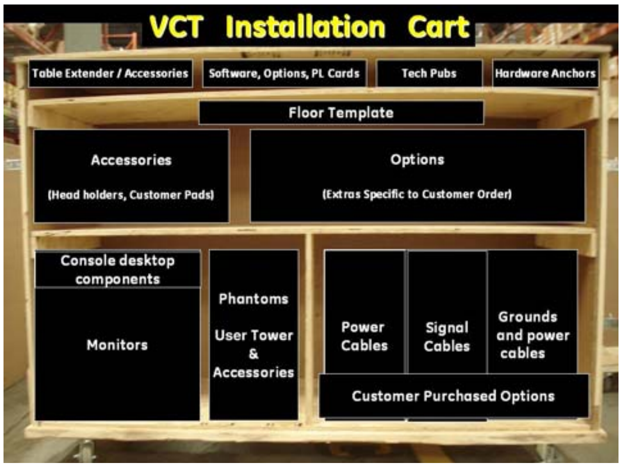



**Figure 2-2 Cable Location in Lean Cart**

| CT750 HD ONLY | | | |
|---|---|---|---|
| | | Part Number | |
| Run No. | Description | Long Cables (Kit 2281840-9) | Short Cables (Kit 2281840-10) |
| 1 | Facility MDP to Room Disconnect (A1) | cust. supplied | cust. supplied |
| 2 | Room Disconnect (A1) to PDU | cust. supplied | cust. supplied |
| 3 | Room Disconnect (A1) to System E-Off | cust. supplied | cust. supplied |
| 4 | PDU to Room Warning Light(s) | cust. supplied | cust. supplied |
| 5 | PDU to Scan Room Door Switch | cust. supplied | cust. supplied |
| 50 | HVDC Power Cable - PDU to Gantry | 2343529 | 2343529-2 |
| 51 | HVAC Power Cable - PDU to Gantry | 2343530 | 2343530-2 |
| 52 | LVAC Power Cable - PDU to Gantry | 2343528 | 2343528-2 |
| 53 | LVAC Power Cable - PDU To Operator’s Console | 5121809 | 5121809-2 |
| 54 | LVAC Power Cable - Gantry to Table | n/a | n/a |
| 55 | Ground, PDU to Raceway | 2371450 | 2371450-2 |
| 56 | Ground, Raceway to Console | 2371450-3 | 2371450-4 |
| 60 | LVAC Power Cable - PDU to Optional UPS | - | - |
| 61 | LVAC Power Cable - UPS Disconnect Panel to PDU | - | - |
| 90 | LVAC Power Cable - PDU to PET | - | - |
| 100 | Signal Cable - Gantry to PDU | 5120646 | 5120646-2 |
| 101 | Signal Cable - Gantry to Console | 5120645 | 5120645-2 |
| 102 | Signal Cable (Ethernet) - Gantry to Console | 2373436 | 2373436-3 |
| 103 | Data Cable (Fiber Optic) - Gantry to Console | 5316594 | 5316594-2 |
| 104 | Signal Cable - Gantry to Table | n/a | n/a |
| 110 | Signal Cable - UPS Control to Room Disconnect (A1) | - | - |
| 111 | Signal Cable - UPS Control to UPS Disconnect Panel | - | - |
| 112 | Cardiac Monitor Cat 5 - Signal cable between console and Gantry Option Board (GOB) (Quantity 2 | 5193967-4 | 5193967-6 |
| 113 | Cable - Gantry to Injector_PathFinder | 5169456 | 5169456-2 |
| 114 | RPM Unit | 5199717 | 5199717-2 |

**Figure 2-3 System Interconnect Cables**

| VCT | | | |
|---|---|---|---|
| | | Part Number | |
| Run No. | Description | Long Cables (Kit 2281840-6) | Short Cables (Kit 2281840-7) |
| 1 | Facility MDP to Room Disconnect (A1) | cust. supplied | cust. supplied |
| 2 | Room Disconnect (A1) to PDU | cust. supplied | cust. supplied |
| 3 | Room Disconnect (A1) to System E-Off | cust. supplied | cust. supplied |
| 4 | PDU to Room Warning Light(s) | cust. supplied | cust. supplied |
| 5 | PDU to Scan Room Door Switch | cust. supplied | cust. supplied |
| 50 | HVDC Power Cable - PDU to Gantry | 2343529 | 2343529-2 |
| 51 | HVAC Power Cable - PDU to Gantry | 2343530 | 2343530-2 |
| 52 | LVAC Power Cable - PDU to Gantry | 2343528 | 2343528-2 |
| 53 | LVAC Power Cable - PDU To Operator’s Console | 5121809 | 5121809-2 |
| 54 | LVAC Power Cable - Gantry to Table | n/a | n/a |
| 55 | Ground, PDU to Raceway | 2371450 | 2371450-2 |
| 56 | Ground, Raceway to Console | 2371450-3 | 2371450-4 |
| 60 | LVAC Power Cable - PDU to Optional UPS | - | - |
| 61 | LVAC Power Cable - UPS Disconnect Panel to PDU | - | - |
| 90 | LVAC Power Cable - PDU to PET | - | - |
| 100 | Signal Cable - Gantry to PDU | 5120646 | 5120646-2 |
| 101 | Signal Cable - Gantry to Console | 5120645 | 5120645-2 |
| 102 | Signal Cable (Ethernet) - Gantry to Console | 2373436 | 2373436-3 |
| 103 | Data Cable (Fiber Optic) - Gantry to Console | 5125259 | 5125259-2 |
| 104 | Signal Cable - Gantry to Table | n/a | n/a |
| 110 | Signal Cable - UPS Control to Room Disconnect (A1) | - | - |
| 111 | Signal Cable - UPS Control to UPS Disconnect Panel | - | - |
| 112 | Cardiac Monitor Cat 5 - Signal cable between console and Gantry Option Board (GOB) (Quantity 2 | 5193967-4 | 5193967-6 |

**Figure 2-4 System Interconnect Cables**

**NOTICE** **Shortening power cables is not allowed unless the appropriate crimping tool and ferrules are used. The crimping tool and ferrules are not shipped with the system.**

**If longer or shorter cables are required, order the correct set.**

**Excess cables cannot be stored under or behind the PDU, gantry or console.**

# Section 2.0 System Interconnect Diagram

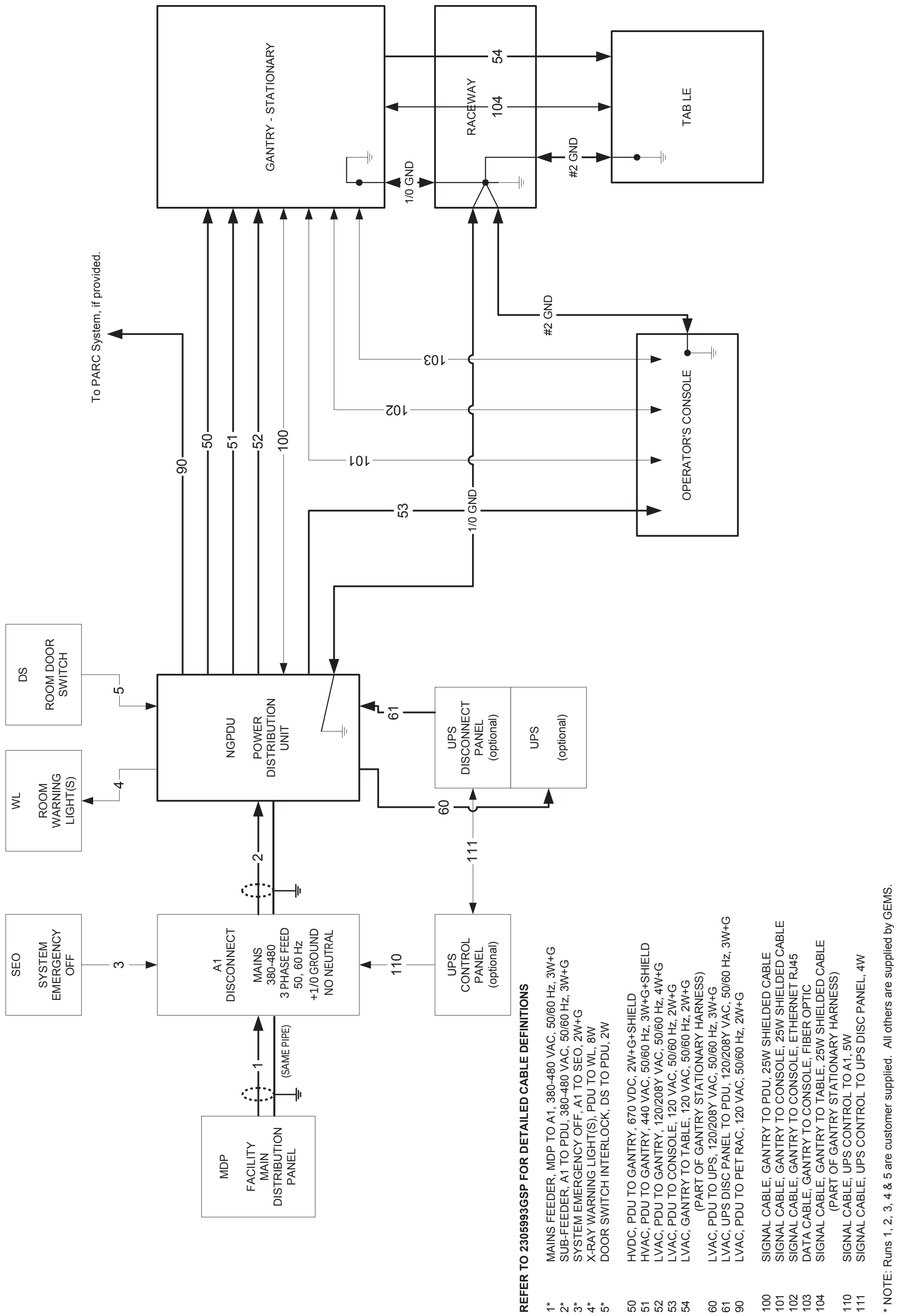



**Figure 2-5 System Interconnect Diagram**

# Section 3.0 Console Connections

## 3.0.1 Monitor and User Tower Placement

1.) Locate and unpack the two monitors from the lean cart (see Figure 2-6).
2.) Carefully place the monitors on the console desktop.
3.) Locate and unpack the media drive tower.
4.) Place the media tower on the console desktop.

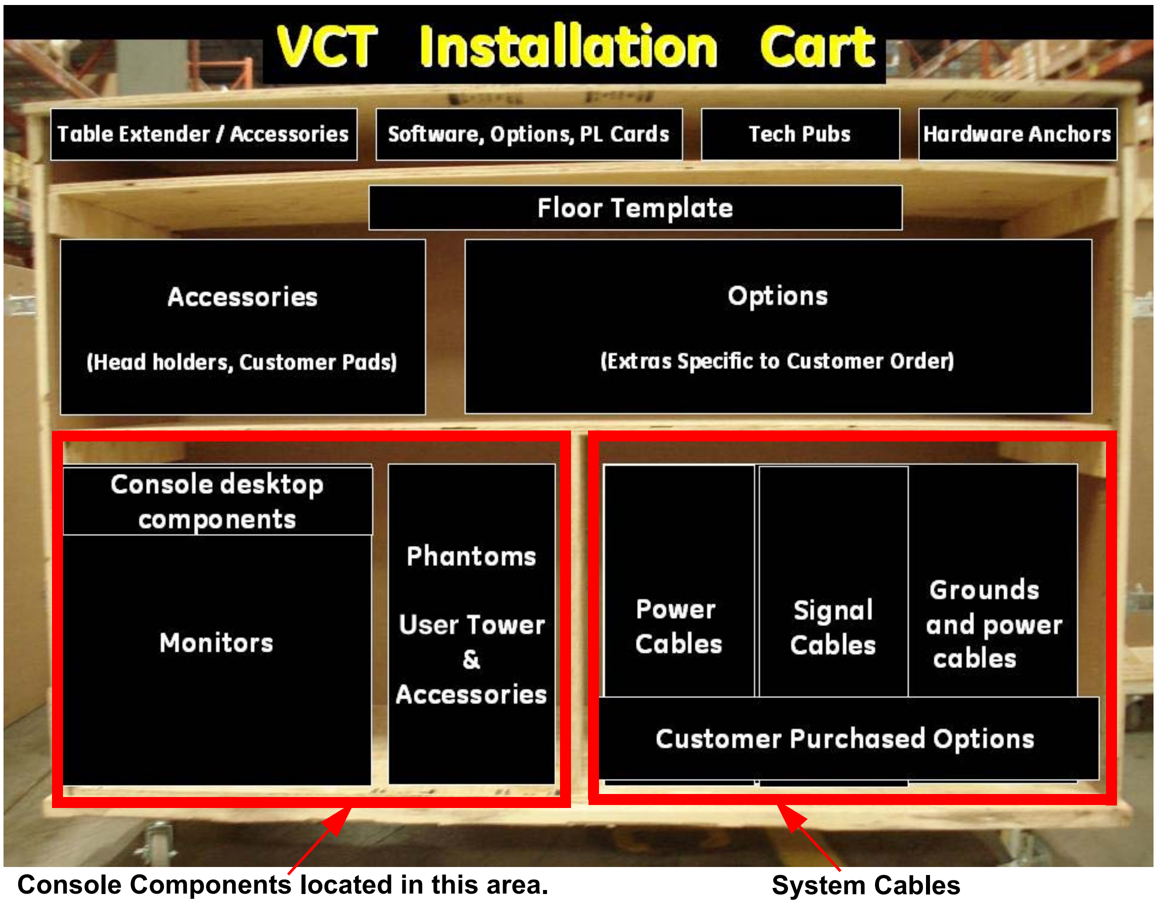



**Figure 2-6 Lean Cart Layout**

**Figure 2-7 Global Console tabletop component placement** (actual components may vary in color and configuration)

Note: Even if the components look different, the connections are the same. (For connection instructions, see Chapter 2, Section 4.)

## 3.1 SCIM, Keyboard, Trackball and Mouse Installation (On Lean Cart in the desktop component tray)

1.) Route the keyboard cable under the SCIM, as shown in Figure 2-8.

**Figure 2-8 SCIM control with keyboard cable routed through SCIM** (actual components may vary in color and style)

**NOTICE** **Potential for equipment damage**

**Never connect a mouse or keyboard with the host computer powered ON. Doing so can destroy components within the host computer.**

2.) The SCIM, keyboard, trackball, mouse, and barcode reader all have USB connectors that connect in the rear bulkhead on the console as shown in Figure 2-9.

3.) The SCIM, keyboard, mouse, and trackball colors may be different than those shown in this direction. Their installation, connection, and operation is the same.

4.) The Media Tower connects using three USB cables in place of the SCSI cable. The optional MOD drive is connected with a USB cable, and can be placed on top of the media tower for easy access.

5.) Power cords for all desktop components connect directly to the left rear power bulkhead. All power cables are shipped with the console. Desktop components cables cannot be extended or replaced.

Note: Return all shipping boxes and packing materials, including unused monitor cables. Place this material back on the lean cart. All installation packaging should be placed on the cart and returned with the dollies.

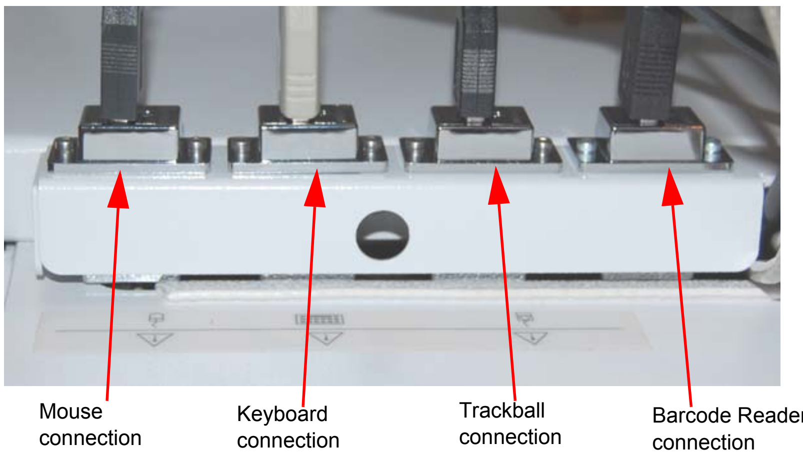



**Figure 2-9 Global Console Rear Bulkhead, showing mouse, keyboard, barcode reader and trackball connections**

6.) Connect the SCIM cable to the SCIM as shown in Figure 2-10. (Note the cable routing.)

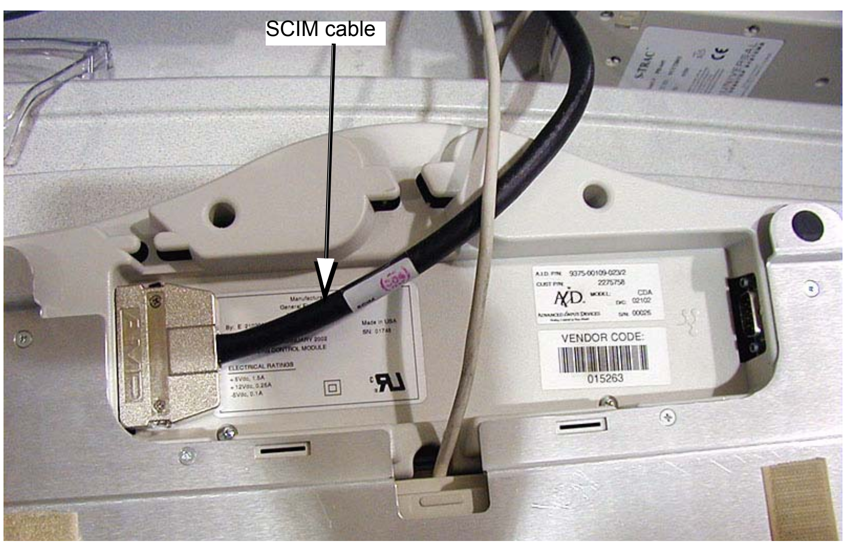



**Figure 2-10 SCIM bottom, showing cables and keyboard mounting bracket**

7.) Make sure the SCIM connector fits snugly. Some plastic molding may need to be removed to allow the cable to fit properly.

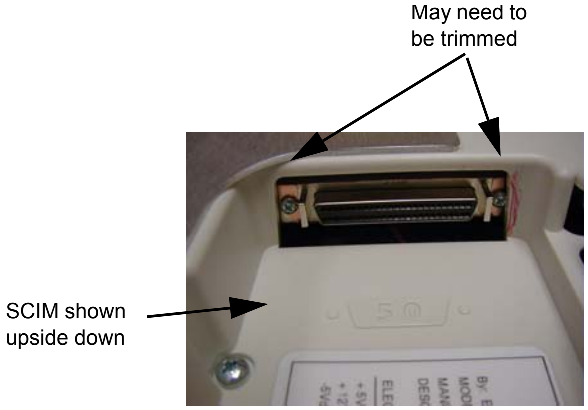



**Figure 2-11 SCIM Connector Area**

8.) Select and install the proper keyboard overlay for your system: (1) with tilt or (2) without tilt.

Verify that none of the buttons get caught and stuck under the overlay. Pay close attention to the prescribed tilt button on systems with the tilt feature.

9.) The keyboard should attach to the SCIM using the supplied Velcro strip and fit snugly against the SCIM when finished, as shown in Figure 2-8 and Figure 2-12.

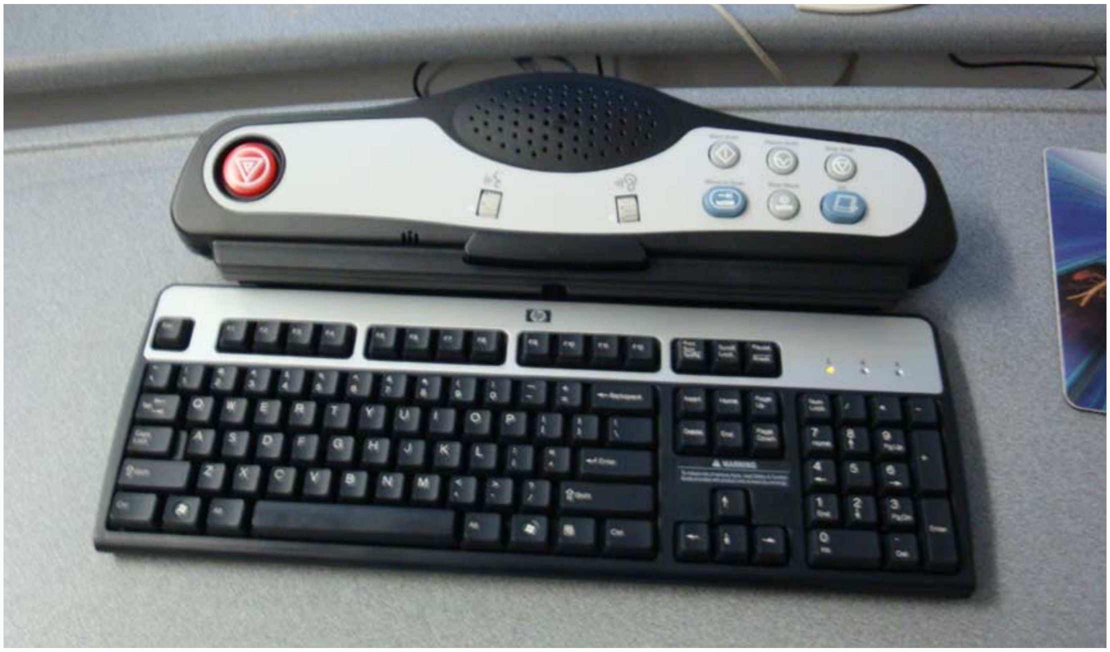
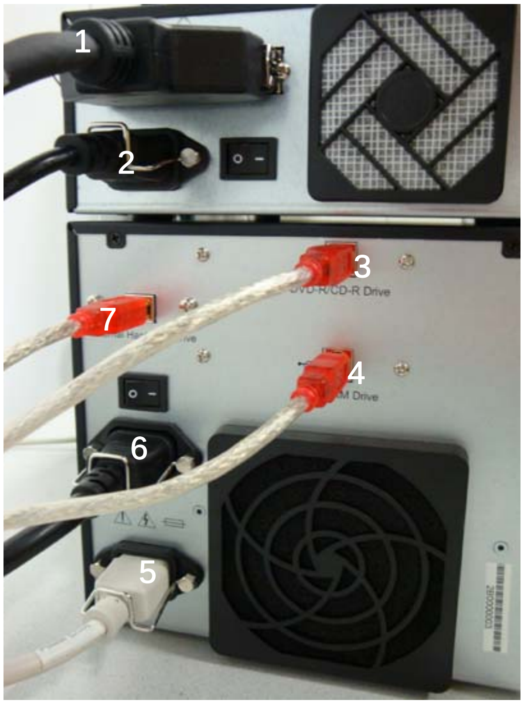


**Figure 2-12 SCIM connected to the keyboard with the US English tilt overlay installed** (actual components may vary in color and style)

## 3.2 Connecting the Media Tower (On Lean Cart in the User Tower & Accessories)




1. SCSI cable Mod Drive to console.
2. Power into the MOD Drive
3. DVD-R/CD-R Drive
4. DVD-RAM Drive
5. Power into the Media Tower
6. Power out to the MOD Drive
7. External Hard Disk Drive

**Figure 2-13 Media Tower Connections shown with optional MOD Drive**

Check the box when each step is completed:

1.) ❒ Connect the three USB cables to the rear of the media tower. Each USB Cable is labeled, plug the labeled cable end into the correct connector.

Note: The cables are included with the console.

2.) ❒ Connect the power cable to the rear of the media tower.

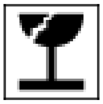
## 3.3 Optional MOD Drive

1.) The power and SCSI cables that are supplied with the option.
2.) Mount MOD drive on top of the media tower.
3.) Connect the short power cable from the media tower to the MOD drive.
4.) Remove the GOC6 rear cover to gain access to the HP.
5.) Locate Channel 1 on the HP.
6.) Connect the SCSI cable to channel 1 and route the cable so that is comes out of the top of the console.
7.) Reinstall the console rear cover and connect the SCSI to the rear of the mod drive.

## 3.4 Connecting the LCD Monitor

With the monitor and computer switched off, connect the video signal cable to the monitor's video input.

**NOTICE** **Equipment damage possible**



**Do not touch the video signal cable connector pins as this might bend them. When connecting the video signal cable, check the alignment of the HD15 connector. Do not force the connector in the wrong way, otherwise the pins might bend.**

Connect the following:

**SCIM:**

- SCSI cable from the console
- Template overlay (Figure 2-12**)**
- Keyboard (see Figure 2-12)
- Route cable under the desk top

**Scan Monitor:**

- Video cable from console
- Power cable J2
- Route through the cable keeper

**Display Monitor:**

- Video cable from console
- Power cable J3
- Route through the cable keeper

**Figure 2-14 Connecting the LCD Monitor**

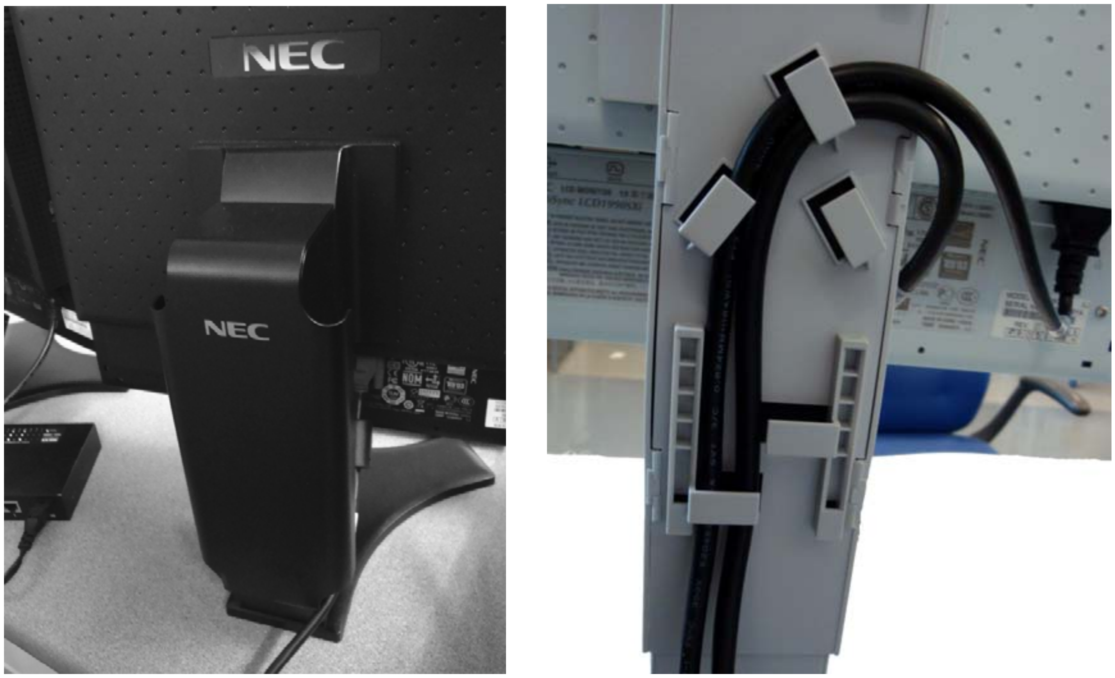



**Figure 2-15 Cable Routing and Keeper**

## 3.5 Power Panel Connections

NOTICE **Console power is two-phase power. Only used assigned outlets.**

1.) Connect the console power cable to the console power panel.
2.) Connect console component power cords as listed in Table 2-3. Plug LAN cables into the rear bulkhead on the console as shown in Figure 2-17.

| HD M3 LAN Cables | Signal Cables | Power Cables |
|---|---|---|
| • J26 Hospital Network | • J20 Cable (Gantry) #101 | • J1 208 VAC Power, Run #53 |
| • J27 TGPU LAN, #102 | • J25 Dual Fiber DIP, #102 | • RUN #056 #2 Ground |
| • J30 HD Cardiac (EKG) | | |

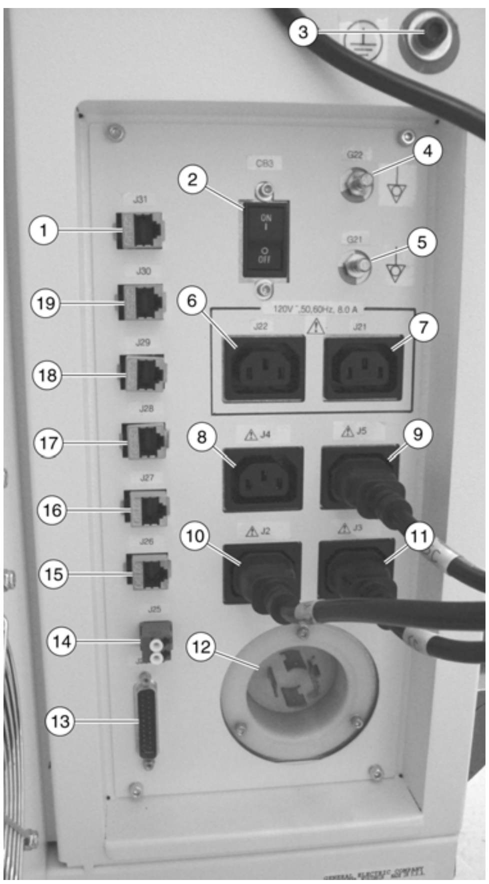
**Table 2-3 Console Component Connections**

Note: The DIP signal cable is keyed to fit only one way.

The console is not shipped with all external power cables connected to their assigned power slots. Connect the monitor and media tower power cables as shown in Figure 2-17.

## 3.6 Console Connections

Plug cables into rear bulkhead on console.



1. J31 RPM (not used)
2. CB3 ON/OFF
3. RUN #056 #2 Ground
4. G22 (not used)
5. G21 (not used)
6. J22 Injector - (Power)
7. J21 RPM - varian power strip
8. J4 (not used)
9. J5 Media tower
10. J2 Scan Monitor
11. J3 Display Monitor
12. J1 RUN # 053 Console Power
13. J20 RUN #101 Scan INTL (Interlock)
14. J25 RUN # 103 DAS Data (Fiber)
15. J26 HSP (not used)
16. J27 TGP RUN # 102 LAN (RJ45)
17. J28 UPS (not used)
18. J29 AW (not used)
19. J30 EKG Cat 5 J45

**Figure 2-16 Console Rear Bulkhead**

Note: Make sure the J26 HSP is connected correctly.

3.) System cable connections are located on the rear of the console. Connect as shown in Figure 2-17 below.

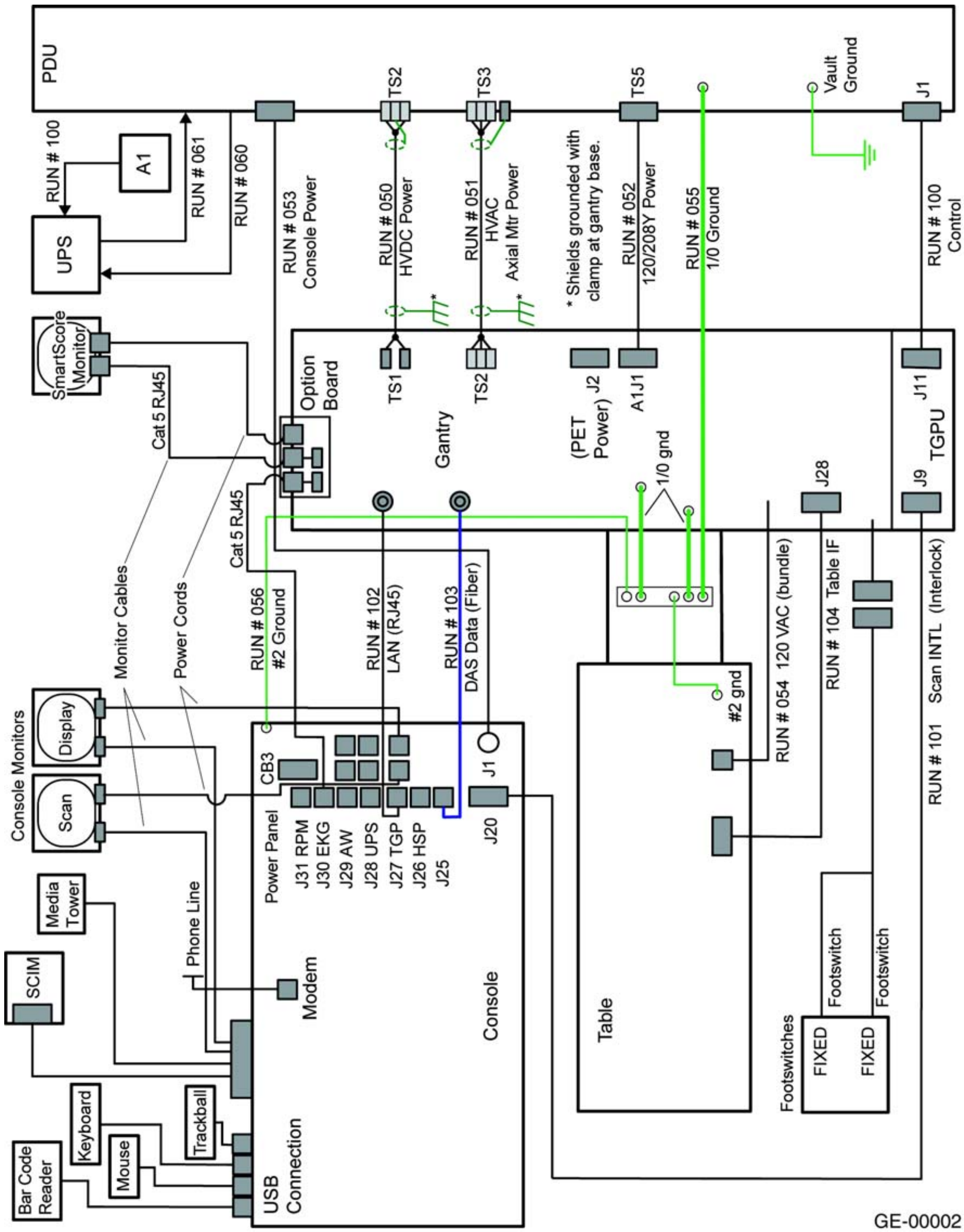
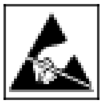



**Figure 2-17 Console Power Connections**

## 3.7 Install Drives

### 3.7.1 Tools and Test Equipment

Active static mat and ground wrist strap



**NOTICE** **ESD can damage hard drives. Use proper ESD precautions when handling hard drives.**

Install drives in the 16 slots shown in Figure 2-19 as follows:

1.) Press the blue button on the front to open the drive front cover.
2.) The drives are shipped on the lean cart in four boxes with four drives in each box. Locate the boxes and place them near the console.
3.) Some drives bays have a filler plug installed. Do not remove filler plugs or install drives in these four slots.

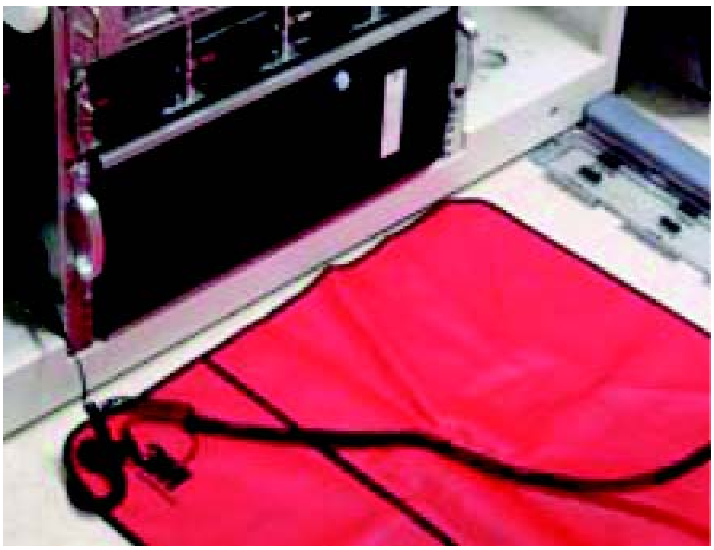


**Figure 2-18 ESD Mat**

4.) With the ESD mat connected and the wrist strap on, remove each drive from the cardboard shipping box. Remove each drive from the static bag, and install one drive in each of the eight slots on the top drive bay.
5.) The drives are properly installed when they click into place. Push the drive only firmly enough to seat the drives. If the drive fails to click into place, remove the drive by pushing the blue button to raise the door, and then restart the installation process. Close the door to secure the drives after installation.
6.) On each side of the drive rack are items blocking access. Slide the latch to release and open the handle or display to gain access to the drive bay.
7.) **FOR VCT**: Install the 8 drives in the lower drive rack.

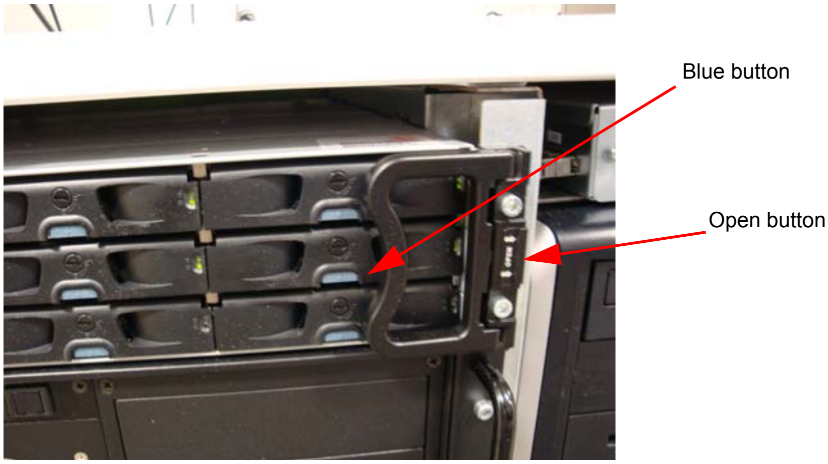



**Figure 2-19 Hard Drive Installation**

**Figure 2-20 Hard Drive Installation (VCT) (Shown with optional components)**

VCT standard configuration will have only one rack of hard drives and one VDIG. VCT performance configuration will have only one rack of hard drives and two VDIGs.

8.) Return the handle to the normal position when the drives are installed.

9.) **CT 750 HD Systems only:** Locate the second drive rack near the top of the console.

### 3.7.2 CT 750 HD Only

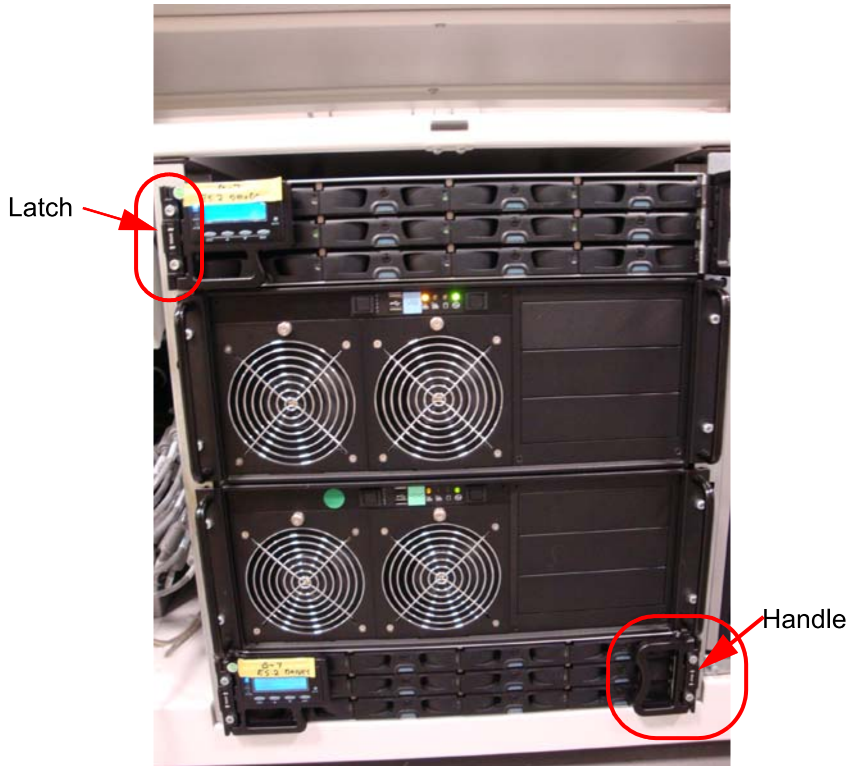



**Figure 2-21 Dual Drives (CT 750 HD only)**

1.) Repeat steps 1–7.

Replace all static bags into the boxes and return the boxes to the lean cart.

## 3.8 USB Barcode Reader Option

1.) Locate the barcode reader box on the lean cart options section.
2.) Consoles with 4 USB ports
3.) Connect the USB cable to port USB A on the back of the console
4.) Dress any excess cable and place under the monitor desktop.

### 3.8.1 Consoles with 3 USB Ports

1.) Remove the console rear cover 4 attachment screws and remove cover
2.) Locate the Host HP on the left side looking from the rear of the console.
3.) Locate and connect the barcode reader USB connector into port 4 on the host.
4.) Route the cable through the cable access opening on the rear of the console and place the barcode reader on the operator desktop.
5.) Dress any excess cable and place under the monitor desktop.
6.) Carefully reinstall the console rear cover. Check that no cables are pinched or disconnected.

# Section 4.0 Install Options

Note: Most shipped options can be located on the lean cart. Only large options such as the UPS and Smart Step will arrive on its own skid.

## 4.1 Install Optional Remote Monitor (on Lean Cart)

Follow the installation procedures in the Remote Monitor box.

## 4.2 Install Respiratory Gating Option (on Lean Cart)

Follow the instructions shipped with the option. The options board should be only mounted on the non-motor side of the gantry. Neatly dress all cables along the gantry base so that the base covers fit properly.

## 4.3 Install Cardiac Gating IVY Monitor and Stand Option IVY 3150B (HD)

Follow the instructions in this manual. This option is attached to the Gantry Option Interface, and should only be mounted on the non-motor side of the gantry. Neatly dress all cables along the gantry base so that the base covers fit properly.

**HD:** The option is connected to the IPC board with cable 114. The unit plugs into the right side option panel. Power is from the gantry. Refer to the option manual shipped with this option.

## 4.4 Install Injector Option (on Lean Cart) VCT /HD

Follow the instructions shipped with the option. If this is a ceiling-mounted option, check that the plate is installed correctly with the holes in the correct location.

Note: VCT 64 injector power will plug into the wall. HD injector power will plug into the console.

## 4.5 Install IVY Monitor and Stand IVY A (on Lean Cart) VCT

Follow the instructions shipped with the Monitor and stand kit. Review the instructions carefully before assembling the stand and accessory basket to avoid repeated steps. Connect to the option interface panel, See Chapter 2 Section 5.1.

## 4.6 Customer Accessories (Head Holders and Extender) (on Lean Cart)

Open the boxes and installed the appropriate language warning labels.

The head holders are shipped with shims that require installation to ensure proper fit. Check that shims are included. Follow the shim procedure in Chapter 3, Section 3. The holder should fit snugly.

## 4.7 UPS Installation (on skid)

If the site has an Uninterruptible Power Supply (UPS), please refer to *UPS Installation for Direction 5272751 for Powerware 9355 UPS*. This manual should be shipped with the UPS. Use caution when removing the UPS from the skid. The UPS weight is 620 lb. (281 kg).

## 4.8 Flat Tabletop used with Respiratory Gating

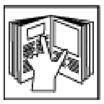
Follow the installation instructions with the Option.

# Section 5.0 Gantry Cable Connections

Note: All system interconnect cables are located on the lean cart on the bottom right corner.

1.) Rear cable entry is standard. Install the rear cable entry box (B7850RC) now, before routing cables to gantry.

**NOTICE**



**CABLE CONNECTION NOTICE: Potential for equipment damage.**

- **Observe correct polarity when connecting the high voltage DC power. Reversing these leads will result in serious equipment damage.**
- **The HVDC positive conductors have red insulation and are labeled ONE. The HVDC negative conductors have black insulation and are labeled TWO. Lead ONE must be connected to lead ONE, and lead TWO must be connected to lead TWO.**
- **Observe correct phase rotation when connecting the axial motor power. Phases one, two and three should be connected top to bottom.**

2.) Install the cables to the gantry power pan. The power pan is located on the rear of the gantry at its base. See Figure 2-23, and Figure 2-32 for connections.

Note: The gantry 120VAC cable may not fit under the gantry frame. Install this cable before gantry placement—or remove the power plug—to route it under the gantry. In doing so, be careful not to damage the cable or power plug.

Leave approximately 3 m (10 ft.) of the dual channel fiber and gantry Cat 5 cables under the gantry frame for troubleshooting (see Figure 2-22)

.

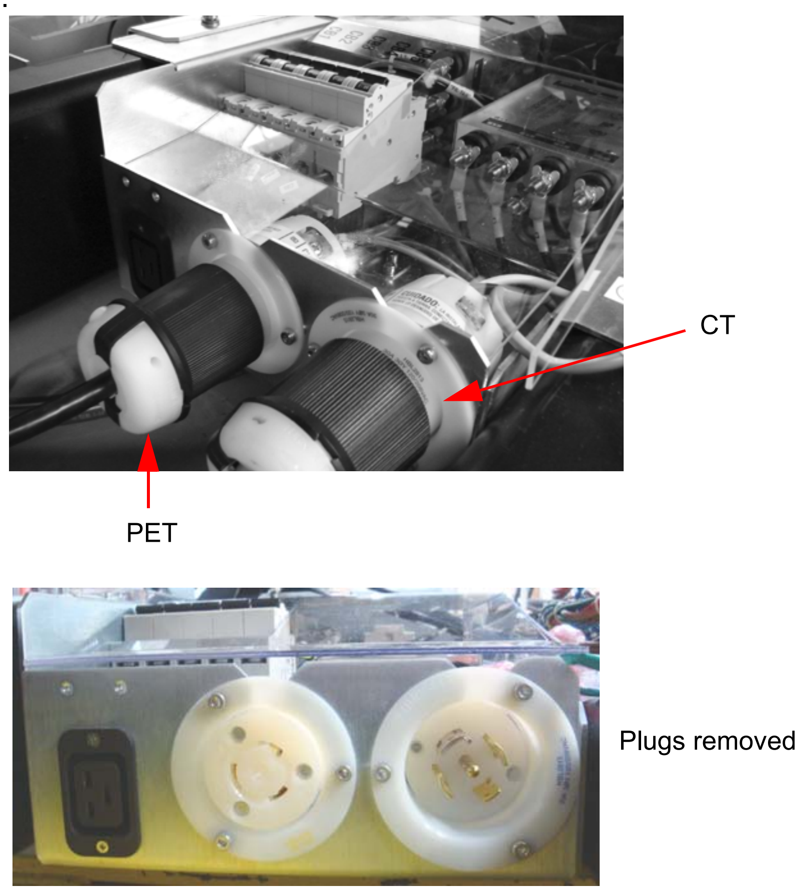



**Figure 2-22 Gantry Power Pan**

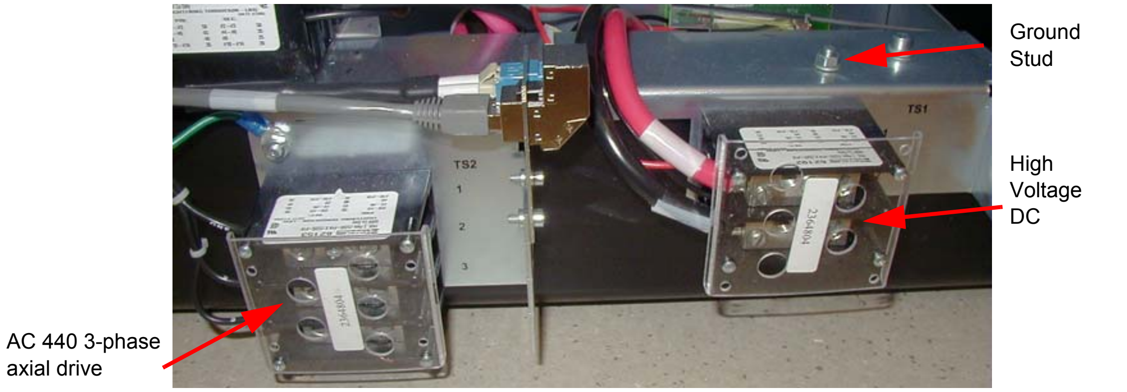



**Figure 2-23 Gantry Power Pan Connections**

3.) From the console:

a.) Install the fiber run #102

b.) Install the console LAN #102

Both cables should click into place when they are connected (see Figure 2-24).

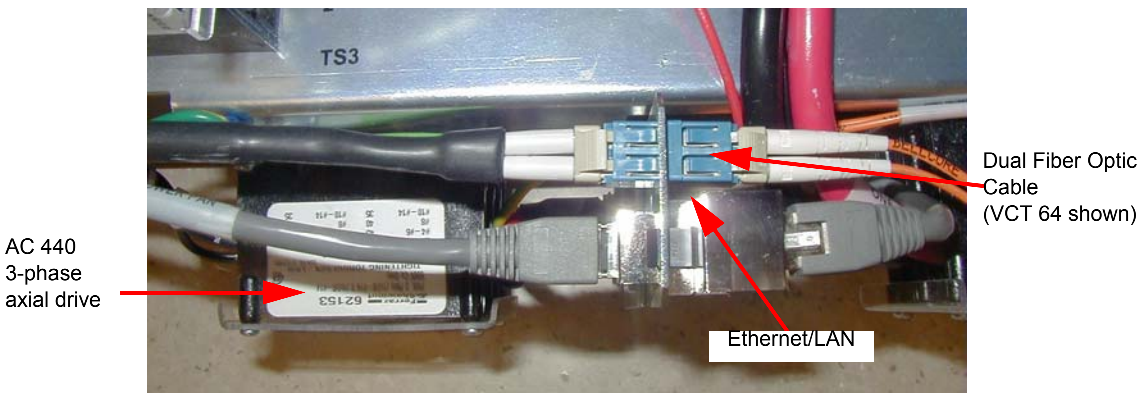



**Figure 2-24 Close-up of Power Pan Bulkhead**

Note: 1.) The fiber cable and the LAN cable must click when locked in place.

2.) HD fiber cable is slightly thicker, but is still fragile! Use care when routing the fiber cable.

## 5.1 Gantry Option Board CT750 HD Only

1.) Separate the shipping bracket from the gantry option panel. Remove the shipping bracket (bright-colored bracket) (see Figure 2-25).

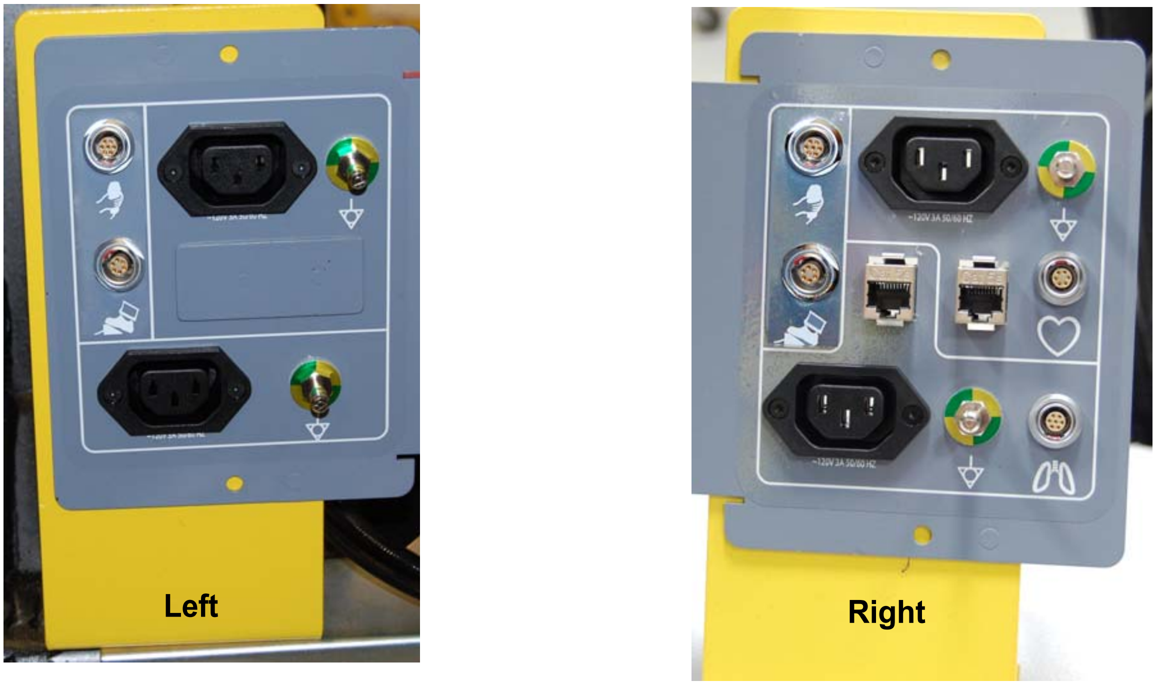



**Figure 2-25 Shipping Brackets (Shown Left and Right side brackets.)**

2.) Install brackets using the same hardware used for shipping brackets.

3.) Locate the option mounting brackets (2) on the Lean cart. Install the options panel to the brackets.

4.) Install the bracket to the gantry.

5.) As needed, make any electrical connections between the IPC and options panel. (see Figure 2-27).

**Figure 2-26 Gantry Option Interface Panel**

6.) Connect all LAN cables and interface cables to the option interface board and the network switch mounted on the gantry. Option interface plates are mounted on the rear of the gantry on both sides. All options are powered from the console or gantry options power panel located on each side of the back of the gantry.

## 5.2 CT750 HD IPC Cable Connections

1.) Pull all cables shipped in the cable collector. (Run 113 and 114) LAN/Cat 5 and option interface cable. Connect to the cables HD 15 (cable type)

.

**Figure 2-27 Option Panel Connection**

2.) Make all necessary electrical connections between the IPC and the interface panel.

## 5.3 CT750 HD Cardiac Monitor Setup (IVY B)

1.) Follow install instructions shipped with monitor to set up the monitor stand with basket.
2.) After the stand is assembled, mount the monitor to the stand by sliding the monitor onto the plate.
3.) Pull down the front pin on plate and slide monitor until it snaps into place.
4.) Secure the monitor using the two nylon set screws under the plate.
5.) Attach the cables. *Do not use the cables shipped with the monitor,* find the 5317480 cable included with the Cardiac Monitor Option Kit (E8007RP).
6.) Connect the IEC power cord, ground wire, LEMO and CAT5 to the gantry option interface panel. (See Figure 2-28).

**Figure 2-28 -Gantry Option Interface**

If the wire power cord cable clamp is too short, replace it with the supplied longer cable.

**Figure 2-29 Connections on Rear of Cardiac Monitor**

7.) Connect the power cord, ground wire, HD15 and CAT5 to the monitor panel.(See Figure 2-29).
8.) The cardiac monitor receives power from the gantry.
9.) Strain relief the cables to the monitor stand, and to the gantry base covers using tywraps. (See Figure 2-30).

**Figure 2-30 Cables Strain Relieved to Stand**

## 5.4 VCT GOB and Option Panel Installation

1.) Run the CAT5 LAN cable for the cardiac option between the console and the gantry.

2.) Install the option interface board. The option interface plates are mounted to the gantry on each side.

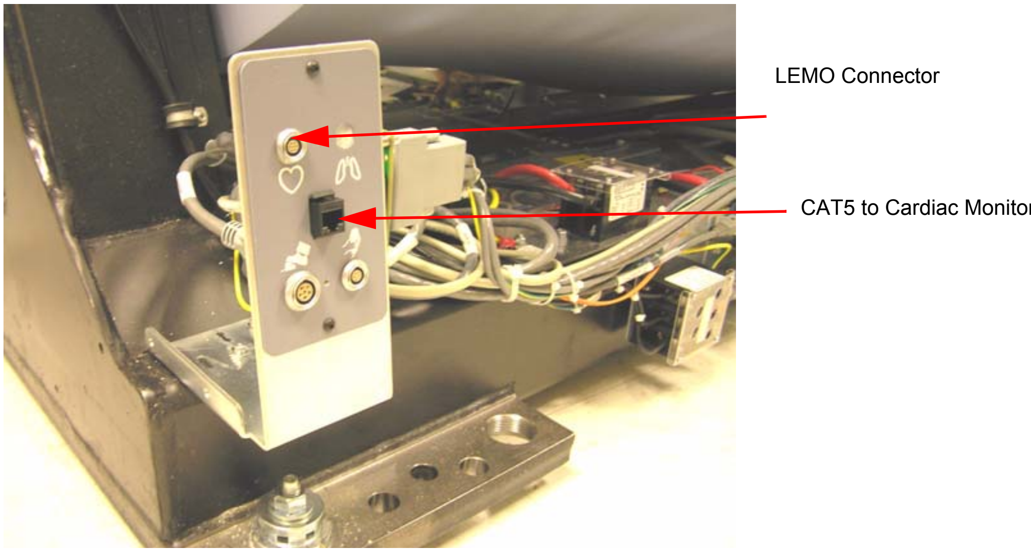



**Figure 2-31 Cardiac Option Board**

Note: Connector is not attached to the plate. Use care when making this connection.

3.) Connect this cable to the gantry base cardiac monitor base plate (GOB board). Different plates are included with the cardiac option kit. Follow the installation procedures in the Cardiac Option kit to install the GOB and cables. (see Figure 2-32).

**NOTICE** **The GOB connectors are not soldered to the board. Do not use the cable to pull on the board.**

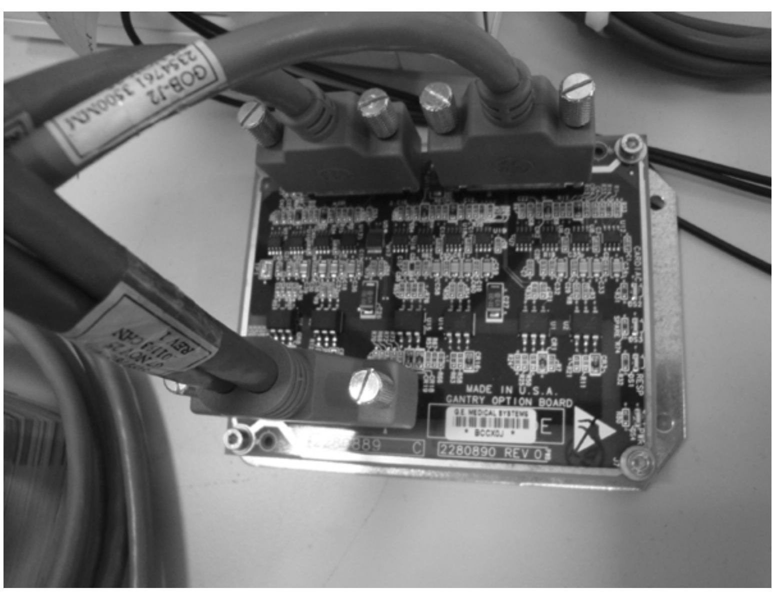



**Figure 2-32 Gantry Option Board**

## 5.5 VCT Cardiac Monitor Setup and Installation (IVY Monitors)

1.) Follow install instructions shipped with monitor to set up the monitor and the stand with basket.
2.) Connect the LEMO and CAT5 to the Gantry option board. (See Figure 2-31.)
3.) Connect the HD 15 (LEMO) to the back of the monitor and the CAT 5 to the back of the monitor.
4.) Power the Cardiac monitor from a nearby wall outlet.

## 5.6 TGPU Connections

Install cables to the gantry TGPU. These two cables can be found on the lean cart in the cable collector boxes. Run 100 gantry TGPU J11 to PDU; run 101 gantry TGPU J9 to console.

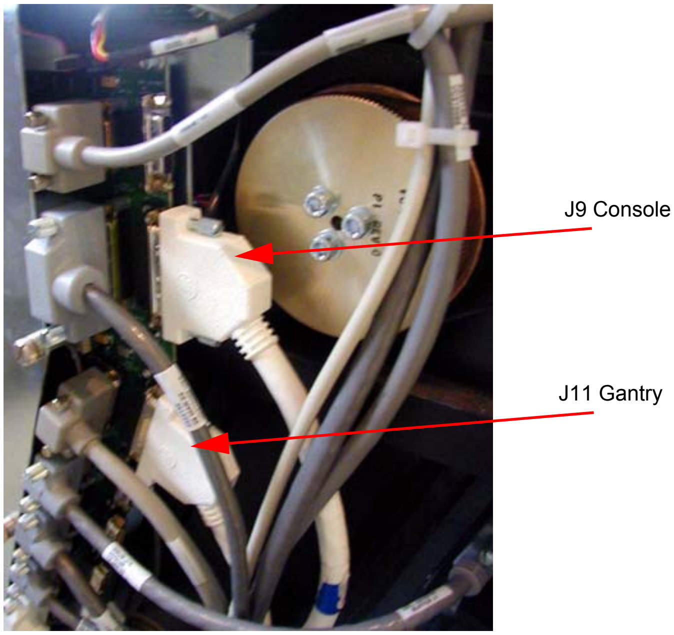



**Figure 2-33 TGPU Connections.**

# Section 6.0 Table Connections

Pull and connect the cables as described in Table 2-4. The table cables are bundled with the gantry frame. Cut the cable ties to release bundles of cables.

Note: The footswitch connector and wiring harness may be run and secured to the ground bar assembly.

| Table | From | CABLE DESCRIPTION |
|---|---|---|
| J1 table power | Gantry | 120 VAC |
| J9 table control | Gantry | Signal Cable Run #104 |
| Table ground | Gantry | Table ground |

**Table 2-4 Table Cable Connections**

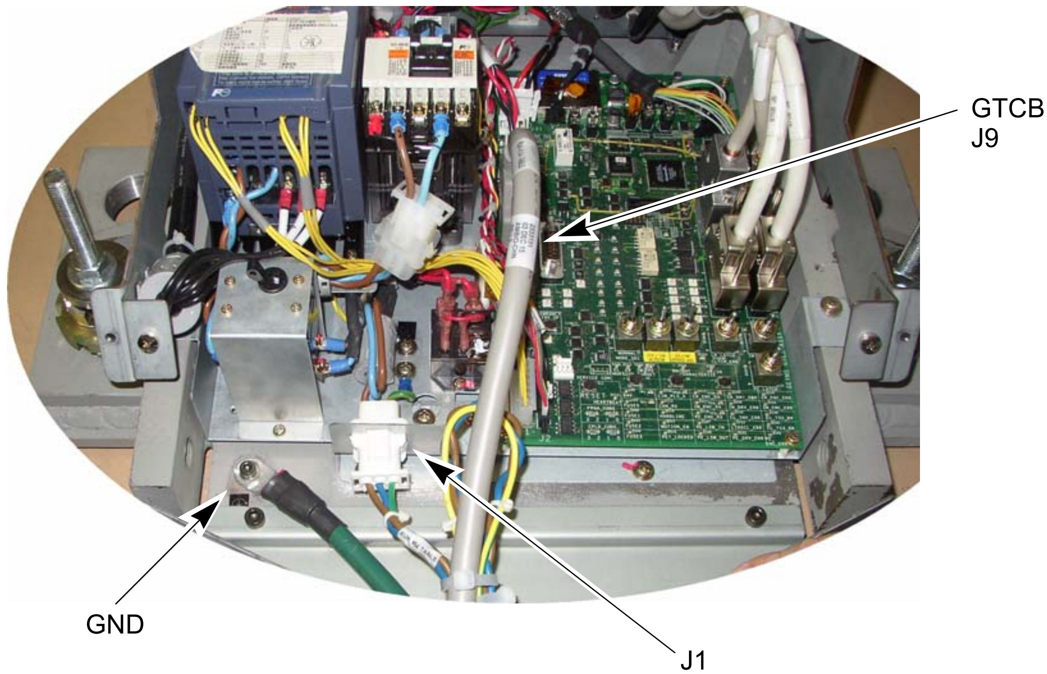



**Figure 2-34 Table Connections**

Note: You need to add the table ground cable, as shown.

☐ Check box when complete.

# Section 7.0 PDU Cable Connections & Configuration

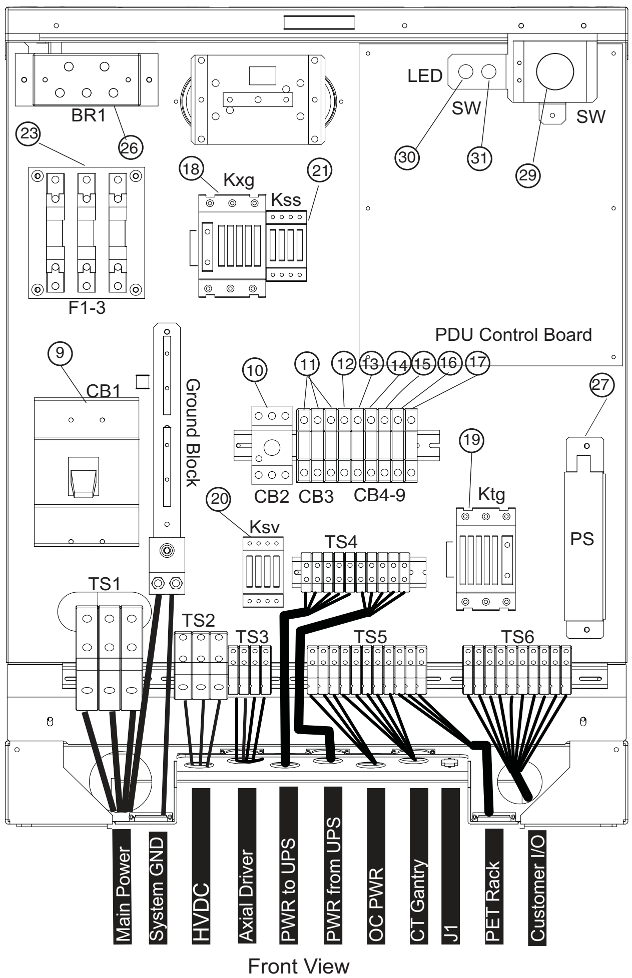
**CAUTION**

**Do not work in an energized PDU. When working on the PDU, follow this simple rule: Always tag and lock out power to the PDU at the main disconnect. Failure to due so can result in electrocution or death.**

**Do not apply power to the PDU until all work has been completed and all PDU covers are in their proper place.**

## 7.1 Introduction to NGPDU

As seen in Figure 2-35, a number of cables must be installed throughout the PDU. Specific details on each connection can be found in the sub-sections that follow. Use Figure 2-35 for reference. The PDU has been designed to have cables routed into the PDU from behind and/or beneath it.




Note: Wire colors may vary

| | | |
|---|---|---|
| HVDC | TS2-1 | Red |
| | TS2-2 | Grn/Yellow |
| | TS2-3 | Black |
| Axial | TS3-1 | Black |
| | TS3-2 | Red |
| | TS3-3 | Orange |
| | TS3-4 | Grn/Yellow |
| Console Power | TS5-0 | Black |
| | TS5-1 | Brown |
| | TS5-2 | Blue |
| | TS5-3 | Grn/Yellow |
| GT Gantry | TS5-4 | Black |
| | TS5-5 | Red |
| | TS5-6 | Orange |
| | TS5-7 | Blue |
| | TS5-8 | Grn/Yellow |

**Figure 2-35 PDU Cable Connections - Front**

## 7.2 Panel - 380 - 480VAC Mains “TS1” Input Power Connection completed by the customer electrician.

**WARNING- A LICENSED ELECTRICIAN SHOULD CONNECT PDU MAIN POWER TO THE PDU. THE MAIN POWER WIRES CONNECTION TO THE PDU SHOULD ALWAYS BE IN CONDUIT. GE HEALTHCARE PERSONNEL SHOULD NOT MAKE THIS CONNECTION.**

1.) If present, remove the TS1 panel front cover.
2.) Strip the wires to fit securely on the power block.
3.) Observe incoming phases (L1, L2, and L3) and insert bare leads into power block.
4.) Insert vault ground into PDU vault ground lug.
5.) Tighten all fasteners securely and replace the TS3 front panel.

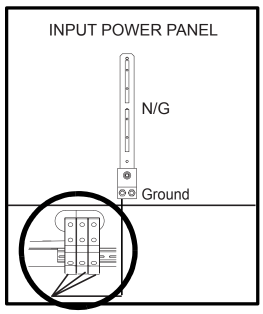



**Figure 2-36 Input Power Panel Connections**

When mains power is available to the PDU, the TS1 power light illuminates.(See Figure 2-35.)

☐ Check box when complete.

## 7.3 Panel - Circuit Breakers

Place the circuit breakers in the off/down position during installation, even with mains incoming power tagged and locked out. After you have completed work on the PDU, you may return the circuit breakers to the ON positions.

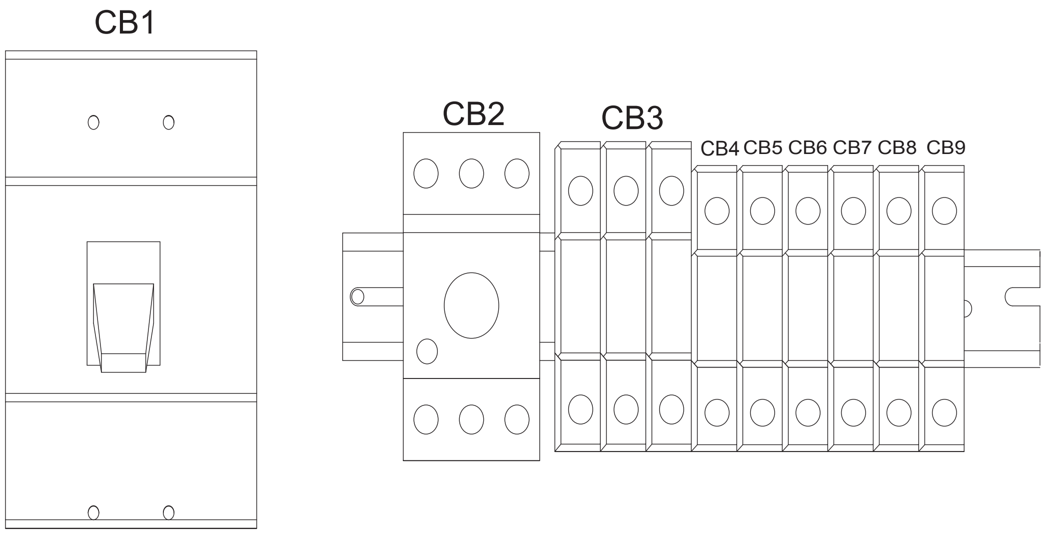



**Figure 2-37 Circuit Breaker Panel**

By design, when CB3 is in the OFF position, circuit breakers 4, 5, 6, and 7 are switched OFF. CB3 is essentially in series with these breakers.

| Circuit Breaker | Description |
|---|---|
| CB3 | Fully Winding Protection (Master power of CB 4, 5, 6, and 7) |
| CB4 | CT Gantry Service Outlets |
| CB5 | CT Gantry rotating loads |
| CB6 | Table & CT Gantry Station Loads |
| CB7 | Operator Console |
| CB8 | PET Gantry |
| CB9 | NGPDU Control Power Supply |

**Table 2-5 Panel Circuit Breaker Descriptions**

☐ Check box when complete.

# 7.4 HVDC Connection

Note: Refer to Table 2-4, System Interconnect Cables, on page 101

Connect the internally shielded HVDC cable to TS2 on the standing panel (See Figure 2-35 for the location of the connector and Figure 2-38 for details). Observe polarities and grounds. Do not cut or shorten cables unless you have all of the appropriate tools and crimper to re-terminate. If short cables are needed, have the PMI order the short cable set.

**WARNING** Excess cable length cannot be stored under or behind the PDU. If cables are to be stored in the cable tray, do not overfill. Consult the local electrician to determine the maximum fill rate for your area.

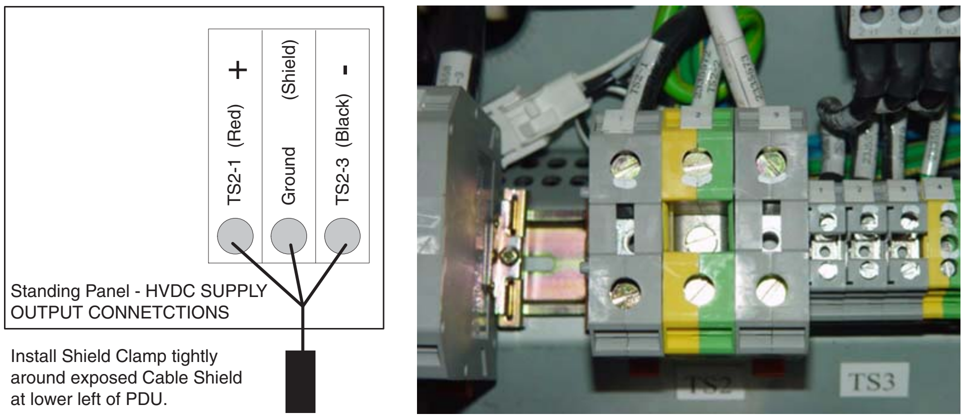



**Figure 2-38 HVDC Connection**

☐ Check box when complete.

# 7.5 440V Connection

Note: Refer to Table 2-4, System Interconnect Cables, on page 101

Connect the internally shielded HVDC cable to TS2 on the standing panel (See Figure 2-35 for the location of the connector and Figure 2-38 for details). Observe polarities and grounds. Do not cut or shorten cables unless you have all of the appropriate tools and crimper to re-terminate. If short cables are needed, have the PMI order the short cable set.

**WARNING** Excess cable length cannot be stored under or behind the PDU. If cables are to be stored in the cable tray, do not overfill. Consult the local electrician to determine the maximum fill rate for your area.

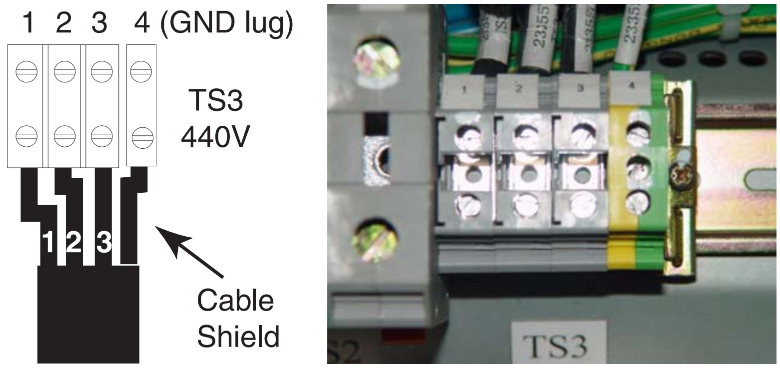



**Figure 2-39 440VAC Connection**

☐ Check box when complete.

## 7.6 Gantry & Console Power Connections (HD/GOC6)

Note: Refer to Table 2-4, System Interconnect Cables, on page 101.

Do not cut or shorten cables unless you have all of the appropriate tools and crimper to re-terminate. If short cables are needed, have the PMI order the short cable set.

**WARNING** Excess cable length cannot be stored under or behind the PDU. If cables are to be stored in the cable tray, do not overfill. Consult the local electrician to determine the maximum fill rate for your area.

Plug the gantry power cable wires into J4 and the Console power cable wires into J5 as shown in Figure 2-40.

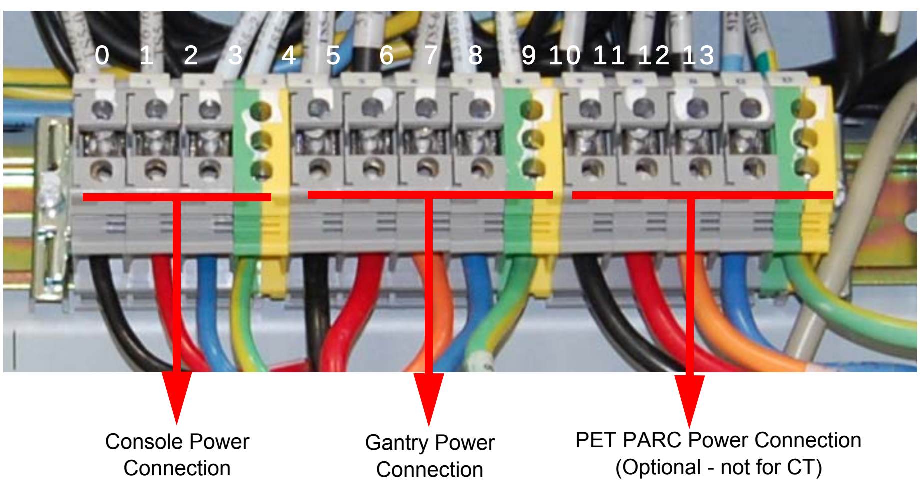



**Figure 2-40 Gantry & Console Power Connections**

☐ Check box when complete.

## 7.7 Console Power Cable Plug Removal

Power cable re-termination is not allowed. Short and long cables are available.

Carefully remove the power plug and **record the color of the wires** in Table 2-6. The terminals are labelled X, Y, W, and G on the plug.

| Terminal | X | Y | W | G |
|---|---|---|---|---|
| Description | Hot | Hot | Neutral | Ground |
| Color (may vary) | Black | Brown | White | Green |
| PDU Connections | TS5-0 | TS5-1 | TS5-2 | TS5-3 |

**Table 2-6 Console Power Cable Termination**

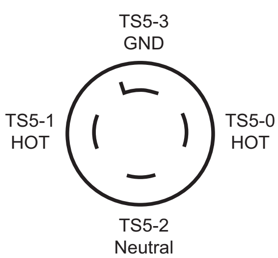



**Figure 2-41 Console Power Plug**

## 7.8 Console Power Connection

**Note:** **The console can only be powered from this connection.**

| FROM PDU Plug | TO console Plug End | |
|---|---|---|
| CB1 -11<br>(J5 Phase X) | Phase X<br>Hot | ☐ Check box when complete |
| N/G-Neutral<br>TS5-2 | Phase Y<br>Hot | ☐ Check box when complete |
| A3 Neutral Bus Bar<br>(J5 -13 Phase W) | Phase W<br>Neutral | ☐ Check box when complete |
| A3 Ground Bus Bar<br>(J5 -22 Ground Green) | Ground (Green)<br>Ground Green screw | ☐ Check box when complete |

**Table 2-7 Resistance Verification Points**

☐ Check box when complete.

## 7.9 PDU Control Cable

The PDU control cable comes pre-terminated and should not be re-terminated in the field. Excess cable length must be stored. Simply plug the cable into J1 on the A panel. Secure it by using the fasteners integrated into cable's connector shell.

☐ Check box when complete.

## 7.10 System Ground Connection

Connect the ground wire (green with a yellow strip) from the table/gantry raceway ground bus to the system ground lug in the PDU. See Figure 2-35, on page 126, and Figure 2-42, below.

**Figure 2-42 PDU System Ground Connection**

☐ Check box when complete.

## 7.11 For HD ONLY

Options that connect to the gantry or the console require an IEC plug for connection. ALL options must be powered from the gantry or the console.

1.) Remote monitor (console)
2.) SmartStep (console/gantry)
3.) Injectors (console)
4.) Respiratory gating (console/gantry)
5.) Cardiac monitor (gantry)

# 7.12 Warning Light & Door Interlock Connections

## 7.12.1 Warning Light Configuration & Connection

1.) Warning light is controlled by signals from the system.

2.) This step is site-specific. The PDU by default is configured for no external warning light connection. If you have external warning lights, see Figure 2-43 for proper connection.

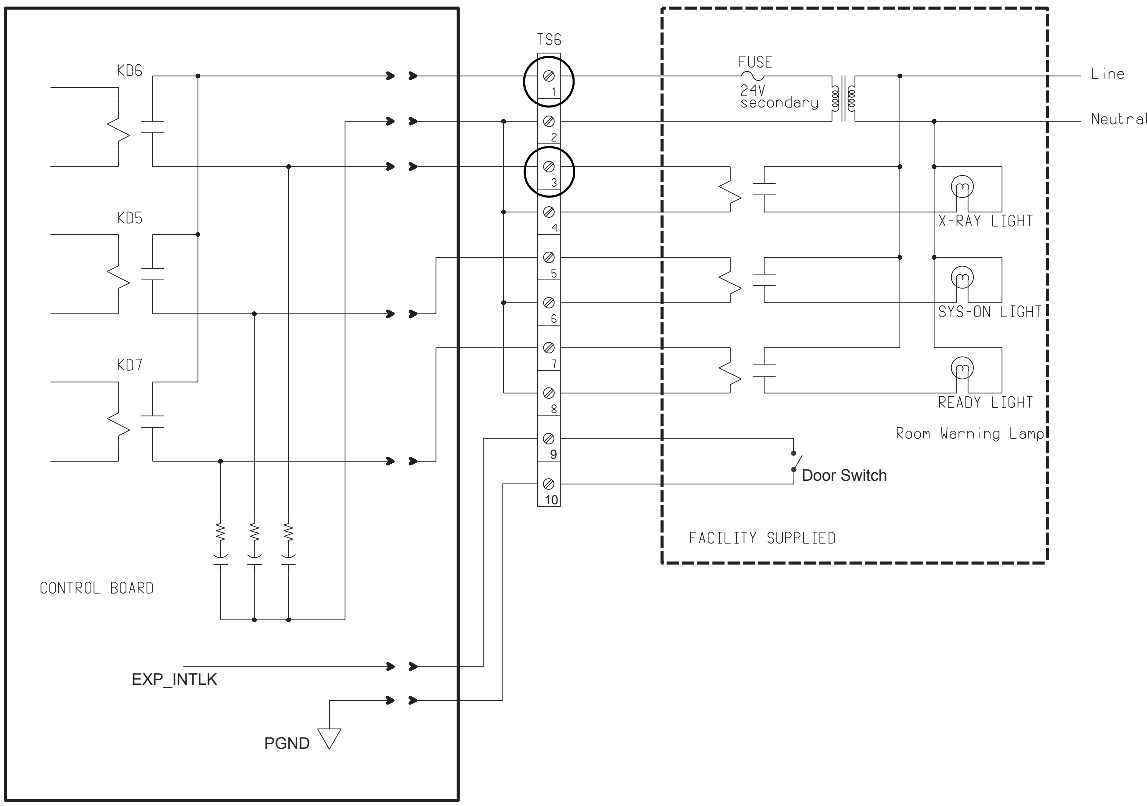



**Figure 2-43 TypicalTS6 Warning Light & Door Interlock Connections**

It is recommended that you use the four (4) wire method of adding an x-ray warning light to a room, as shown in Figure 2-43. When using this method, you:

- Minimize EMC interference.
- Increase contact life of the relay used in the PDU.

☐ Check box when complete.

**WARNING** If a door interlock is used, remove the jumper (terminals 9 and 10).

### 7.12.2 Door Interlock Connections

Door interlocks are used to prevent x-rays from being generated when the scan room door is open.

The door interlock circuitry in the PDU is shipped from the factory engaged. This means the system cannot generate x-ray until disengaged. A short must exist between pins 9 & 10 for x-ray to be generated. Using a small piece of wire, short pins 1 and 2 together. See Figure 2-44.

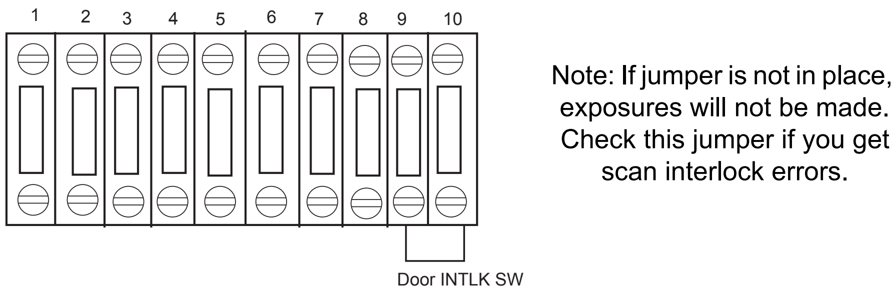



**Figure 2-44 Without a Door Interlock**

To use the system with a door interlock, wire a normally open switch between pins 1 & 2 that is attached to the interlock.

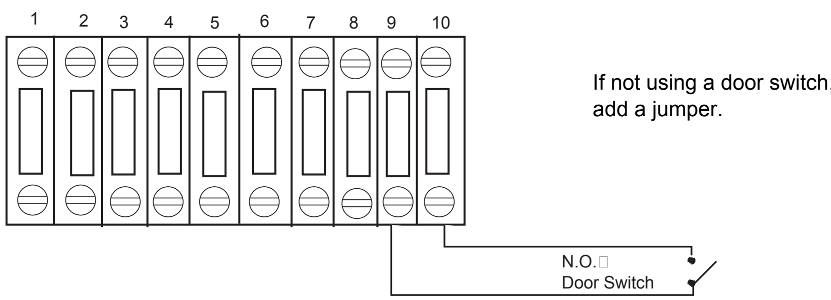



**Figure 2-45 With a Door Interlock**

☐ Check box when complete.

# Section 8.0 System Ground Connections

As seen in Figure 2-46, the table/gantry raceway ground bus is used to centralize all system grounding. The system ground is tied to vault ground at the PDU, through its chassis.

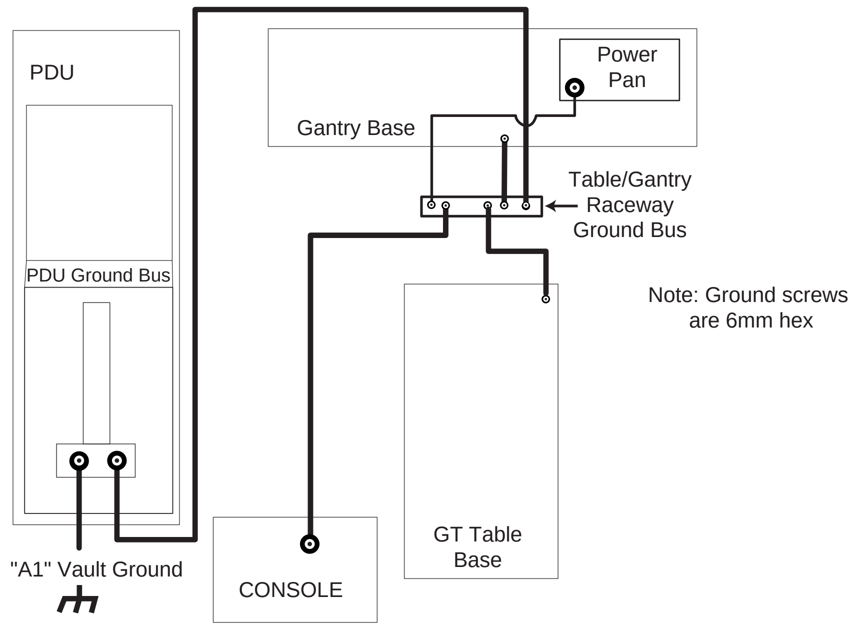



**Figure 2-46 CT System Ground Connections**

The gantry is tied to system ground at a number of points. It is important that all of these ground connections are securely made. See Figure 2-47.

Check that the terminals will not be loosened, by moving or swinging the cables rather strongly by hand.

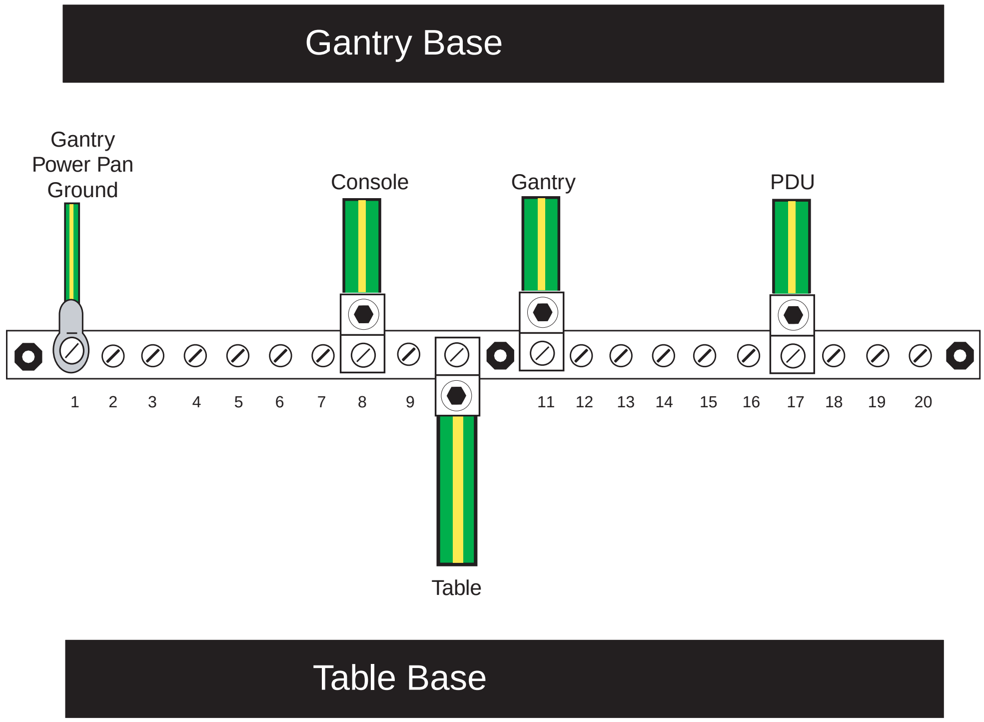



**Figure 2-47 Table/Gantry Raceway Bus - Grounds**

Various types and sizes of wire are used to ground the system. Please use the type and sizes specified in Table 2-8.

| AWG # | Connection To | Connection To |
|---|---|---|
| #1/0 | PDU | Power Main |
| #1/0 | Gantry (Power Pan) | Raceway |
| #2 | Console | Raceway |
| #1/0 | Gantry | Raceway |
| #2 | Table (frame) | Raceway |
| #1/0 | PDU | Raceway |

**Table 2-8 System Ground Connections**

☐ Check box when complete.

# *Chapter 3*
# *System Continuity & Ground Checks*

**NOTICE** **Potential for Data Loss and/or Equipment Damage**

**To prevent potential data loss and equipment damage, please do the following:**

- **Record data collected from procedures in this chapter into Form F4879 when directed, located in Chapter 6 of this book.**
- **Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.**

## Section 1.0 System Continuity (Mechanical Contractor)

### 1.1 Time and Personnel

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 1 (FE or mechanical supplier) | | 20 minutes labor on-site | |

### 1.2 Tools and Test Equipment

- Digital VOM with the capability to read 0.5 ohms
- 30 ft of #18 wire
- 600 VAC meter leads

### 1.3 Procedure

Reference Figure 3-1.

**WARNING** **USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES; LOCK OUT WALL POWER.**

1.) Remove all System Power at the A1 Mains Disconnect Panel. Follow Lockout/Tagout procedures.
2.) Remove the PDU input power panel cover.
3.) Verify, with a voltmeter, that mains power is disconnected.
4.) Verify that less than 1 ohm of resistance exists between the following ground connections:

| From | To | |
|---|---|---|
| Wall ground connection | PDU Cabinet | ☐ Check box when complete |

**Table 3-1 Mains Connections to PDU**

5.) Verify that less than 1 ohm of resistance exists between the following connections:

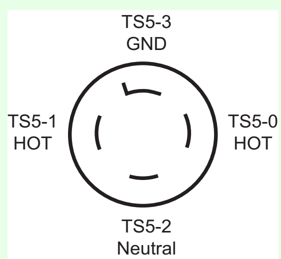
| From | Signal Name (color) | To | |
|---|---|---|---|
| PDU TS2-1 | +HVDC (Red) | Gantry HV Power Pan TS1-1 | ☐ Check box when complete |
| PDU TS2-2 | HVDC Ground (Green/Yellow) | Gantry Power Pan Chassis | ☐ Check box when complete |
| PDU TS2-3 | -HVDC (Black) | Gantry HV Power Pan TS1-2 | ☐ Check box when complete |
| PDU Ground Bus | HVDC shield | Gantry HVDC cable shield | ☐ Check box when complete |
| PDU TS3-1 | Axial drive 440vac (Black) | Gantry HV Power Pan TS2-1 | ☐ Check box when complete |
| PDU TS3-2 | Axial drive 440vac (Red) | Gantry HV Power Pan TS2-2 | ☐ Check box when complete |
| PDU TS3-3 | Axial drive 440vac (Orange) | Gantry HV Power Pan TS2-3 | ☐ Check box when complete |
| PDU TS3-4 | Axial drive ground (Green/Yellow) | Gantry Power Pan Chassis | ☐ Check box when complete |
| PDU Ground Bus | Axial drive shield | Gantry 440 VAC cable shield | ☐ Check box when complete |
| PDU TS5-0 | 120vac Phase B (Black) | Console Power Plug:   | ☐ Check box when complete |
| PDU TS5-1 | 120vac Phase A (Brown) | | ☐ Check box when complete |
| PDU TS5-2 | 120vac Neutral (Light Blue) | | ☐ Check box when complete |
| PDU TS5-3 | Ground (Green/Yellow) | | ☐ Check box when complete |
| PDU TS5-4 | 120vac Phase A (Black) | Gantry LV Power Pan A1J1 FIlter - L1 (by power plug) | ☐ Check box when complete |
| PDU TS5-5 | 120vac Phase B (Red) | Gantry LV Power Pan A1J1 FIlter - L2 (by power plug) | ☐ Check box when complete |
| PDU TS5-6 | 120vac Phase C (Orange) | Gantry LV Power Pan A1J1 FIlter - L3 (by power plug) | ☐ Check box when complete |
| PDU TS5-7 | 120vac Neutral (Light Blue) | Gantry LV Power Pan A1J1 FIlter - N (by power plug) | ☐ Check box when complete |
| PDU TS5-8 | Ground (Green/Yellow) | Gantry Power Pan Chassis A1J1 Filter Ground Stud | ☐ Check box when complete |

**Table 3-2 Resistance Verification Points**

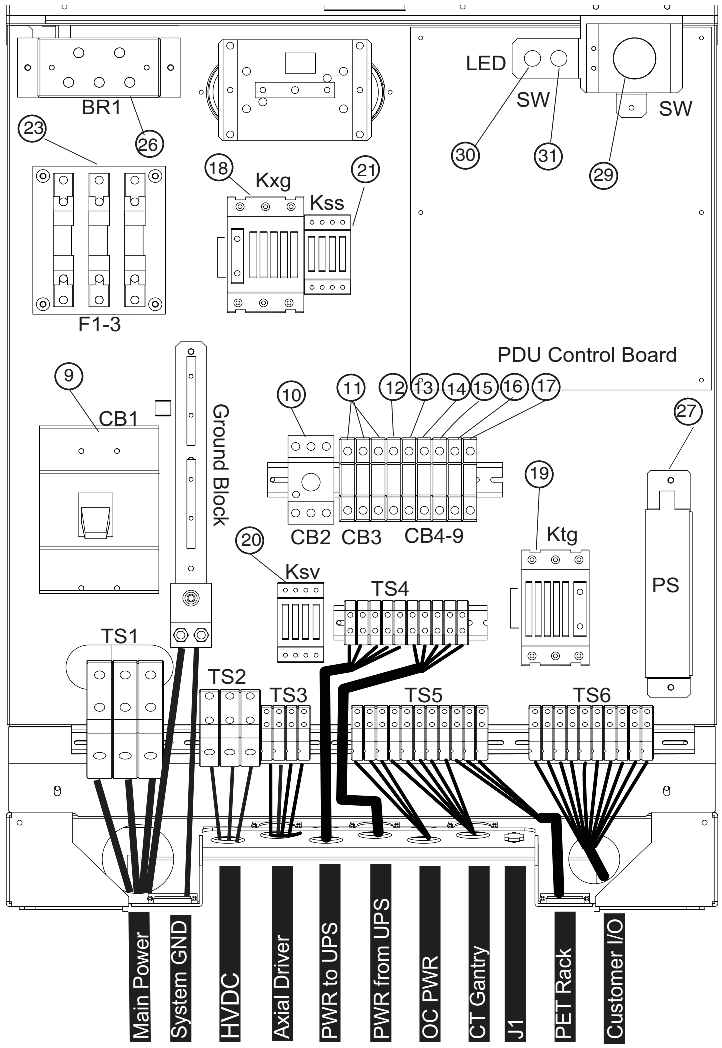



**Figure 3-1 Front View of NGPDU with covers removed**

**WARNING** **TURN OFF ALL PDU CIRCUIT BREAKERS.**

6.) Set an ohmmeter to the lowest scale. Check between the following points for shorts to ground. Verify no continuity exists between the following points:

| FROM PDU | TO A1 Breaker Box | |
|---|---|---|
| TS2-1 (+HVDC) (Red) | vault ground | ☐ Check box when complete |
| TS2-3 (-HVDC) (Black) | vault ground | ☐ Check box when complete |
| TS3-1 (440vac output) (Black) | vault ground | ☐ Check box when complete |
| TS3-2 (440vac output) (Red) | vault ground | ☐ Check box when complete |
| TS3-3 (440vac output) (Orange) | vault ground | ☐ Check box when complete |

**Table 3-3 No Continuity Verification Points**

7.) Leave the metal cover off the PDU input power panel until you complete the checks in the next section.

# Section 2.0 Site Ground Continuity Check

1.) Use an ohmmeter to verify the presence of **less than 1.0 ohm of resistance** between each of the following points:

| FROM | TO | |
|---|---|---|
| PDU Ground Bus | Vault Ground | ☐ Check box when complete |
| PDU Ground Bus | Table/Gantry raceway ground point | ☐ Check box when complete |
| Table/Gantry raceway ground point | Gantry Chassis | ☐ Check box when complete |
| Table/Gantry raceway ground point | Table Chassis | ☐ Check box when complete |
| Table/Gantry raceway ground point | Operator Console Chassis | ☐ Check box when complete |
| All Display or Computing Options (if any) | Operator Console Chassis | ☐ Check box when complete |

**Table 3-4 Resistance Verification - Site Ground**

2.) Install remaining covers on:
- Gantry
- Table
- Raceway
- Console
- PDU

# Section 3.0 Shim Installations

This procedure applies to the following table types:

| | |
|---|---|
| GT 2000 | Shim axial head holders, foot extender and phantom holder. |
| GT 1700 | Shim axial head holders, foot extender and phantom holder. |
| HP 1600 | Shim axial head holders, foot extender and phantom holder. |
| PET Tables | Shim axial head holders, foot extender and phantom holder. |

## 3.1 Time and Personnel

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 1 (FE or mechanical supplier) | 10 min. | 15 min. labor on-site | 5 min. |

## 3.2 Tools and Test Equipment

- Standard FE Tool Kit
- Shim Kit

**NOTICE** **Understand and Follow All General Table Safety Procedures.**

## 3.3 Preparation

Check head holder for a tight fit. If the head holder fit is loose, follow this procedure and shim for.

- Axial head holder
- Foot extender
- Phantom holder

**Introduction**:

- Some Axial Head Holders have a large free-play in the horizontal direction which could potentially lead to motion and therefore image artifacts.
- Installation of the 2327335 rubber shim kit can minimize this motion.

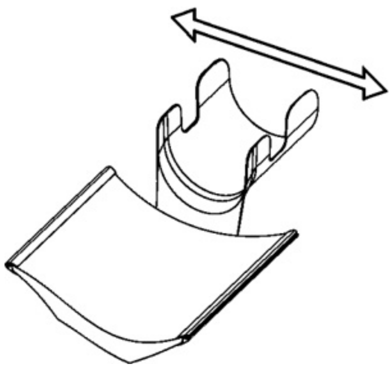


**Figure 3-2 Axial Head Holder**

**Notes Before Selecting Shim Thickness:**

- While selecting the best shim size, do not attach the rubber shim yet using the adhesive on the back. It is best to use a piece of tape to hold on the shim in order to see if the size is correct.
- Selecting a shim size that is too thick may result in:
  - Difficulty latching the head holder properly. The head holder must latch so that a patient is not injured.
  - Damage to the plastic latch or the plastic screws that secure it.

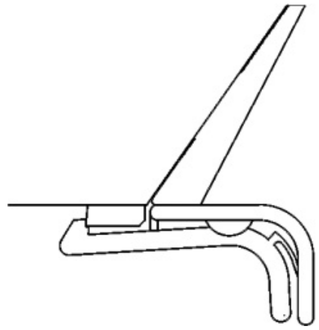


**Figure 3-3 Correct - Head Holder is latched onto first step of plastic latch mechanism (The head holder does not need to be latched onto the second step)**

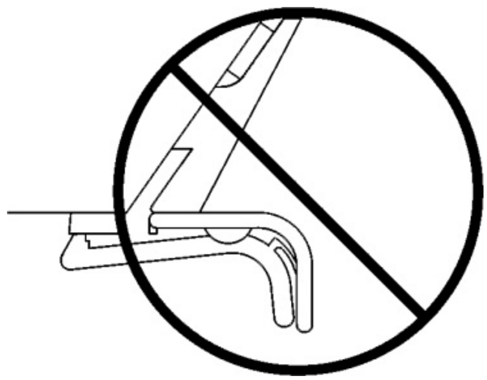


**Figure 3-4 Wrong - Head Holder is NOT latched after installing shims**

## 3.4 Procedure

1.) First place the two 4.0mm shims (thickest size) onto both edges of the head holder as shown (use a piece of tape to temporarily secure them)
- The shim must be placed with the tab facing out
- The thickness is printed on the shim

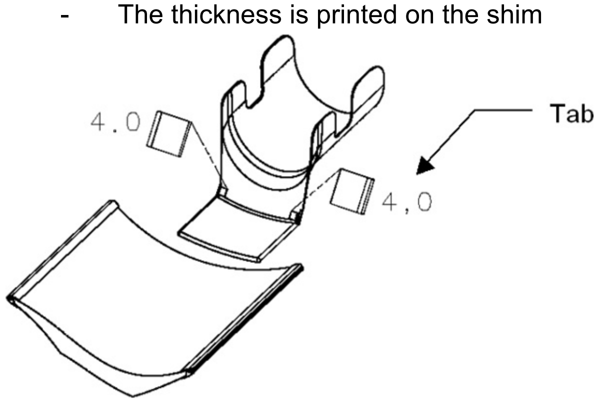



**Figure 3-5 Headholder Tab**

2.) Insert the head holder into the cradle
3.) Check if the head holder is latched onto the cradle at the first step of the plastic latch mechanism. (The head holder does not need to be latched onto the second step)
4.) Check if the head holder has a small free-play in the horizontal direction

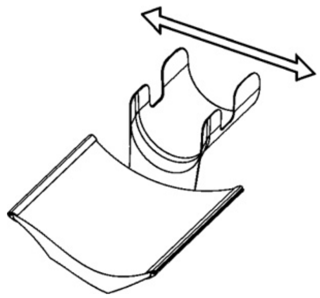


**Figure 3-6 Axial Head Holder**

5.) If the rubber is too thick, repeat steps 1-4 using a thinner shim (3.5, 3.0…0.5mm) until the head holder is latched (without excessive force) and fits securely in the cradle.

If the thinnest shim (0.5mm) is too tight, the tab can be cut off to reduce the thickness

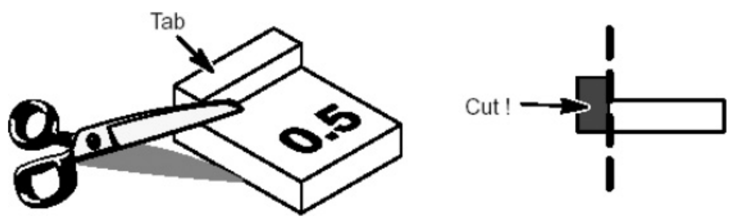



**Figure 3-7 Cut the shim for Headholder**

6.) Clean off the surfaces where the shims will mount using alcohol.

7.) Peel off the paper from the back of the selected shims and attach with the tabs facing out. Hold each shim with your fingers for a few seconds to attach it to the head holder.

## 3.5 Finalization

Review latching of the head holder with the customer after installation.

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# Chapter 4
# System Covers: Installation & Alignment

## Section 1.0 Process Overview

Cover install process overview:

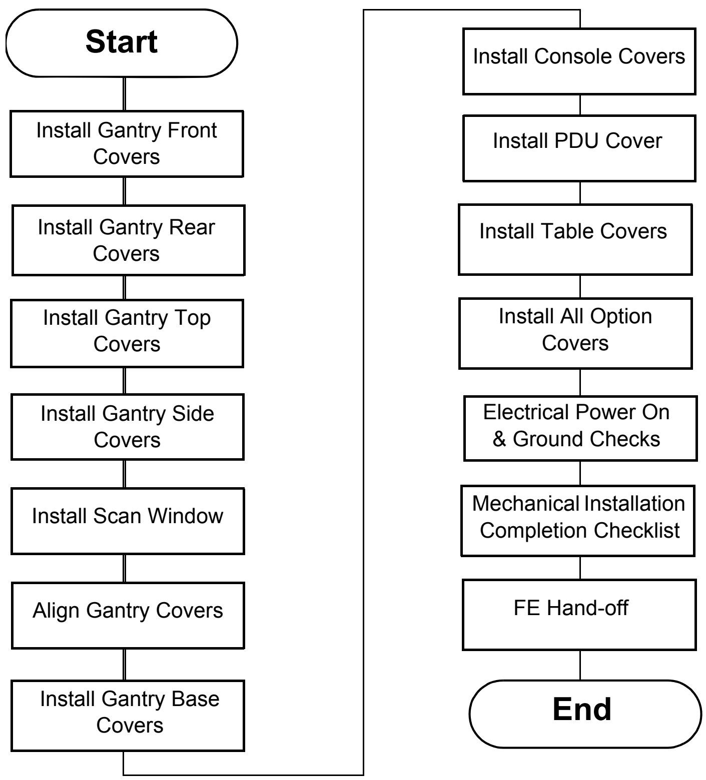



**Figure 4-1 Cover Installation Flowchart**

## 1.1 Gantry Front Cover Installation

1.) Move the gantry to the vertical position and the front cover next to the gantry.

2.) Lift the cover onto the stud and attach the front cover:

   a.) Align the studs on both sides of the front cover with each associated receiver. The receiver is located on the gantry frame.

**Figure 4-2 Cover Stud and Mounting Bracket Receiver**

   b.) Insert the stud on one side into its associated receiver and attach the rubber retaining straps.

   c.) Insert the stud on the other side into its associated receiver and attach its rubber retaining straps.

Note: You may find it helpful to “lift up” on the cover to align the stud while attaching the rubber retaining straps.

3.) Reattach the upper and lower Cantrell brackets on both sides:

   a.) Remove the upper Cantrell brackets from the service position and rotate them into position over their associated retaining pins (see Figure 4-3 and Figure 4-4).

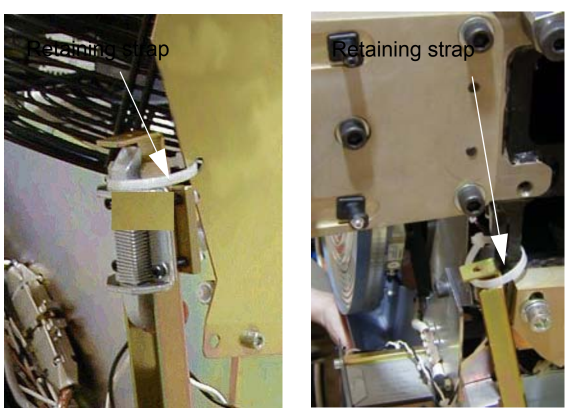



**Figure 4-3 Service Position of Upper and Lower Cantrell Brackets.**

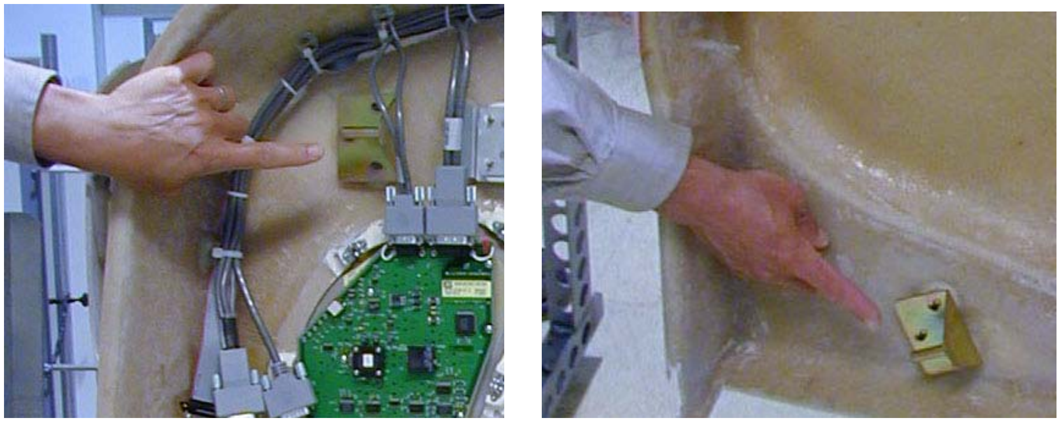


**Figure 4-4 cover Retaining Pins (top and bottom)**

b.) Press down firmly on the bracket and “snap” it into place. The locking mechanism on each upper bracket should lock the bracket securely into place. Do this on both sides (see Figure 4-5).

**Figure 4-5 Locking the cover brackets into place**

c.) Remove the lower Cantrell brackets from the service position (Figure 4-3), and rotate them into position over their associated retaining pins.

d.) Press down firmly on the bracket and snap it into place (see Figure 4-5).

4.) Remove the dolly; disassemble it and store it safely away for later use.

5.) Reattach the cables to the cover.

6.) Reinstall the Mylar (scan) window.

## 1.2 Gantry Rear Cover Installation

1.) Position the cover in the back of the gantry.

2.) Attach the rear cover:

a.) Align the studs on both sides of the rear cover with the receivers located on the gantry frame.

b.) Insert the stud on one side into its associated receiver and attach the rubber retaining straps.

c.) Insert the stud on the other side into its associated receiver and attach its rubber retaining straps.

Note: You may find it helpful to “lift up” on the cover to align the stud while attaching the rubber retaining straps.

3.) Reattach the upper and lower Cantrell brackets on both sides:

a.) Remove the upper Cantrell brackets from the service position and rotate them into position over their associated retaining pins.

b.) Press down firmly on the bracket and “snap” it into place. Do this on both sides.

Note: The locking mechanism on each upper bracket should lock the bracket securely into place.

c.) Remove the lower Cantrell brackets from the service position and rotate them into position over their associated retaining pins.

d.) ]Press down firmly on the bracket and “snap” it into place.

Note: Adjustment of the Cantrell brackets can cause misalignment of the top and side covers. The upper and lower Cantrell brackets do not require adjustment during normal use.

4.) Remove the dolly, disassemble it, and store it safely away.

5.) Reattach the cables to the cover.

6.) Reinstall the Mylar (scan window. Carefully bend the scan window and place it into the channel (groove) provided in the covers.

## 1.3 Gantry Top Covers

CAUTION

**Potential for Electric Shock.**

**Gantry Power Is Present.**

**Before you remove top covers, always make sure the three (3) power switches have been turned off.**

CAUTION

**Potential for Personal Injury.**

**Top Cover is Heavy.**

**Follow the Removal Procedure.**

### 1.3.1 Installation

The top cover consists of two (2) pieces. Install the front and rear gantry covers, if not already installed.

1.) Lift the top cover so that the cover is chest high. Fit the cover guides on the top of the cover into the rail slots on the gantry covers.

2.) Slide the top cover up the front and rear covers onto the top cover ramp. The cover should snap into place in the top cover bracket.

3.) Insert the cover-locking tabs (at the bottom of the top cover) into the slots on the gantry covers.

4.) Check cover alignment. Adjust as needed.

5.) Reconnect the fan power.

6.) Repeat above steps 1-5 for the other cover.

# 1.4 Gantry Side Covers

## 1.4.1 Installation

1.) To install a side cover, place it over the top cover and let the two (2) side cover latches slide behind the metal tabs, located on the top cover. See Figure A-3 on page 168.

2.) Use hex wrench to secure the side cover to front cover by turning the bolts a quarter turn. See Figure A-3 on page 168.

# 1.5 Scan Windows

## 1.5.1 Installation

Note: The front and rear covers must be installed before installing the scan window.

1.) Shape the scan window as shown in Figure 4-6, and nest the scan window at the bottom of the opening between the front and rear covers, (Figure 4-7) with the rivets in the 6 o'clock position. Remember the rivets must be in the 12 o'clock position when the mylar window is fully installed.

2.) After you complete the initial seating of scan window, let the window slowly unfold, and work both sides of the window into position, starting at the bottom and finishing at the top.

3.) Make sure you position the window with the rivets at the 12 o'clock position, and the mylar window slit at either the 3 or 9 o'clock position.

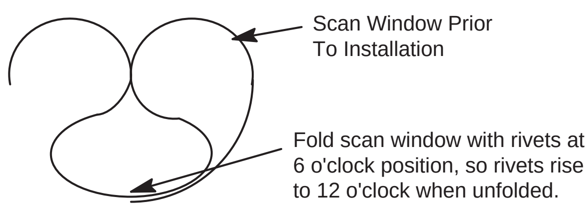



**Figure 4-6 Install Scan Window**

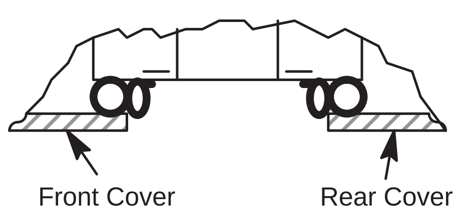



**Figure 4-7 Scan Window Nested Between Front and Rear Cover**

# 1.6 Align Gantry Covers

## 1.6.1 Overview

This section explains the adjustment capability available for the Gantry covers and guidelines for performing adjustments.

## 1.6.2 Tools Required

3mm, 5mm, 8mm, hex wrenches

## 1.6.3 Adjustment Guide

### 1.6.3.1 Front/Rear Cover Adjustment

The front and rear covers are held in a fixed vertical position by the cover locking mechanisms at the vertical middle of each cover on each side. The only adjustment capability is found on each of the cover Cantrells (see Figure 4-8, one near each corner of the front and rear covers. Adjustments are best done at zero tilt.

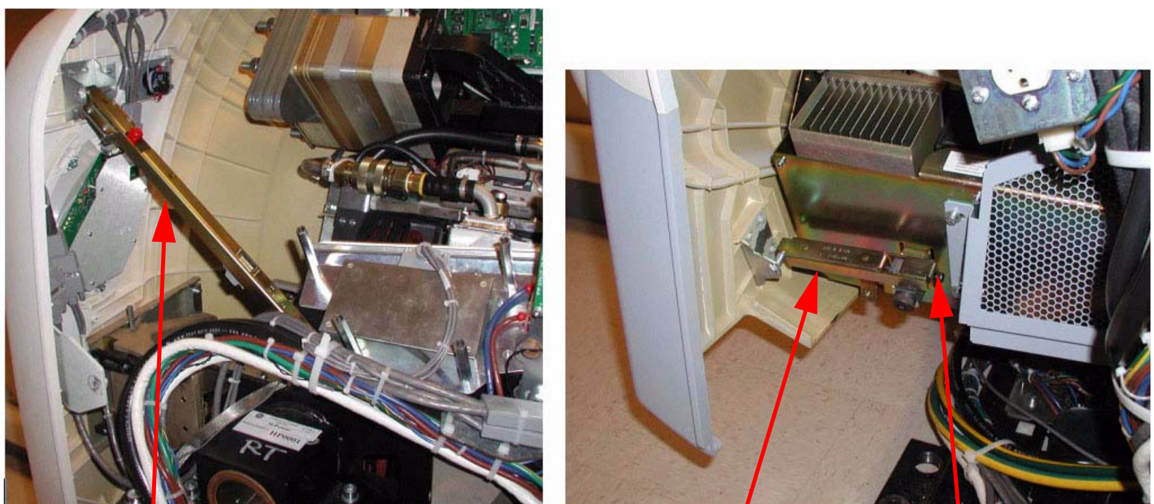



**Figure 4-8 Cantrells**

1.) Remove the gantry side covers using a 8mm hex wrench to release the bottom latches. Refer to, for further details.
2.) Disable gantry rotation, 120VAC and HVDC via the service switch panel.
3.) Disconnect the fan power via the white molex connector (one for each top cover). Remove the top covers by lifting up and pulling to the side of the gantry.
4.) Remove the scan window to avoid creases or kinks during cover movements.
5.) Using a 5mm hex wrench, adjust the bolt on the gantry end of the cover cantrells (see Figure 4-8) to lengthen or shorten the cantrell arm that will push or pull the cover corner. The distance between the inside edges of the front and rear covers will be typically 27" + 1/8” (689mm). Distance should remain the same as measured along the inside edges from top to bottom with both covers hanging level as seen with a level on the inside vertical edge.

### 1.6.3.2 Scan Window

Position the scan window in the opening to verify that it seats all the way around with no gaps. The scan window should seat easily if the front and rear covers have been adjusted properly. If not, go back and readjust such that the scan window seats properly. There is no easy solution other than adjusting and looking for fit.

### 1.6.3.3 Top Cover Adjustment

The top covers rest in a top bracket that is mounted to the stationary gantry frame and subsequently just rest on top of the front/rear covers. There are also two alignment fingers on the outer edge of the top covers that need to sit in brackets on the front/rear covers.

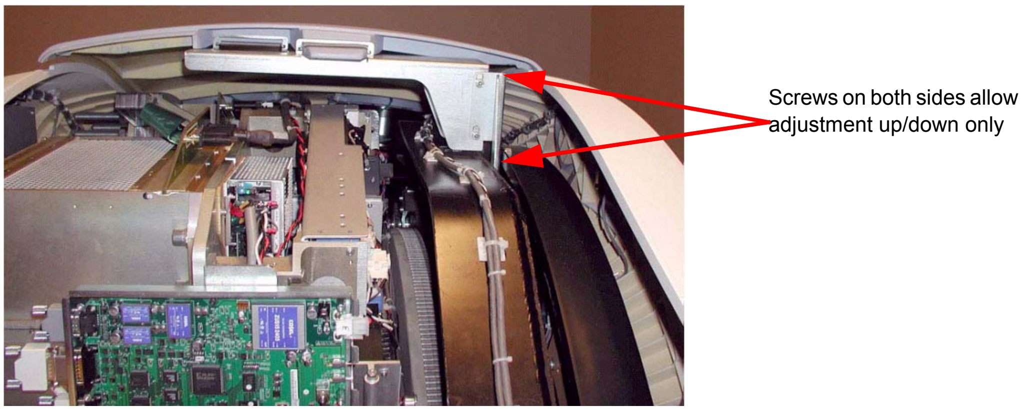



**Figure 4-9 Top Cover Bracket**

1.) Set the top covers back on the gantry and look for proper fit. The top covers should rest on the front/rear covers all along the edges. If they do then skip the rest of this section.
2.) If the top covers do not rest on the gantry at the top then lift them back off and use a 5mm hex wrench to loosen the bolts that hold the top cover bracket for adjusting vertical alignment to allow the top covers to rest on the front and rear covers.
3.) The brackets on the front/rear covers for the fingers on the outer edge of the top covers are adjustable using a 3mm hex wrench and sliding the bracket till it lines up with the top cover finger. (See Figure 4-10)

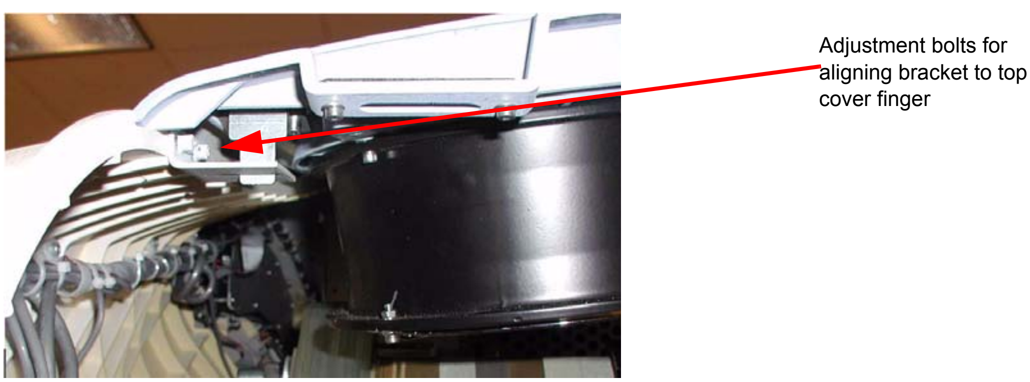



**Figure 4-10 Top Cover Fingers**

4.) Verify the top cover tapered edges fit over the lip along the top edge of the front/rear covers. If not, go back to Section 1.6.3.1 and readjust the cantrells as necessary.
5.) When done adjusting the top covers, reconnect the fan power molex connections.

#### 1.6.3.4 Side Covers

The side covers merely hang from the top covers by two fingers that are not adjustable. When set into the top covers, the side covers should hang straight down with the tapered edges fitting over top of the lip of the front/rear covers.

## 1.7 Install Gantry Base Covers

### 1.7.1 Tools Required

3mm and 8mm hex wrenches

### 1.7.2 Installation (VCT 64)

Note: Tighten means torque to 2.3 Nm

1.) Figure 4-11 shows the gantry base covers. There are two side covers, two rear covers and a single front cover.
2.) Start with the rear covers (Shown as items 6 and 7). Using an 8 mm hex wrench twist the 3 twist locks counter-clockwise to release the locks and pull each cover out from the back of the gantry.
3.) Remove the side covers (shown as items 4 and 5) if needed by removing the one screw on the side (3 mm hex wrench) and 2 screws the interface panel on the back side of the gantry, then slide the side cover away from the gantry to the side.

Note: The side covers slide around the Interface Panels. No need to remove cabling or move the interface panels.

4.) Remove the base front cover (shown as item 3) if necessary by removing one screw on each end to release the cover from the brackets. Lift the cover up and over the foot switch cover and away from the gantry.
5.) To reinstall the gantry base covers, reverse the above steps, putting on the covers in the order of front, sides then rear covers

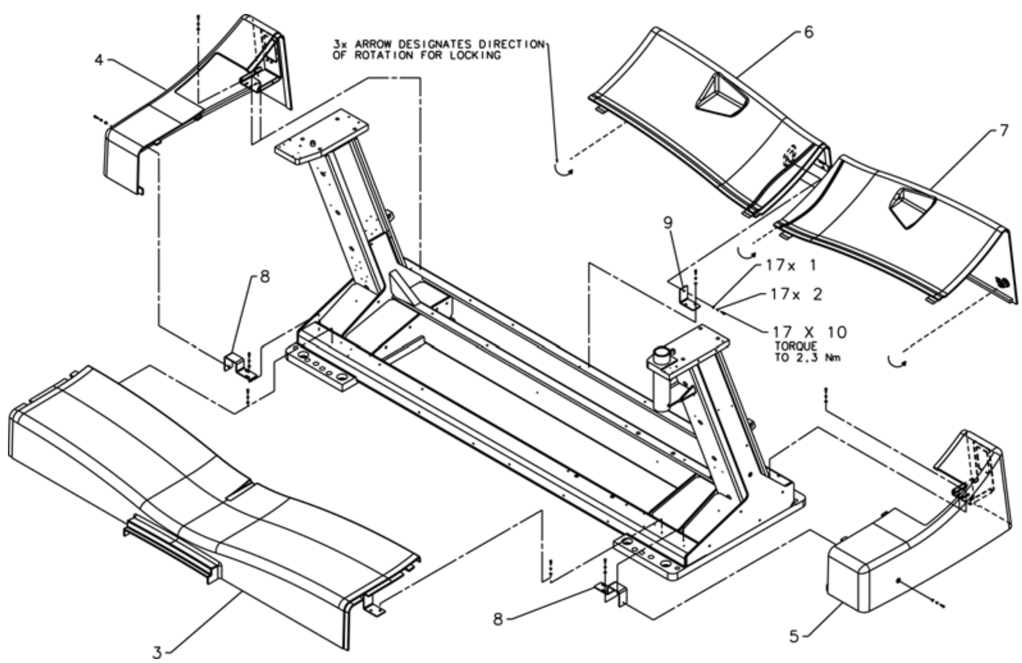



**Figure 4-11 Gantry Base Covers Installed**

### 1.7.3 Installation (HD)

1.) Position base front assembly (item 3) on gantry base with bracket slots aligned to gantry holes. Center cover left to right and attach with (4) hardware items 1, 2, and 10 as shown and tighten.

2.) Assemble (2) brackets (item 8) to gantry base using (4) hardware items 1, 2, and 10. Finger tighten hardware with brackets moved outward to end of slots. If not already replaced, remove (2) shipping brackets and replace with brackets (items 11 and 12) and reattach IPC panels. Install side cover assemblies (items 4 and 5) an base pushing brackets (item 8) inward until properly aligned with front cover.

3.) Assemble last bracket (item 9) loosely to gantry base with (2) hardware items 1, 2, and 10. Install rear cover assembly (item 6) to base properly aligned to side cover (item 4). Attach rear cover assembly to bracket with hardware items 1, 2, and 10 tightening all fasteners. Lock latch as shown.

4.) Place rear cover (item 7) on gantry base aligned to side cover assembly (item 5). Lock both latches as shown.

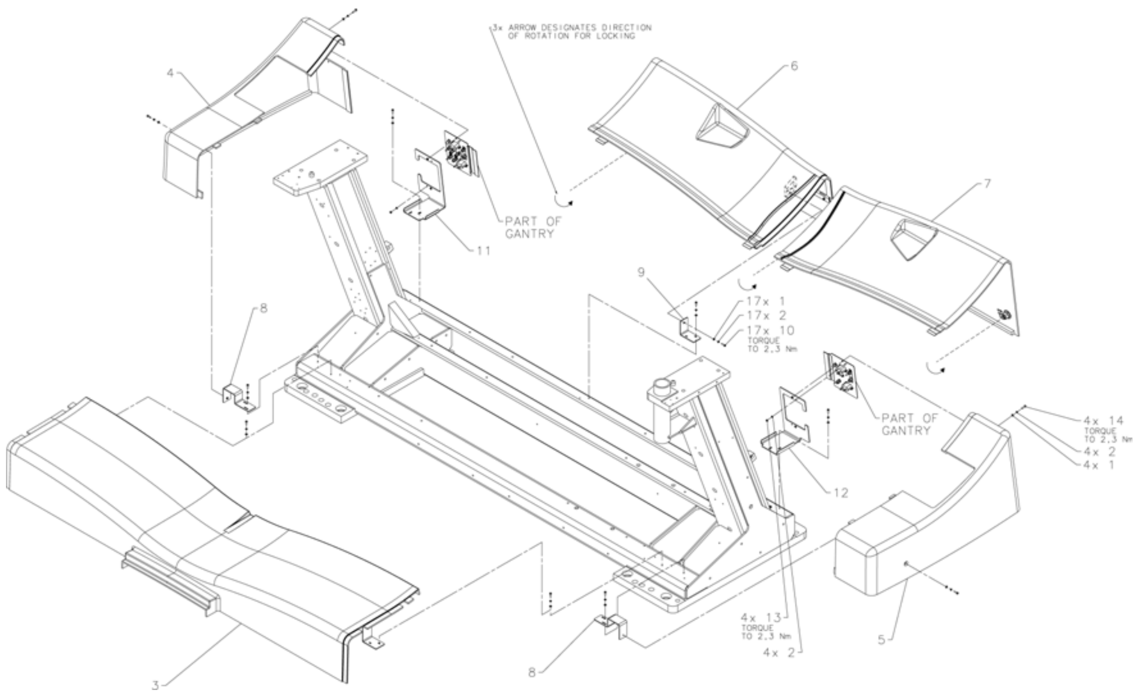



**Figure 4-12 HD Base Covers Installed**

### 1.7.4 Footswitch Covers Installation

5.) Install the footswitch cover using three (3) screws (see Figure 4-13).

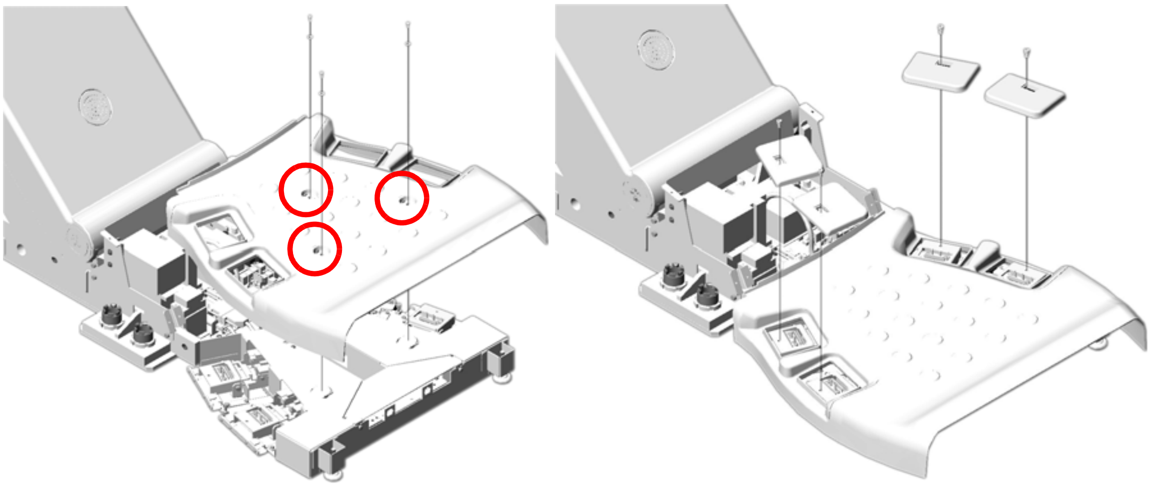


**Figure 4-13 Footswitch Cover Installation**

6.) Install cover caps on each pad.

**Figure 4-14 Footswitch Pad Caps**

7.) Install the four (4) foot pads onto the footswitch assembly.

**Figure 4-15**

# Section 2.0 Install Console Covers

## 2.1 Time and Personnel

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 1 (FE or mechanical supplier) | | 15 minutes labor on-site | |

## 2.2 Tools and Test Equipment

- Medium flat blade screw driver

## 2.3 Preparation

Locate the front and side console covers.

## 2.4 Procedures

### 2.4.1 Console Front Covers

1.) Position the front cover between the keyboard desktop side rails. This may be snug.
2.) Notice that there are three tabs on the top of the console chassis that hook onto the cover. Raise the cover and place it onto the three tabs.
3.) Push down on the cover to seat it.
4.) Use the two captured flat blade screws in the cover to secure it to the console chassis.

Note: The captured flat blade screws are quarter-turn screws. Do not over tighten them.

### 2.4.2 Console Side Covers

1.) Remove the side covers from their packaging. Each cover hooks onto two tabs on the top of the console chassis.
2.) Raise the cover and place it onto the two tabs.
3.) Push down on the cover to seat it.
4.) Use the two captured flat blade screws in the cover to secure the cover to the console chassis.

Note: The captured flat blade screws are quarter-turn screws. Do not over tighten them.

# Section 3.0

1.) Remove screws (2) on tape switch.
2.) Remove back under-side covers (2) plus black screws.
3.) Undo the 2 red/black connectors.
4.) Remove all six (6) 4mm hex-head screws.

## 3.1 Install Panels

### 3.1.1 Top Panel #1

Install two (2) 4mm hex-head screws. Leave them loose until the bottom screws are installed.

Install 6nd wire using one (1) 4mm hex-head screw.

### 3.1.2 Bottom Panel #1

Install white washer between grey base and panel. Insert Phillips screw into the bushing.

Tighten top screws. (Torque 8 lb-in)

Note: Second panel over laps the first panel

### 3.1.3 Top Panel #2

Install two (2) 4mm hex-head screws. Leave them loose until the bottom screws are installed.

Install 6nd wire using one (1) 4mm hex-head screw.

### 3.1.4 Bottom Panel #2

Insert white washer between grey base and panel. Insert Phillips screw into the bushing.

Tighten Phillips screws. Tighten top screws.

## 3.2 Re-install Side Panel

Install with two (2) Phillips screws. Reconnect cable.

## 3.3 Table Side Covers Install

1.) With the side covers toward table front, align the tabs on the cover with the slots on the table.
2.) Slide the cover toward the table until it stops.
3.) Slide the cover toward the back of the table to lock the cover in place.
4.) Install the two (2) 4mm hex-head screws on each end to secure cover. (1700 Table: Install one (1) screw.)

## 3.4 Table Side Covers Removal

1.) Remove the two (2) 4mm hex-head screws that secure the side cover. (1700 Table: Remove one (1) screw)
2.) Slide the cover toward the gantry until the locking tabs disengage and the cover is free.
3.) Pull the cover away from the table to remove.
4.) Store in a safe place.

# Section 4.0 Install All Option Covers

Follow the instructions that came with each of your optional components.

# Section 5.0 Electrical Power On & Ground Checks

**WARNING THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.**

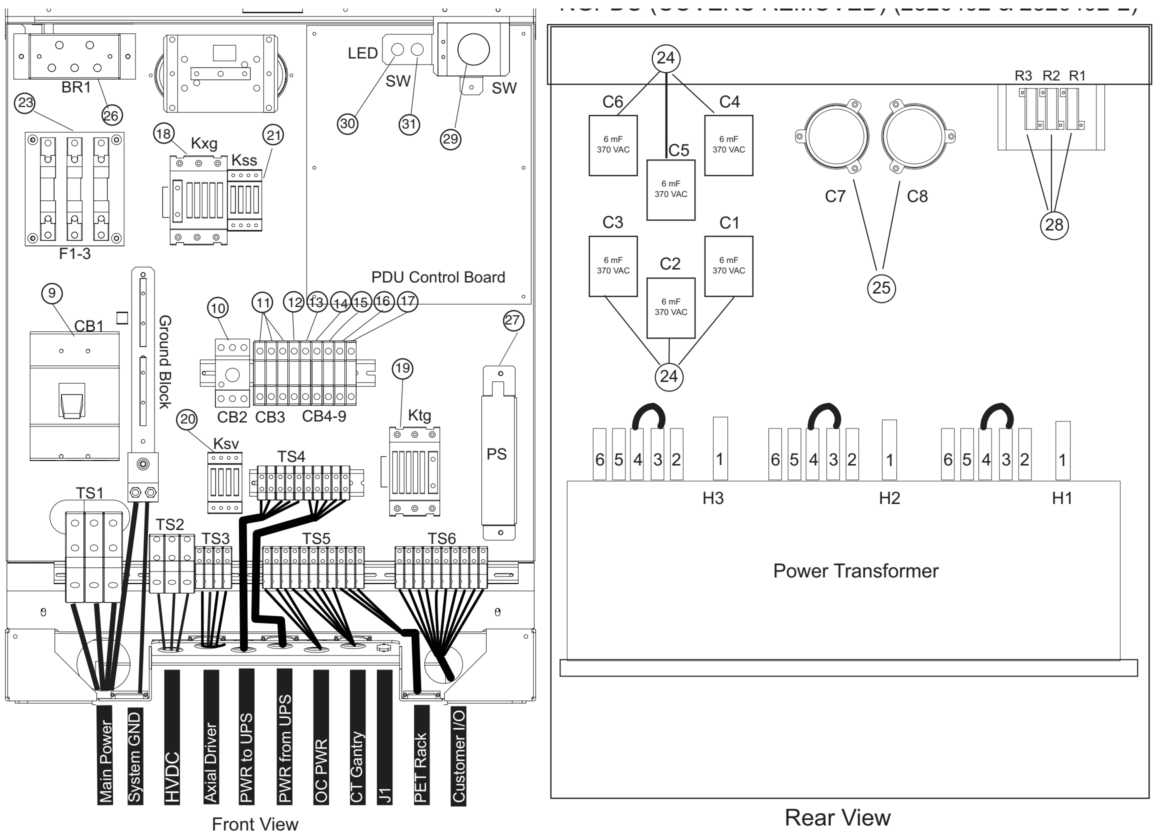
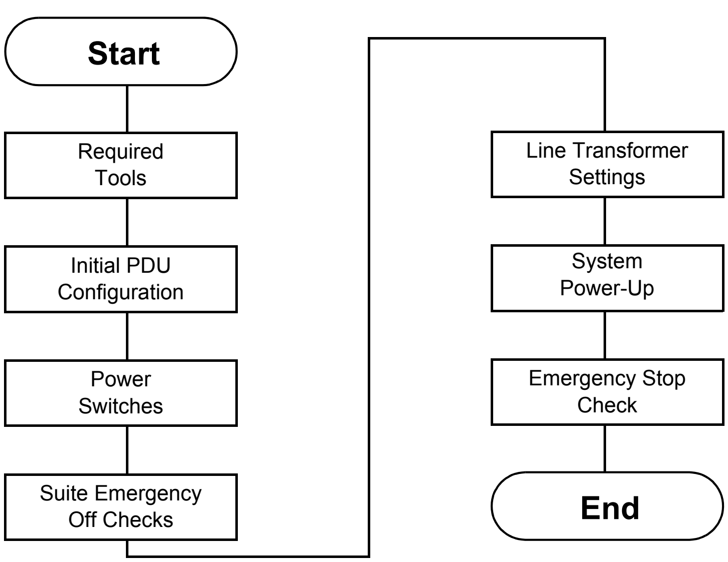



**Figure 4-16 NGPDU**

## 5.1 Introduction and Flowchart



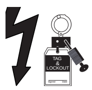
# 5.2 Electrical Power On & Ground Checks Process Overview

# 5.3 Required Tools

- Multimeter with a rating of at least 1000 volts
- Multimeter leads with a rating of at least 1000 volts

# 5.4 Initial PDU Configuration

**WARNING** **THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.**



## 5.4.1 Circuit Breakers

Set all PDU, gantry, console, and table circuit breakers to OFF.

## 5.4.2 Relay Board

1.) Set SW 2 to the Auto-Off position.
2.) When the system is powered, three lamps should be “ON”. (Refer to Figure 4-17.)

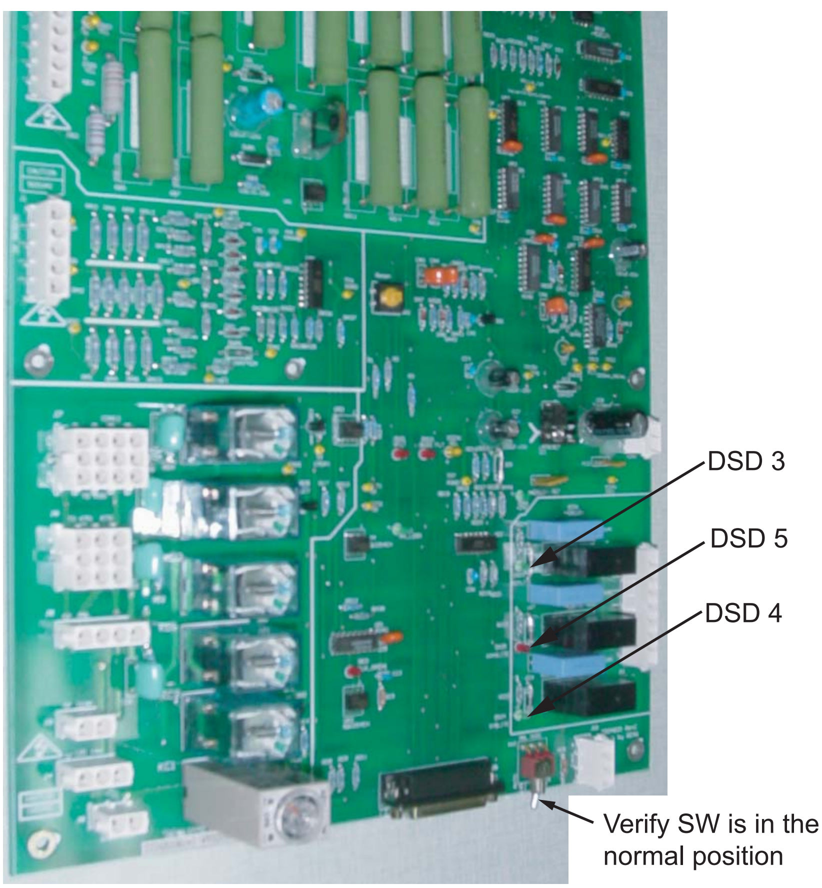



**Figure 4-17 NGPDU Control Board**

### 5.4.3 Power Switches

Turn OFF all system power switches at their subsystems.

- Gantry power pan breaker
- Gantry slip ring ground fault breaker on stationary base right side & all service switches
- Table base power
- Console power

### 5.4.4 Hardware and Connection Check

Use this step to check mechanical connections and tighten anything that may have shaken loose during shipment. Verify **all** hardware and connections in the PDU are securely fastened.

- PDU
- Gantry
- Table
- Console

### 5.4.5 Covers

Install, or verify the presence of, all the lexan safety covers for the PDU.

## 5.5 Suite Emergency Off Checks


**WARNING** **VERIFY ALL PERSONNEL HAVE CLEARED THE SYSTEM BEFORE YOU TURN ON WALL POWER.**

1.) Turn wall power ON to the PDU.

Note: Do not stand in front of the main disconnect to turn on power.

**2.) Press the suite emergency off button and verify it turns off wall power to the PDU.**

(Typically, this red palm button is located on the wall close to the console, within the scan suite.)

3.) Verify that all “Emergency Off” buttons are working properly.

4.) Leave power “OFF”.

## 5.6 Line Transformer Settings

### 5.6.1 Requirements

1.) **The PDU is shipped configured for 480VAC.**
2.) **Complete only if your site uses a voltage other than 480VAC.**
3.) If PDU is configured for 480VAC, go to 5.7. Otherwise, proceed to Section 5.6.2.

**WARNING**

**MAKE SURE YOU TURNED OFF, TAGGED AND LOCKED THE MAIN WALL POWER BEFORE YOU CHANGE TAPS. FAILURE TO DISCONNECT POWER AT MAIN INPUT MAY RESULT IN ELECTROCUTION. TURN OFF WALL POWER TO CONNECT OR MOVE METER LEADS, OR TO REMOVE OR INSTALL COVERS.**

### 5.6.2 Line Input Conditions

1.) Monitor the No Load Line to Line Voltage at L1, L2, L3, during the workday. Do not record this data during “brown out” conditions.
2.) After you determine the nearest nominal line, verify the tap connections match (refer to Table 4-1 and Figure 4-18 for tap locations).

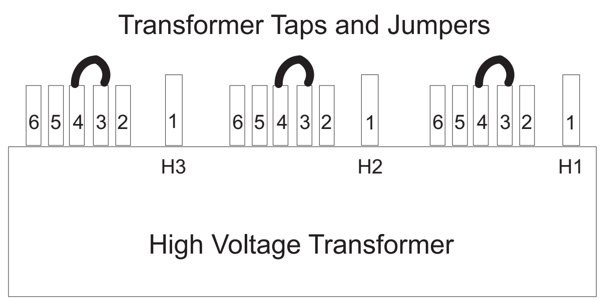
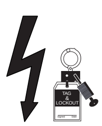



**Figure 4-18 PDU Tap Positions (Rear)**

Note: **Taps should be shipped as shown for 480 VAC only. For all others, you must move the taps. The tap check should be completed by the mechanical installer.**

3.) Verify that the No Load Line to Line Voltage never falls outside the corresponding minimum and maximum values listed in Table 4-1.
4.) Use a 0-750 AC voltmeter of 3/4% accuracy to measure the line-to-line voltages at L1, L2, & L3.
   - Verify the highest line-to-line voltage does not exceed 1.02 times the lowest voltage.
   - **Example:** If the lowest voltage equals 474, the highest voltage should not exceed 474 x 1.02 = 483.5 volts.

**WARNING**



**THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.**

| **No Load Line to Line Voltages** | | **Tap Connections (All 3 phases must have same the configuration)** | | |
|---|---|---|---|---|
| **Nominal** | **Maximum Range (10%)** | **Phase A Connection** | **Phase B Connection** | **Phase C Connection** |
| 480V* | 432 to 528* | 3-4* | 3-4* | 3-4* |
| 460V | 414 to 506 | 3-5 | 3-5 | 3-5 |
| 440V | 396 to 484 | 3-6 | 3-6 | 3-6 |
| 420V | 378 to 462 | 2-4 | 2-4 | 2-4 |
| 400V | 360 to 440 | 2-5 | 2-5 | 2-5 |
| 380V | 342 to 418 | 2-6 | 2-6 | 2-6 |
| 240V** | 216 to 264** | 1-4** | 1-4** | 1-4** |
| 220V** | 198 to 242** | 1-5** | 1-5** | 1-5** |
| 200V** | 180 to 220** | 1-6** | 1-6** | 1-6** |
| * Factory Default<br>** 2326492-3 PDU only | | | | |

**Table 4-1 PDU Line Tap Connections**

Record system voltages here:

Phase A: _______ Phase B: _______ Phase C: _________

## 5.7 System Power-Up

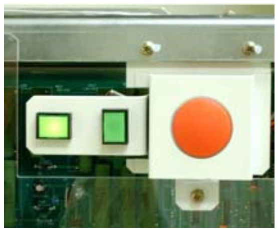
**CAUTION** **Verify all personnel have cleared the system before you turn on wall power.**

1.) Turn ON the A1 breaker panel.

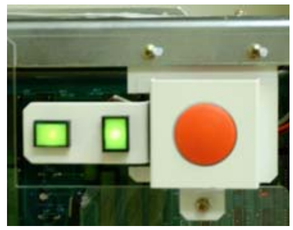
Note: Do not stand in front of the main disconnect to turn on power

2.) Turn ON all system power switches and breakers (PDU, gantry, table, console).

- All PDU breakers
- Make sure that the on/off button (on the front PDU panel) is ON for console power.



PDU Power Switch Off



PDU Power Switch On

- Gantry power pan breaker
- Gantry slip ring ground fault breaker on stationary base right side & all service switches
- Table base power
- Console power (Check internal breaker.)

### SUB-SYSTEM POWER-UP

1.) Turn ON switch S3 in the table (120VAC 24-hour power).

2.) Turn the gantry **120 - 208VAC** to ON. (Light should turn on.)

3.) Turn **AXIAL DRIVE ENABLE** ON. (Light should turn on.)

4.) Turn **HV DC ENABLE** ON. (Light should turn on.)

5.) Push the Service Switch Panel reset button. (See Figure 4-19.)

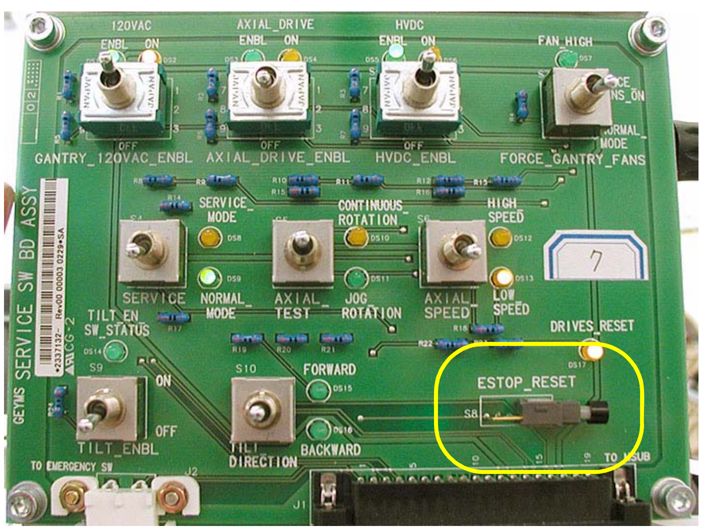



**Figure 4-19 Service Switch Panel**

## AXIAL ENABLE SWITCH TEST

1.) Unplug all top cover fan plugs.
2.) Turn OFF axial drive enable switch **AXIAL_DRIVE** on the Service Switch Panel.

Note: For the initial condition, do NOT leave the tube at the 2:30 position.

3.) Clear the gantry area for rotation.
4.) Press the alignment light push button.
5.) Verify that the gantry did not rotate.

## ROTATION SAFETY CHECKLIST

1.) Manually rotate the gantry 360 degrees.
   - Listen for any interference between the rotating and stationary parts. (Correct any interference problems.)
   - Listen for any loose parts. (Tighten parts as needed.)
2.) Turn ON all enable switches.

**WARNING** **MAKE SURE THERE ARE NO OBSTRUCTIONS AROUND THE GANTRY. PRESSING THE ALIGNMENT LIGHT PUSHBUTTON WILL CAUSE THE GANTRY TO ROTATE.**

3.) Press the alignment light push button.
4.) Verify that the gantry rotates.
5.) Turn off the laser light.
6.) Perform a 2-second X-ray OFF scan.

**NOTICE** **During the scan, it may be necessary to enter the scan room to obtain a better listening position. If so, keep a finger on one of the four E-STOP buttons (on the gantry), to quickly stop the gantry, if necessary.**

   a.) From the console, click on the SERVICE DESKTOP icon.
   b.) Select DIAGNOSTICS.
   c.) Select DIAGNOSTIC DATA COLLECTION
   d.) Set the scan time to 4.00 seconds and rotating X-ray Off.
   e.) Select ACCEPT.
   f.) Leave the door open. (This makes it easier to hear any loose or interfering parts.) The gantry should spin for approximately 45 seconds
      * Listen for any interference between the rotating and stationary parts. (Correct any interference problems.)
      * Listen for any loose parts. (Tighten parts as needed.)
   g.) After completing the 4-second scan, repeat Step a through Step f, with the following scan times:
      * 2.0 second scans
      * 1.0 second scans
      * 0.7 second scans
      * 0.5 second scans

7.) Confirm all enabled switches are on then install removed covers.

## 5.8 Install PDU Covers

### 5.8.1 Time and Personnel

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 1 (FE or mechanical supplier) | | 10 minutes labor on-site | |

### 5.8.2 Tools and Test Equipment

- Medium +blade screw driver
- Medium -blade screw driver

### 5.8.3 Procedure

1.) Confirm that the plastic safety shield is still in position and secured to the PDU.
   - If it is not, install the shield using the remover hardware.
   - Position the front cover so that the bottom is resting on the two guide pins located on the bottom of the PDU chassis.

2.) Raise the cover into place and use the two thumb screws on the top of the front cover to secure it. Screws should be tight, but do not over tighten them.

3.) Place the top cover on the PDU.

4.) Slide the cover toward the front of the PDU until the cover latches.

Using a +blade screw driver, tighten the screws. Do not over tighten them.

### 5.8.4 Emergency Stop Check

1.) Use the gantry push-buttons to advance the cradle about 0.5m (2ft) from the home position.

2.) Press one of the E-STOP buttons on the gantry.

3.) Make sure the TABLE POWER shuts off, and the green LED flashes.

4.) Depress one of the table elevation buttons, to verify the emergency stop disabled table elevation.

5.) Depress one of the cradle drive buttons, to verify the emergency stop disabled the cradle drive.

6.) Press one of the **RESET** buttons to turn on X-RAY DRIVES POWER. (120 VAC LED stops flashing.)

7.) Press the other E-STOP button on the gantry.
   a.) Make sure the Table Power shuts off.
   b.) Manually move the cradle to the home position to make sure the cradle clutch released.
   c.) Make sure the cradle latches securely in the home position.

8.) Press one of the **RESET** buttons to turn on X-RAY DRIVES POWER.

9.) Press one of the four table tape switches to make sure the table down motion stops. Repeat with the three remaining table tape switches.

10.) Press the console emergency stop switch; make sure the Table Power shuts off.

11.) Press one of the **RESET** buttons to turn on X-RAY DRIVES POWER. (See Figure 4-20).

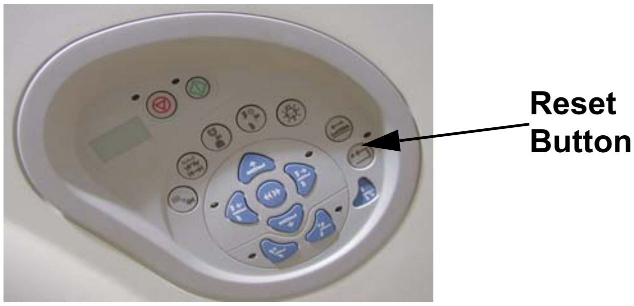



**Figure 4-20 Reset buttons on Gantry and Service Switch bank.**

Note: Emergency Stop buttons are located on the front and rear of the gantry (8 in all), as noted in Figure 4-21. They are also located on both sides of the table base (4 in all). Additionally, an emergency stop button is provided on the Operator Console SCIM (see Figure 4-22).


**Figure 4-21 Gantry Emergency Stop Button Positions**

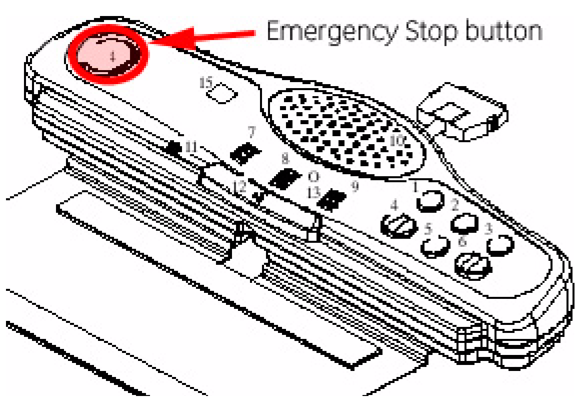



**Figure 4-22 SCIM Emergency Stop Button**

# Section 6.0 Mechanical Installation Completion Checklist

### System-Level

- ❑ FE Service cabinet moved to the location shown on the site print
- ❑ All covers installed and aligned
- ❑ All options installed on the table and gantry
- ❑ All packing materials and boxes returned to the lean cart
- ❑ All service items placed in the service cabinet.

### Optional and Regional

- ❑ Seismic mounting installed, if required in your area.

### Site Clean Up

- ❑ All customer items placed on a cabinet or on a counter and labeled customer material.
- ❑ All system service tools placed in the GE service cabinet.
- ❑ System software and options left on the lean cart in the software tray
- ❑ System cleaned and nicks repaired
- ❑ Installation site cleaned and all trash properly disposed.

### Dolly Return

- ❑ Return of dollies and lean carts arranged for and pick-up made.

### Paperwork

- ❑ Mechanical installation section of the GE Form e4879 completed
  - ✓ Room information recorded on the GE Form e4879
  - ✓ Table gantry alignment completed per the installation manual
  - ✓ Table gantry anchoring completed per the installation manual
- ❑ GE Healthcare personnel notified that the mechanical installation is completed
- ❑ All installation issues have been addressed and or documented so FE can follow-up as needed.

Note: This is the END of the Mechanical Installation.

# *Appendix A*
# *Gantry Cover Removal and Dolly Setup*

## Section 1.0 Gantry Cover Removal

### 1.1 Time & Personnel

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 2 (FE or mechanical supplier) | | | |

### 1.2 Tools and Test Equipment

- Front and rear cover dollies

### 1.3 Procedures

#### 1.3.1 Scan Window Removal

1.) Grip the window by the metal band at the top and pull firmly but carefully downward.
2.) Continue to pull until the top of the scan window makes contact with the bottom portion of the scan window.

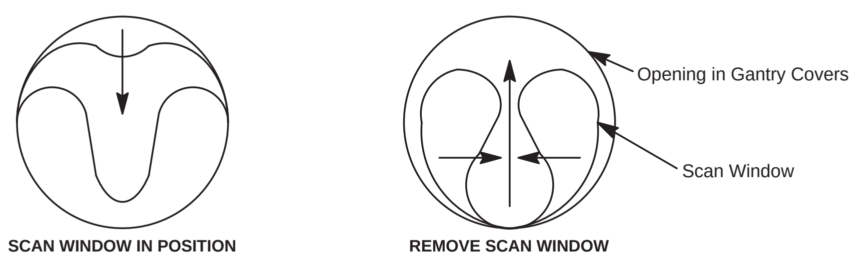



**Figure A-1 Scan Window Removal**

3.) Hold the scan window as shown in Figure A-2 and remove it from between the front and rear covers.

**Figure A-2 Removing Gantry Scan (Mylar) Window**

4.) To prevent damage, place the scan window in a secure place.

### 1.3.2 Side Cover Removal

1.) Lower table to the home (lowest) position.

**CAUTION**

**Potential for injury if the covers are removed and power is "ON".**

**Always remove the right side cover first, and turn OFF power at the Service Switch Panel.**

2.) Use an 8 mm Hex wrench to unlatch the side cover from the front cover. See Figure A-3.

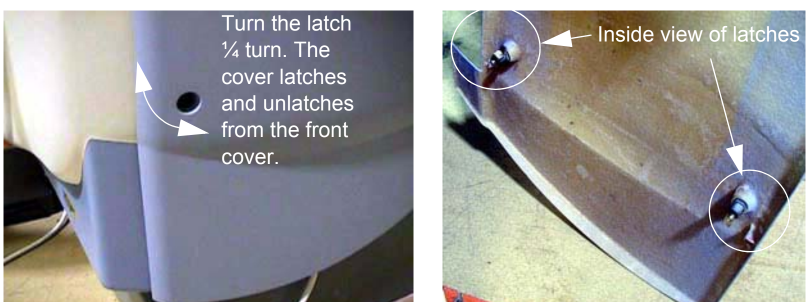



**Figure A-3 Side Cover Latches**

3.) Remove the right side cover by lifting it upward to release the two (2) latches, located on the top edge of the cover. Once removed, the MSUB/TGP left side should be exposed.

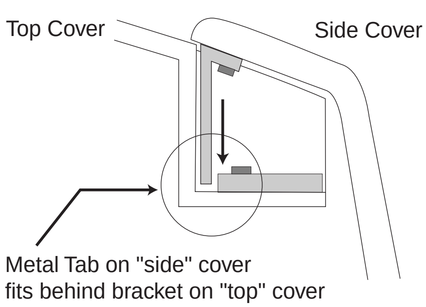



**Figure A-4 Side and Top Cover Clasp**

### 1.3.3 Top Covers Removal

1.) On a new installation, the top covers are shipped ty-wrapped in place. Use a pair of side cutters to remove them, and discard the ty-wraps.
2.) The top cover fans are connected to the gantry using a Molex connector. Locate and disconnect the fan power plug.
3.) Take the end of the top cover closest to you and tilt upwards.
4.) Grip the cover on both sides. The top covers have edge guides that center the cover on the raised ridges of the front and rear covers. This allows the cover's tab to disengage from the mounting bracket (see Figure A-3).
5.) Slide the cover down using the gantry front and rear covers as support until the cover is at chest level.

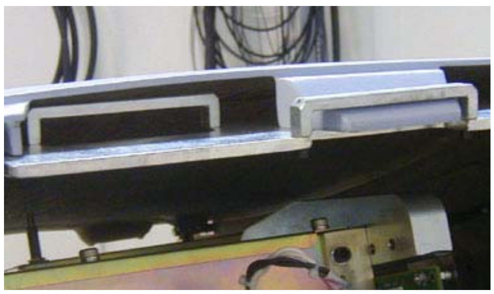


**Figure A-5 Top cover tabs and bracket**

6.) Lift the cover clear, then place the flat end on the floor.
7.) Repeat steps 1-6 for the other top cover.

### 1.3.4 Gantry Front Cover

**NOTICE** **Potential for cover damage.**

**Front and rear cover removal and installation can be safely accomplished by one (1) person using the dollies provided with the system. Failure to use these dollies significantly increases the likelihood of damage to the covers. Do not lean the covers against the walls.**

**FRONT COVER DOLLY SETUP**

**DANGER** **DO NOT USE DOLLIES ON UNEVEN SURFACES SUCH AS STEPS OR ELEVATOR THRESHOLDS. THE DOLLIES ARE DESIGNED FOR FLAT, LEVEL FLOORS WITHIN THE SCANNING SUITE ONLY. MISUSE CAN RESULT IN PERSONAL INJURY OR DAMAGE TO COVERS OR OTHER FACILITY ITEMS.**

**CAUTION** **Rotating arms on the stand are supposed to be stiff. If they fall freely, tighten the tensioning nuts. Loose rotating arms reduce the stability of the dollies when supporting the front cover. Do not lubricate.**

1.) Arrange dolly sections for assembly. The base and post can be assembled only one way. Refer to Figure A-6 and Figure A-7.
- The base uses two (2) palm screws to clamp the four (4) legs in the open (usage) mode.
- The base also uses the same palm screws to prevent the legs from falling in storage mode.
- The top post can be inserted in either base and is keyed for proper engagement.
- The top post locking pin prevents the sections from separating during use.

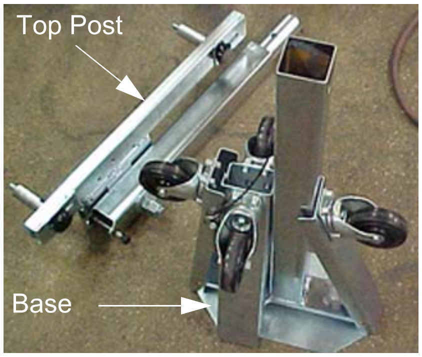



**Figure A-6 Front Cover Dolly in Storage Mode**

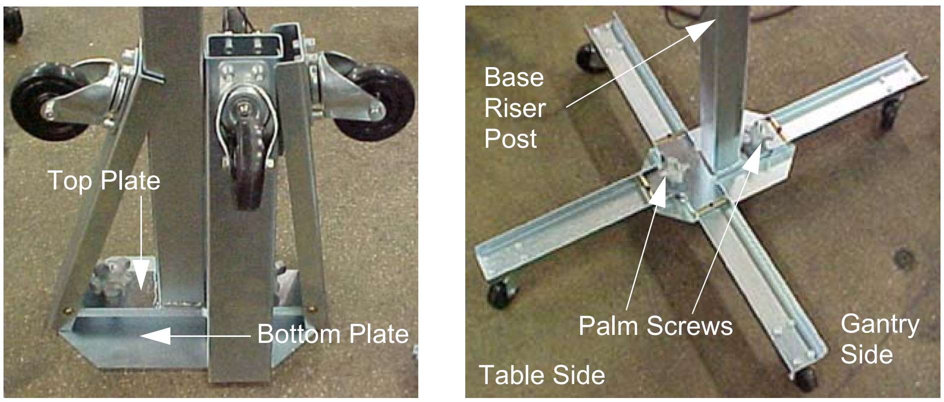



**Figure A-7 Front Cover Dolly Base Assembly**

2.) Unfold the base legs by loosening both palm screws to the top of their travel.

3.) Carefully unfold the legs so that the castors touch the floor.

4.) Tighten the palm screws to clamp the legs between the base top and bottom plates.

Note: Lifting the base by the riser post while leaving the castors on the floor eases palm screw tightening (see Figure A-7).

**WARNING**

**ENSURE BOTH PALM SCREWS ARE TIGHTENED SECURELY AND THE LEGS ARE CLAMPED TIGHTLY BETWEEN THE BASE TOP AND BOTTOM PLATES. FAILURE TO DO SO RESULTS IN INSTABILITY DURING FRONT COVER HANDLING.**

5.) Insert the top post into the base riser post. Align the key for complete engagement.

6.) Insert top post locking pin to secure both top and bottom sections.

7.) Reverse above steps to disassemble.

Note: For base storage, only one (1) palm screw needs to be tightened. This engages the bottom base plate and the leg ends, preventing the legs from unfolding during transport and storage.

## REMOVAL

8.) Make sure that the three (3) power switches have been turned off.

9.) Assemble the front cover dolly.

a.) Tighten the two (2) shoulder bolts to the gantry securely. This makes cover installation easier (see Figure A-8).

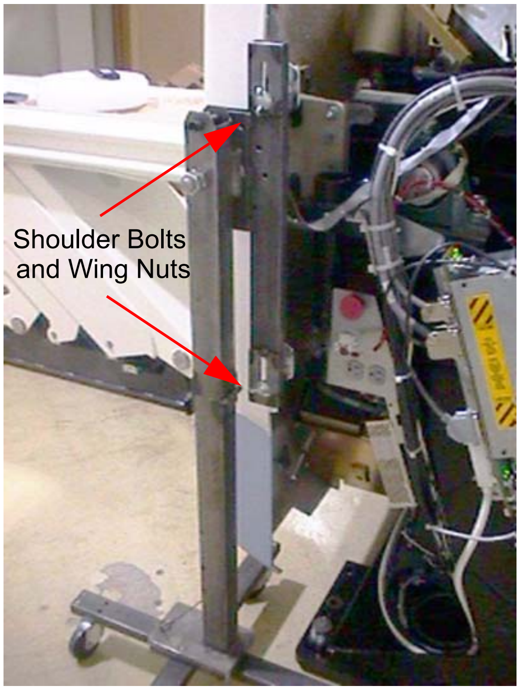



**Figure A-8 Front Side Dolly**

b.) Attach the side dolly to the shoulder bolts and secure assembly with two (2) wing nuts.

c.) Repeat steps a and b to assemble the other side dolly.

10.) Detach the front cover cables J3 and J2 and BKHD J1.

11.) Remove the front cover:

a.) Disengage the upper and lower Cantrell brackets on both sides of the cover.

1.) Using steady but firm pressure, lift each of the lower Cantrell brackets from their associated retainers (see Figure A-9).

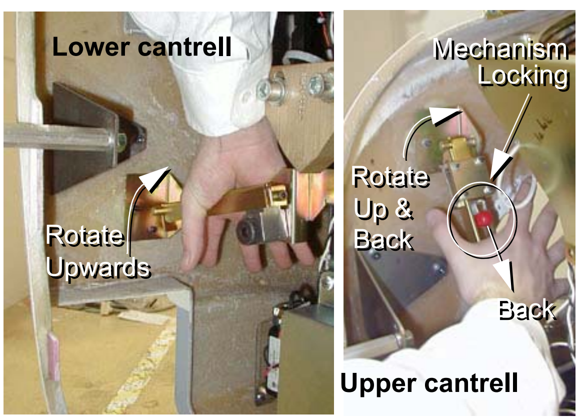



**Figure A-9 Releasing cover brackets**

2.) Disengage the locking mechanism on the upper Cantrell brackets by using your thumb to slide the trigger (red lever) back. This releases the locking mechanism and allows you to rotate the Cantrell bracket upward with steady and firm pressure.

b.) Disengage the rubber retaining straps on both sides.

c.) Lift and rotate the cover locking arm to the unlocked position.

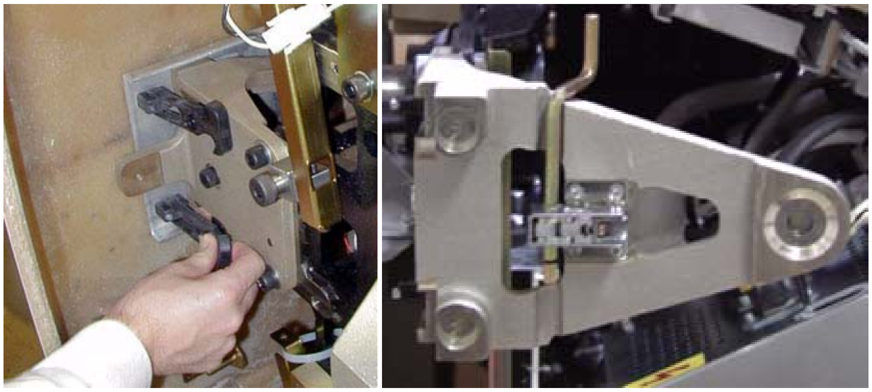


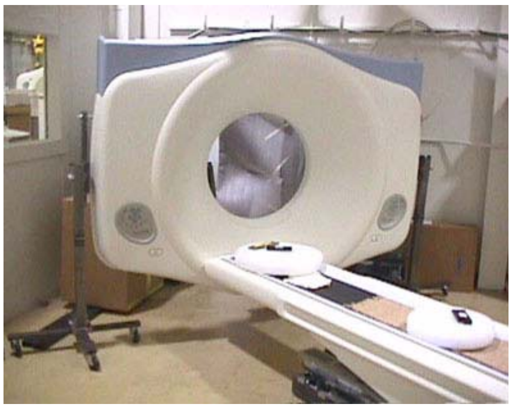
**Figure A-10 Rubber retaining straps and Cover Locking mechanism**

12.) Rotate the front cover away from the gantry.

a.) Move the front cover away from gantry giving enough space (about 1524 mm [5 ft]) between the front cover and the gantry.

b.) Pull the locking pin and rotate the front cover away from the gantry. Place the locking pin in one of the side dolly perforations (see Figure A-11).


**Figure A-11 Releasing Front Cover Dolly Hinge**



1. **Upper Left:** Rotate cover to make room for step 2.
2. **Upper Right:** Rotate cover to clear the table. Roll the cover to foot end of table.
3. **Lower Left**: Rotate the cover upside down to provide clear work area.
   Note: Remove Gantry Display and Control Panel in position 2.

**Figure A-12 Front Cover Removal Sequence**

13.) Rotate the cover horizontally and move it back and over the table to a safe location. Once in a safe location, you may over-rotate the cover full vertically but upside down.

## 1.4 Gantry Rear Cover Removal

1.) Assemble the rear cover dolly.

a.) Tighten the two (2) shoulder bolts to the rear cover.


**Figure A-13 One side of the Rear cover dolly**

b.) Fit the side dolly through the shoulder bolts and secure the assembly with two (2) wing nuts (see Figure A-13).

c.) Repeat steps a and b for the other side dolly.

**CAUTION** **Potential for injury if the covers are removed and power is ON.**

2.) Disconnect cables on the right side of the rear cover.

3.) Remove rear cover.

a.) Disengage the upper and lower Cantrell brackets on both sides of the rear cover.

1.) Using steady but firm pressure, lift each of the lower Cantrell brackets from their associated retainers (see Figure A-9).

2.) Disengage the locking mechanism on the upper Cantrell brackets by using your thumb to slide the trigger (red lever) back. This releases the locking mechanism and allows you to rotate the Cantrell bracket upward with steady and firm pressure.

b.) Disengage the rubber retaining straps on both sides.

# Section 2.0 CT UMI Tilting Dolly

## 2.1 Installation Procedure

1.) Remove dolly side rails.
2.) Remove two inboard caster towers, and install and pin them at gantry end of dolly.
3.) Elevate all caster towers until their casters contact the floor.
4.) Remove gantry end HSA dolly end, replace dolly pins in the frame receptacles, retain HSA dolly end.
5.) Retrieve and attach pickers to gantry end of table with 16 millimeter (mm) by 40 mm cap screws.
6.) Remove and retain spacers and nuts from 16mm x 240 mm SHCS. Retain for return.
7.) Remove foot end HSA dolly end, replace dolly pins in the frame receptacles, retain HSA dolly end.
8.) Remove and retain 5/16 inch diameter pivot lock pin. Note that this pin is slightly smaller than all other pins used on the dolly, and the only pin that will work well at the pivot.
9.) Remove and retain the sheet metal component at each side of the base weldment and both screws.
10.) Pivot frame open and surround table, maintaining approximate equal clearance between the frame and table on each side. The caster towers may require adjustment to allow ease of movement.
11.) Pivot frame back to approximately square.
12.) Attach and pin lifting arms to pickers.
13.) Carefully and squarely align the 16 mm x 240 mm SHCS to the table. Caster tower adjustments may be required.
14.) Thread the 16 mm x 240 mm SHCS into table until their threads extend inboard of the base weldment.
15.) Remove side cover bracket.
16.) Attach and pin arm separator in place.
17.) Insert pivot lock pin.
18.) Attach and pin HSA dolly ends. Some height adjustments may be required. Insure dolly pins are fully seated.
19.) Remove, stow, and pin caster towers. They may need adjustments to ease pin insertion.
20.) Attach and pin side rails. Be alert for pinch points of fingers between the side-rail and lifting arm. Best results are often achieved by pinning one side-rail before inserting the second.
21.) Manually rotate jack screw to take up play at pickers.
22.) Elevate HSA dolly ends for transport.

## 2.2 Tilting the Dolly

1.) Verify the blue handle/foot end is 14.5 inches away from the rectangular tunnel in the leg. If not:
   a.) Remove the shroud on the patient's left side
   b.) Detach the transit arm travel limiter
   c.) Verify the table transit lock is positioned correctly
   d.) Reattach the shroud on the patient's left side

2.) Lower HSA dolly ends until table is in contact with floor.
3.) Unpin, remove, and retain side rails, replace pins place pins in a convenient location on HSA dolly end.
4.) Attach and pin all caster towers in active positions. Height adjustments may be required to ease pin insertion.
5.) Elevate caster towers until their casters are in contact with floor.
6.) Remove HSA dolly ends and replace dolly pins in receptacles in frame.
7.) Elevate caster towers to appropriate height. A good target is to elevate until the dolly pins clear the floor, and are supported by their rings.
8.) Rotate jackscrew to tilt table.
    a.) Regularly observe the clearance between the table and the floor; adjust caster towers to maintain an effective distance of approximately one inch, depending on the surfaces and thresholds to be traversed.
    b.) Regularly observe the clearance between the pivot pin of the frame and the table. Excessive rotation of the table may cause the rounded “sweep” of the table to contact the frame, and subsequently damage the table.

# Section 3.0 Gantry Auxiliary (Mini) Dolly Installation

## 3.1 Time & Personnel

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|---|---|---|---|
| 2 (FE or mechanical supplier) | | | |

## 3.2 Tools and Test Equipment

•

## 3.3 Procedures

1.) At each corner, select one of the three open holes.

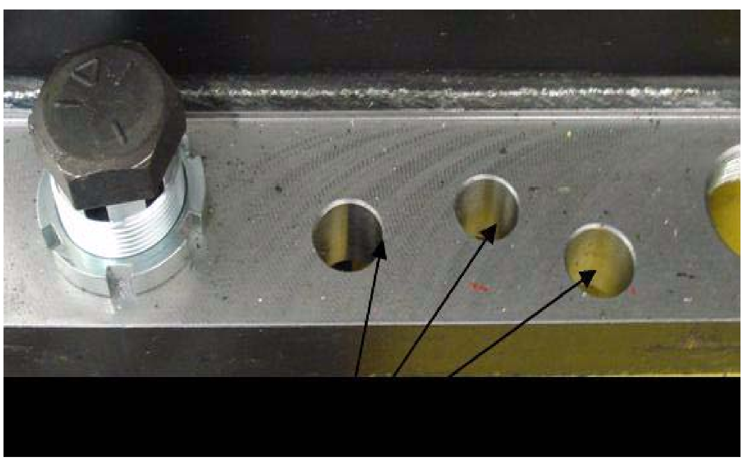


**Figure A-14 Select an open hole**

2.) Using the blue dollies, raise the gantry high enough to install a wheel at each corner.
3.) Insert the brass bushing into the selected open corner hole.

**Figure A-15 Insert the brass bushing**

4.) Install the wheel bolt through the brass bushing and the gantry base.

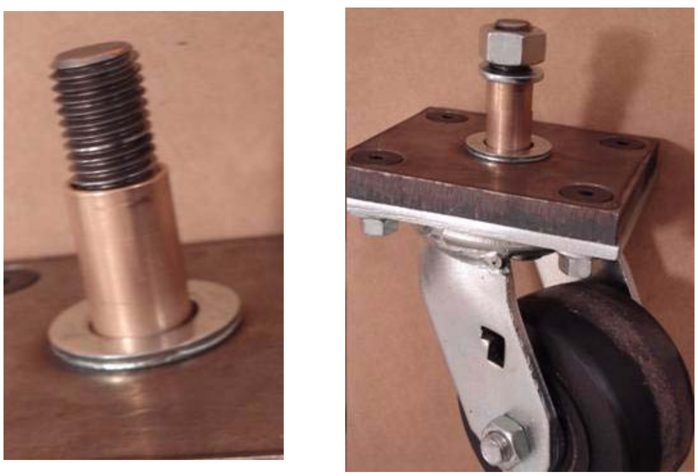


**Figure A-16 Wheel bolt through brass bushing (left) and through gantry base (right)**

5.) Tighten the nut assembly.

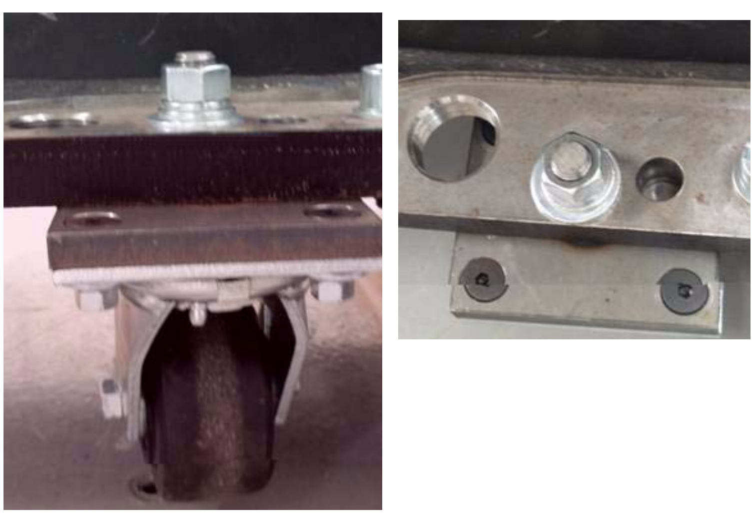


**Figure A-17 Tighten nut (two views)**

6.) Repeat steps 1-5 for each remaining corner.

7.) Lower and remove blue dollies.

## INSTALLING THE BUMPER BRACKET

1.) Attach the bumpers in the same holes as the blue dollies.

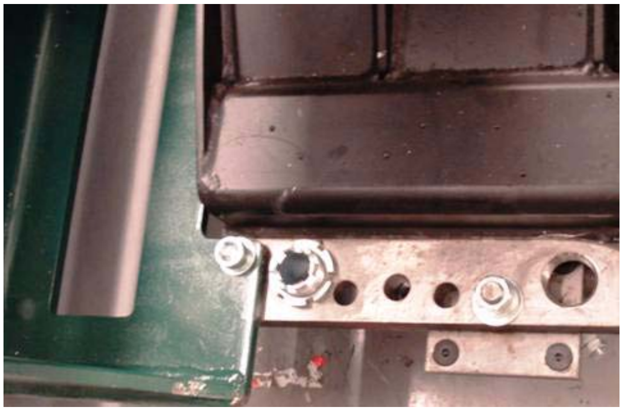


**Figure A-18 Position the bumper bracket**

2.) Tighten all mounting hardware.

**Figure A-19 Install bolts and tighten**

Note: Bumpers protect the gantry from damage and should always be used when moving the gantry into elevators.

## INSTALLING THE HANDLES

1.) The mounting plate uses the same holes as the blue dollies.

2.) Attach the handles to the plate as shown in Figure A-20.

**Figure A-20 Position the mounting plate and handles**

3.) Attach the handles to the plate as shown both Figure A-20 and Figure A-21.

**Figure A-21 Install bolts and tighten**

4.) Tighten all mounting hardware.

Note: Use the handles to push, pull, and maneuver the gantry in tight corridors or when pushing the gantry into an elevator.

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# Appendix B
# Pictorial Representation of Required Tools

Use the following guide as a reference, if you are unsure of a tool listed in Section 2.6, on page 32.(Continued)

| Tool Name | Picture | Example Part Number* |
|---|---|---|
| Adapter |  | Sears Industrial: $^{3}/_{8}$" to $^{1}/_{2}$" (9-4258) |
| Ball-Peen Hammer |  | Sears Industrial: 1lb/2lb (9-38465) |
| Canned Air |  | Miller Stephenson: Aero Duster (MS-222N) |
| Clamp on Amp Meter |  | Sears Industrial: 9-WTAD105 |
| Combination Wrench Set |  | Sears Industrial: U.S. Standard & Metric (9-44048) |
| Cordless Screwdriver |  | Sears Industrial: 9-MU65401 |
| Deep Well Socket |  | Sears Industrial: $^{3}/_{4}$" X $^{3}/_{8}$" (included with 9-34496) |
| Dental Pick |  | |

* Part Numbers given for reference only. GE Healthcare does not endorse any tool brand name.

**Table B-1 Required Tools**

| Tool Name | Picture | Example Part Number* |
|---|---|---|
| Diagonal Cutting Pliers |  | Sears Industrial: Small (9-45077) |
| Drill |  | Sears Industrial: $^3/_8$" or $^1/_2$" (9-27859) |
| Drill Adapter |  | Sears Industrial: 3" X $^3/_8$" (9-APSZ24) |
| Drill Bit Set |  | Sears Industrial: U.S. Standard (9-66084) |
| DVM |  | Sears Industrial: 9-82028<br>Sears Industrial: 9-FL873 |
| Extension for Ratchet Wrench |  | Sears Industrial: 3" X ½" (9-44133) |
| Gloves |  | Sears Industrial: Large (9-40502) |
| Hammer Drill |  | Sears Industrial: ½" (9-27205) |
| Hex Bit Set |  | Sears Industrial: ¼" (9-SK45508) |

* Part Numbers given for reference only. GE Healthcare does not endorse any tool brand name.

**Table B-1 Required Tools (Continued)**

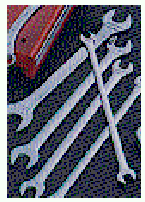
| Tool Name | Picture | Example Part Number* |
|---|---|---|
| Hex Key (Allen Wrench) Set |  | Sears Industrial: U.S. Standard (9-46284) |
| Level |  | Sears Industrial: 4’ (9-39856) |
| Masonry Bit |  | |
| Open-End Wrench (Thin or Standard Tappet) |  | Snap-on: 10mm (SRSM10) & 21mm (LTAM2124) |
| Pozi Screwdriver |  | |
| Ratchet Wrench |  | Sears Industrial: $^{3}/_{8}$" (9-43175) |
| Reciprocating Saw with Blades |  | Sears Industrial: 9-MU650921 |
| Safety Glasses |  | Sears Industrial: 9-18650 |
| Safety Shoes |  | |
| Screwdriver Set |  | Sears Industrial: Phillips & Straight (9-41505) |
| * Part Numbers given for reference only. GE Healthcare does not endorse any tool brand name. | | |

**Table B-1 Required Tools (Continued)**

| Tool Name | Picture | Example Part Number* |
|---|---|---|
| Socket Set |  | Sears Industrial: Standard $^{3}/_{8}$" (9-34496) |
| Sockets |  | Sears Industrial: 1 $^{1}/_{8}$" X $^{1}/_{2}$" (9-47516) |
| Step Ladder |  | Sears Industrial: 6’ (9-WN6006) |
| Tongue & Groove Pliers |  | Sears Industrial: Large (9-CL440) |
| Torpedo Level |  | Sears Industrial: 9" (9-39829) |
| Torque Wrench |  | Sears Industrial: $^{3}/_{8}$" (9-WR3470) |
| Universal Joint |  | Sears Industrial: $^{3}/_{8}$" (9-4435) |
| Vacuum Cleaner |  | Sears Industrial: 8 Gal (9-17780) |
| * Part Numbers given for reference only. GE Healthcare does not endorse any tool brand name. | | |

**Table B-1 Required Tools (Continued)**

# Appendix C Regulatory Clearance Quick Reference Guide

## Section 1.0 Regulatory Code Description

**Egress**: 29 CFR 1910 Subpart E (OSHA) and NFPA 101 (Life Safety Code) define the minimum requirements for means of egress. The requirement most applicable to equipment installation and room layout is minimum width of exit access. Under OSHA 1910.37(f)(6), the minimum width of exit access shall in no case be less than 711 mm (28 in) from any potentially occupied point in the room.

Under NFPA 101 (2006 edition) 7.3.4.1, the minimum width of any means of egress is 914 mm (36 in). However, NFPA allows this to be reduced to 711 mm (28 in) around furniture or equipment, provided that a 914 mm (36 in) clearance would otherwise be available without moving permanent walls.

**Electrical Clearance**: 29 CFR 1910 Subpart S (OSHA) and NFPA 70E (Standard for Electrical Safety in the Workplace) define minimum clearance requirements for the workspace around electrical equipment. Under both OSHA 1910.303(g)(1) and NFPA 70E (2004 edition) 400.15, a minimum clear space of 914 mm (36 in) depth (with minimum 762mm (30 in) width and 1981 mm (78 in) height) must be provided in front of electrical equipment with parts operating at 600 volts or below and likely to require examination, adjustment, servicing, or maintenance while energized.

This safety clearance requirement applies to all GEHC equipment. Although 914 mm (36 in) is the minimum clearance for most installations, the standards require an increased minimum clearance distance where parts operate above 150 volts (but still below 600 volts) under the following circumstances:

- If the wall or surface directly facing the electrical equipment is grounded (e.g. brick, concrete, or tile) or includes grounded protrusions (such as medical gas ports, metal door or window frames, water sources and metallic sink structures, metallic cabinetry, electrical disconnects or emergency off panels, air conditioners or vents), then a 1067mm (42 in) clearance depth is required.
- If the possibility exists of exposed and unguarded live parts on both sides of the workspace (for example if a power distribution unit were positioned on the wall directly facing the GEHC equipment), then a 1219 mm (48 in) clearance depth is required.

# Section 2.0 Terms and Definitions

**Egress**: The path of exit from within any room. U.S. regulations require a minimum of 711 mm (28 in) of continuous and unobstructed space, including trip hazards along the path of exit.

**Workspace**: The dimensional box required for safe inspection or service of energized equipment. It consists of depth, width, and height. The depth dimension is measured perpendicular to the direction of access. The U.S. regulation minimum is 914 mm (36 in), but additional conditions can increase the minimum dimension requirement. GE defines this as the envelope of the component superstructure with the external covers in place.

**Service Access Width**: The width of the workspace in front of the equipment. A minimum of 762 mm (30 in), or the width of the equipment, whichever is greater.

**Head Clearance**: The height dimension of the workspace. The height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s), 1981 mm (78 in) or the height of the equipment, whichever is greater.

**Grounded Wall**: Any wall that can be electrically conductive to earth ground. Masonry, concrete, and tile are considered conductive. Additional commonly found aspects of a wall should also be considered grounded. This is not an all-inclusive list:

- Medical gas ports and plates
- Metal doors and window frames
- Water sources and metallic sink structures
- Metallic wall-mounted cabinets
- A1 main disconnect panel
- Equipment Emergency Off panels
- Industrial equipment (such as air conditioners and vents)
- Expansion joints
- Surface raceway
- Exposed wall conduits
- Floor outlet boxes
- Floor HVAC boxes
- Floor medical gas

Common wall components NOT constituting grounded elements include:

- Standard wall outlet
- Light switches
- Telephones
- Communication wall jacks
- Ceiling tile grids

# Section 3.0 Minimum Regulatory Working Clearance by Major Subsystem

Note the following when referring to the tables below:

- These requirements apply to equipment operating at 600 V or less, where examination, adjustment, servicing, or maintenance is likely to be performed with live parts exposed.
- The customer MUST maintain the required regulatory clearance distances and may NOT use these areas for storage. This applies during normal system operation as well as during service inspection and maintenance.
- Direction of Service Access refers to a direction perpendicular to the surface of the equipment serviced.

| Workspace Requirement | Minimum Clear Space | Additional Conditions |
|---|---|---|
| Direction of service access: Front and rear console | N/A<br>(No exposed live part hazards) | |
| Service access width:<br>(Front-Back or Workspace) | | Refers to the width of the working space in front of equipment. 762 mm (30 in.) min or the equipment width, whichever is greater. |
| Head clearance | 1981 mm (78 in.) | Refers to the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or any overhead obstructions. 1981 mm (78 in.) or the height of the equipment, whichever is greater. |
| Note: Distances are measured to the finished covers. | | |

**Table C-1 Console Subsystem**

| Workspace Requirement | Minimum Clear Space | Additional Conditions |
|---|---|---|
| Direction of service access: Front of PDU | 914 mm (36 in.)* | 1219 mm (48 in.) if exposed live parts of 151 - 600 volts are present on both sides of the workspace with the operator between.<br>1067 mm (42 in.) if the opposite wall is grounded and exposed live parts of 151 - 600 volts are present. |
| Service access width: Left-Right of workspace | 762 mm (30 in.) | Refers to the width of the working space in front of equipment. 762 mm (30 in.) min or the equipment width, whichever is greater. |
| Head clearance | 1981 mm (78 in.) | Refers to the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or any overhead obstructions. 1981 mm (78 in.) or the height of the equipment, whichever is greater. |
| Note: Distances are measured to the finished covers. | | |

**Table C-2 NGPDU – Minimum Workspace Clearances**

| Workspace Requirement | Minimum Clear Space | Additional Conditions |
|---|---|---|
| Direction of service access: All sides | 914 mm (36 in.)* | 1219 mm (48 in.), if exposed live parts of 151 - 600 volts are present on both sides of the workspace with the operator between.<br>1067 mm (42 in.), if the opposite wall is grounded and exposed live parts of 151 - 600 volts are present. |
| Service access width: Left-Right of workspace | 762 mm (30 in.) | Refers to the width of the working space in front of equipment. 762 mm (30 in.) minimum or the equipment width, whichever is greater. |
| Head Clearance | 1981 mm (78 in) | Refers to the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or any overhead obstructions. 1981 mm (78 in.) or the height of the equipment, whichever is greater. |
| Note: Distances are measured to the finished covers. | | |

**Table C-3 Gantry –Minimum Workspace Clearances**

| Workspace Requirement | Minimum Clear Space | Additional Conditions |
|---|---|---|
| Direction of service access: Table head or foot | N/A | |
| Direction of service access: Table sides | 914 mm (36 in.)* | *Can reduce to 711 mm (28 in.), provided the local team obtains written and signed approval from the local AHJ (Authority Having Jurisdiction). GE must have the signed document on file. |
| Direction of service access: Table foot<br>[CT ONLY] | 711 mm (28 in.)* | 457 mm (18 in.) minimum for Front Gantry Cover removal, only if an unobstructed egress space of 711 mm (28 in.) exists around the equipment for room exit, and no trip hazards exist along the path of egress. |
| Service access width: Left-Right of workspace | 762 mm (30 in.) | Refers to the width of working space in front of equipment. 762 mm (30 in.) minimum or the equipment width, whichever is greater. |
| Note: Distances are measured to the finished covers. | | |

**Table C-4 Table –Minimum workspace Clearances**

# Section 4.0 Room Dimensions

| System Configuration | Suggested Room Size | Typical Room Size | Minimum Room Size |
|---|---|---|---|
| VCT or CT750 HD with GT 1700 | 7772 mm x 4420 mm[1] (25 ft., 6 in. x 14 ft., 6 in) | .6096 mm x 4115 mm[1] (20 ft. x 13 ft., 6 in) | 6096 mm x 3556 mm[2] (20 ft. x 11 ft., 8 in) |
| VCT or CT750 HD with GT 2000 | 7772 mm x 4420 mm[1] (25 ft., 6 in. x 14 ft., 6 in) | 6706 mm x 4115 mm[1] (22 ft. x 13 ft., 6 in) | 6706 mm x 3556 mm[2] 22 ft. x 11 ft., 8 in. |

[1]All regulatory requirements apply.

[2]All regulatory requirements apply, with the addition of no energized left-side service.

**Table 4-2 Suggested Room Size Dimensions**

## 4.1 Suggested Room Size

The suggested room configuration offers the most flexibility for future upgrades. It provides both ample workspace and space to add millwork and still meet all regulatory requirements. When local regulations require a sink in the scan room, this room size also provides sufficient space for a sink. This room size accommodates the needs of larger hospitals and medical teaching facilities, where patients may require transportation to the scan area in beds, gurneys, and larger wheelchairs and where they may require the assistance of larger medical care teams. Likewise, this room offers adequate access for crash carts and other emergency medical equipment on both sides of the table.

The suggested size supports all service activities, including tube change, and accommodates all future two-step installations.

## 4.2 Typical Room Size

The typical room configuration represents what is most commonly found at installation sites. It offers adequate workspace and may provide adequate space for a sink, but allows only LIMITED space to add millwork and still meet all regulatory requirements. The size of this room accommodates the needs of larger clinics and medium-sized hospitals, where patients may require transportation into the scan area using gurneys and wheelchairs, and where they may require the assistance of small medical care teams. It provides access for crash carts and other emergency medical equipment on only one side of the table.

The typical room size supports all service activities, including tube change, and may offer compatibility with SOME future upgrades and two-step installations.

## 4.3 Minimum Room Size (Limited Access)

The minimum room configuration represents the smallest functionally acceptable space for this product and represents the type of room often found at doctor’s offices and smaller clinics and outpatient facilities. Due to its limited size, and to functional and regulatory requirements, this room usually provides only LIMITED workspace, and leaves to NO space to add in-room millwork and sinks and still meet the necessary regulatory and service requirements. This room can accommodate the transportation of patients into the scan area using wheelchairs, and provides access for crash carts and other emergency medical equipment on only one side of the table.

Sites considering a minimum room size may not have been designed with the structural requirements necessary to support the system and consequently may require upgrading prior to installation.

Customers considering a minimum room size should discuss their workspace requirements and future upgrade plans with their PMI, as the size and layout of these rooms often eliminates them from any future upgrade considerations and offers NO compatibility with future two-step installations.

If using the square meters (square footage) to determine regulatory compliance, please note that the front and rear cover clearances are wider than the regulatory clearance along the table length, and that the cover park position is behind the table in the home position.

Note: Sites must provide sufficient space to allow the removal of the rear cover, which is on wheels, from behind the gantry during service operations.

## 4.4 Regulatory Caution

Site prints are required for all system installations including relocation and moves. CT room layout as shown on your site print shall meet all regulatory requirements as described in the installation manual. Additional room components such as cabinets reduce room size. Equipment not shown on the site print may void the caution statement, making the room non-compliant. Actual site measurements before installation will be taken to determine room size and compliance.

For minimum room size sites, where the gantry distance between the wall and side cover is between 357 mm – 686 mm (14 in. – 27 in.) the left side cannot be considered or used for egress. In this configuration, egress around the left side of the gantry will be restricted. Unobstructed egress route shall be provided around the table foot end or behind the back of the gantry. If millwork is required or will be installed later, consider room placement locations that allows for regulatory compliance and clearances.

## 4.5 Operational Caution

In a minimum room layout (356 mm to 686 mm [14 in. to 27 in.]) the customer should consider workflow, customer access for patient care, and critical-care operations space requirements. Additionally, this room provides only limited equipment access on the gantry left side when loading patients or when positioning patient equipment in the room between the gantry and the wall.Detailed customer installation tasks are detailed in the product PIM. Chapters 1-4.

# Section 5.0 Egress

An egress is the single path of exit from within any room. Egress must conform to these requirements:

- OSHA requires a minimum width of 711 mm (28 in.) of continuous and unobstructed space, free of trip hazards, along the entire path of the exit.
- The path of exit may contain ONLY OSHA-approved ramps.
- All doors used to improve egress MUST lead to an exit.
- The path of exit CANNOT include the surface floor raceway in areas that would require lifting to remove the front or rear gantry covers for service.
- There must be a clear, unobstructed route out of the room, either around the back of the gantry or around the back of the table.
- Determine the amount of clearance by measuring from the table center line out to any obstruction.

Note: Gantry front cover removal requires the use of tilting dollies that move side-to-side to reach a park position of 457 mm (18 in.) at the foot of the table. If the planned egress route is not at the back of the table, ensure that this distance exists at the back of the table to park the gantry front cover. Maintain a continuous width of 3200 mm (126 in.), equaling 1600 mm (63 in.) clearance on each side of the table to allow removal of the front cover.

If not planning to park the gantry front cover at the back of the table, be sure to maintain at least 305 mm (12 in.) of clear space along the back of the table to allow movement of the gantry front cover along the back of the table.

# Section 6.0 How to Measure

Figure 4-23 offers guidance on the proper way to measure to check minimum regulatory clearances

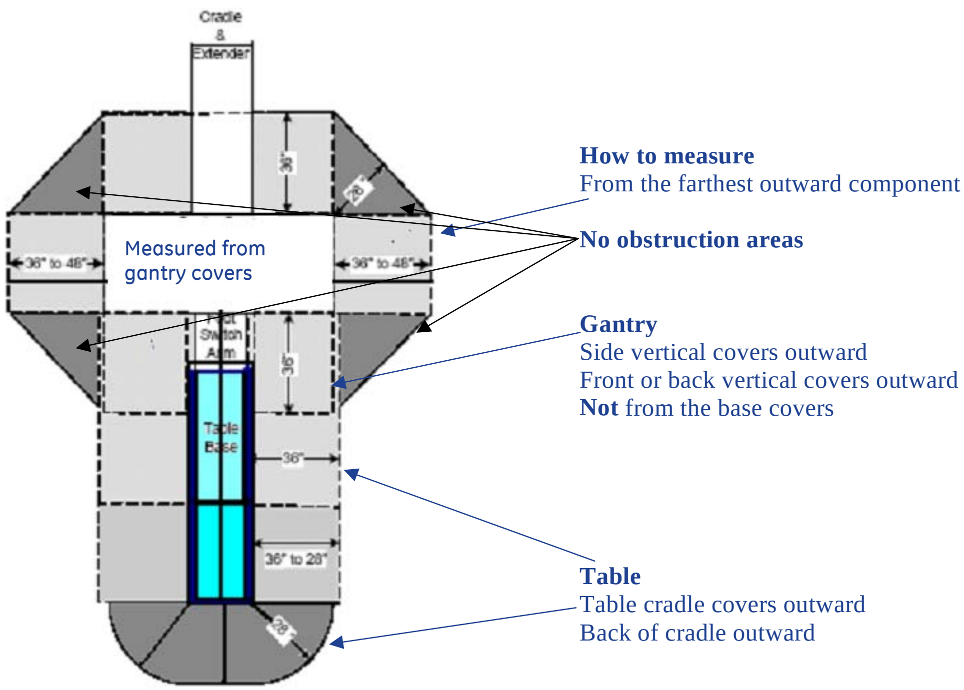



**Figure 4-23 Measuring Minimum Regulatory Clearances**

# Section 7.0 Minimum Room Layouts

Note: For minimum room size layouts, the *short left side* catalogue number is B7700SA.

**Room A** provides less than 914 mm (36 in.), but greater than 711 mm (28 in.) of space, measured from covers to the left-side wall; it offers sufficient service, egress, and workspace around the gantry.

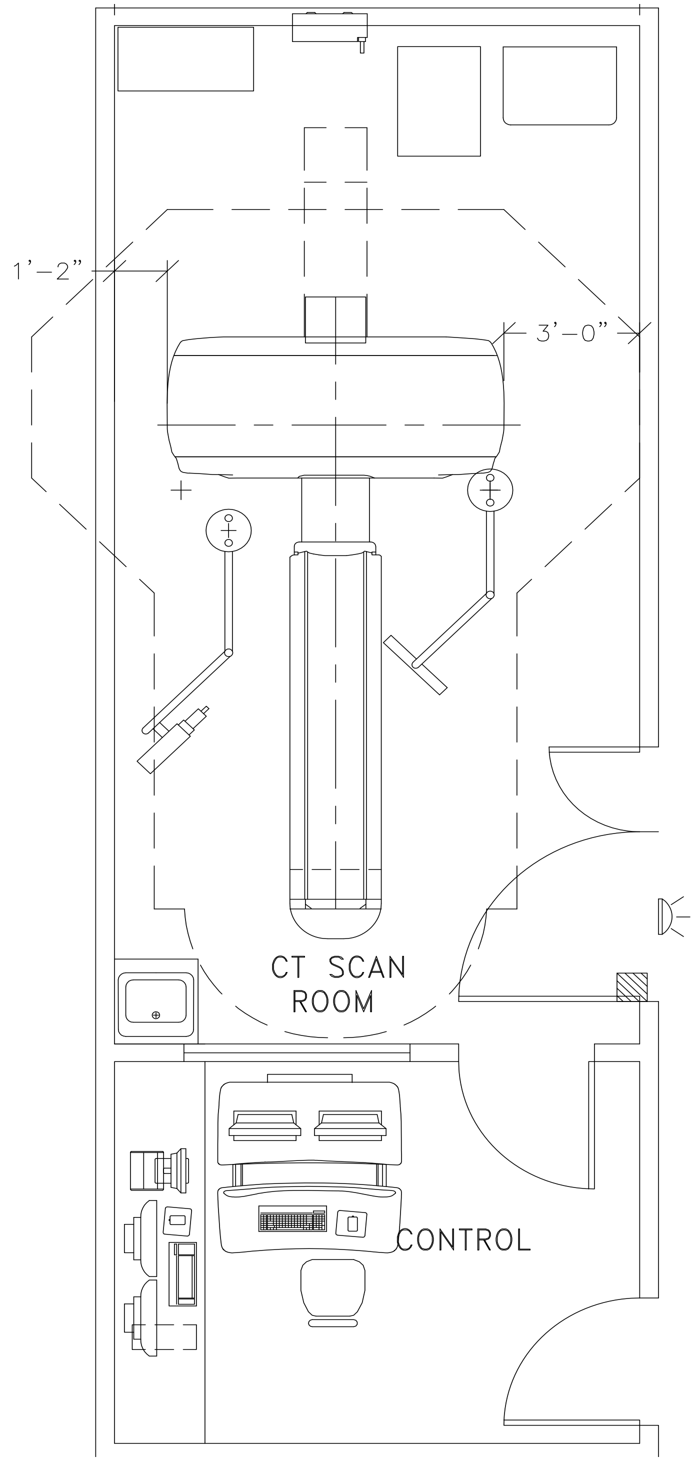



**Figure 4-24 Minimum room Size–Layout A**

**Room B** provides less than 711 mm (28 in.), but greater than 256 mm (14 in.) of space on the gantry left side, measured from covers to left-side wall, compromising service, egress, and workspace on the gantry's left side.

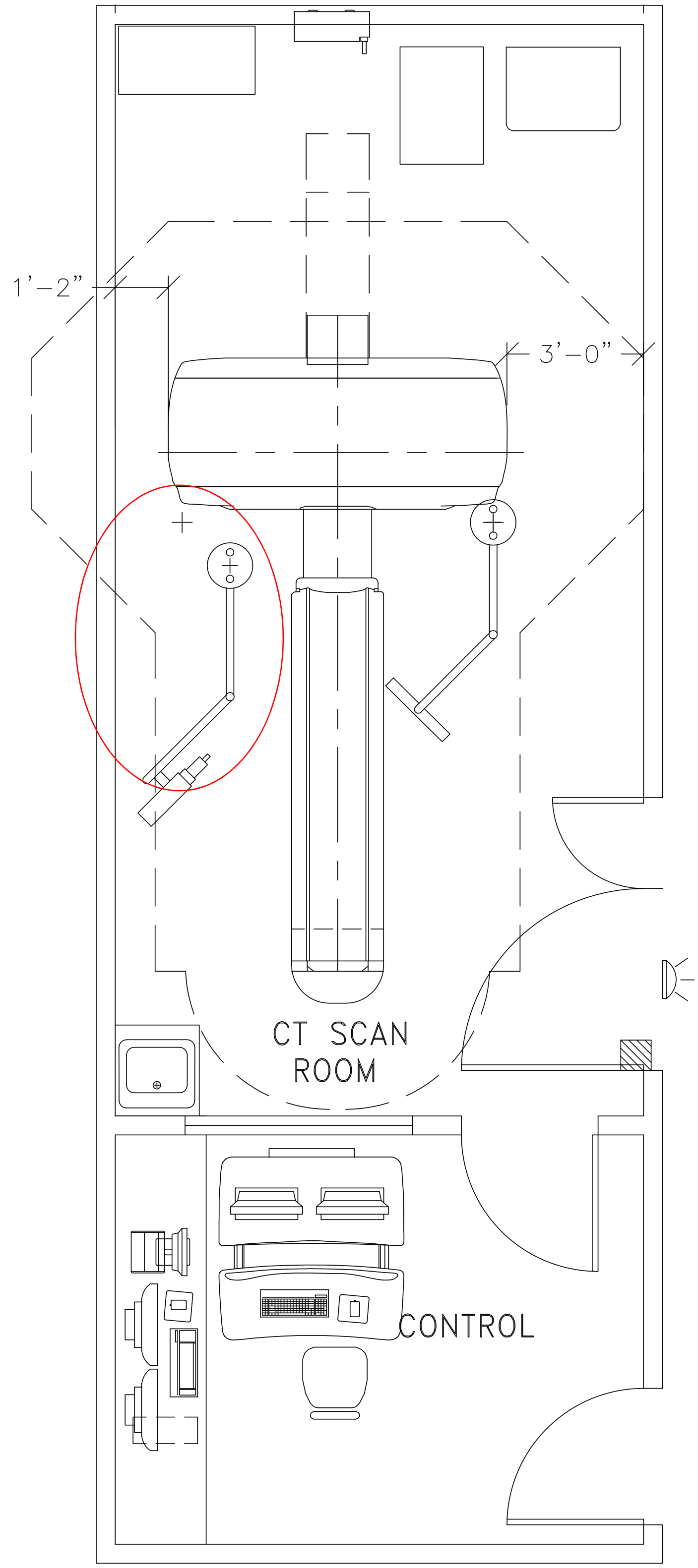



**Figure 4-25 Minimum Room Size–Layout B**

# Section 8.0 Suggested and Typical Room Layouts

Figure 4-26 and Figure 4-27 show the recommended and typical room layouts, both with and without in-room cabinets. You need to know the locations for medical gas, surface ductwork, or other items that make a grounded wall.

Note: Your room layout may meet the recommended or typical room requirements but appear different than Figure 4-26 or Figure 4-27. Your salesperson can provide a detailed room layout for your site

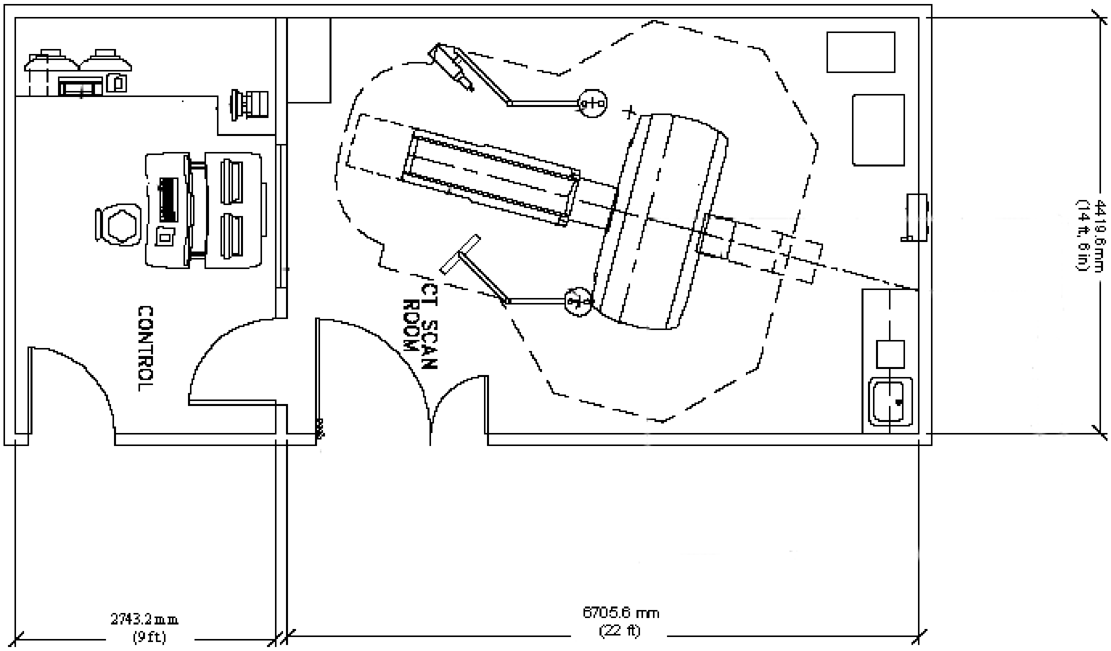



**Figure 4-26 Room Layout with Cabinets**

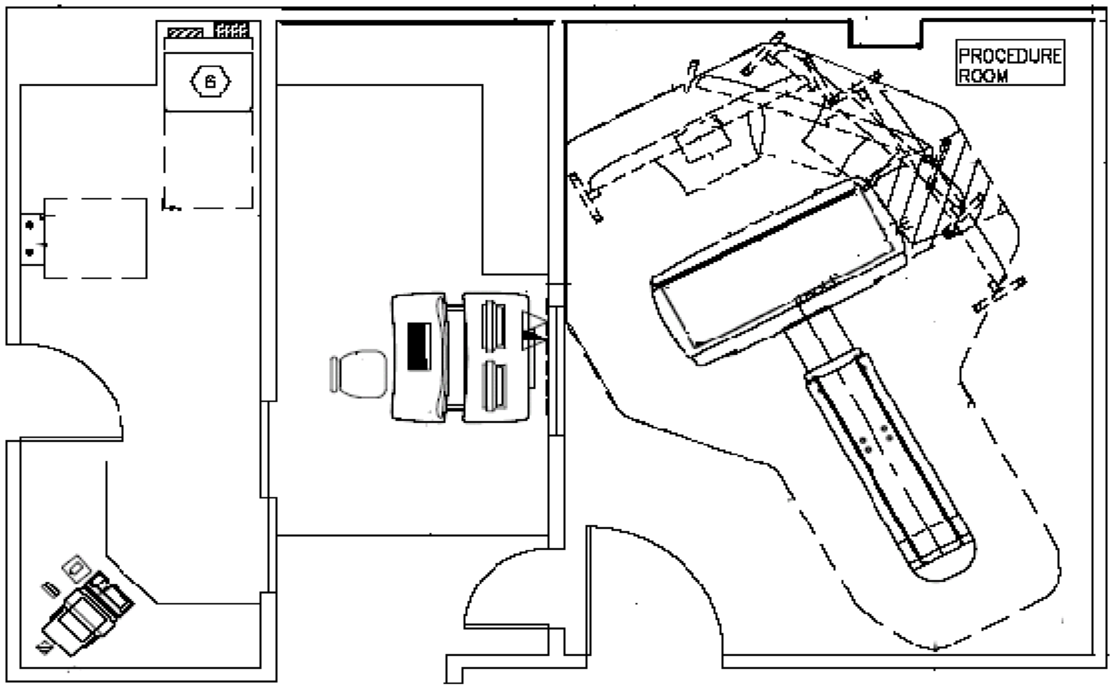



**Figure 4-27 Room layout without Cabinets**

Note: The room depicted in Figure 4-27 shows ample space for the removal of both front and rear covers.

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